DTU Compute
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Designing a Vocabulary Acquisition Support Tool for Adult Danish Learners

Master Thesis



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Abstract

Adults learning a second language meet new vocabulary across scattered everyday contexts, in class, in conversation, in what they read, and retaining a word depends on capturing it where it is met and returning to it deliberately over time. That sequence rests with the learner, who has to notice which words are worth keeping, record them before they are lost, and decide what to practise and when; words met in passing therefore seldom reach sustained study, particularly once the structure of a class is removed. Recent advances in generative AI change what a vocabulary tool can be built around: practice material for any word a learner records can be generated on the spot, rather than drawn from a set authored in advance.

This thesis addresses that problem through Research through Design. Semi-structured interviews with four adults enrolled in Danskuddannelse 3 were analysed through framework analysis to establish where their vocabulary practice breaks down and to produce the problem statement the design answers: how a system can be designed to support learner autonomy in retaining self-encountered vocabulary. Three candidate concepts were developed and compared, and one was realised over three build cycles, each closing in validation with learners. Five learners then used the deployed web application unsupervised for seven days in a single-group pretest–posttest study combining a self-regulation questionnaire, automatically recorded usage, and a standard usability questionnaire.

The resulting system pairs two loops: a vocabulary loop that carries a word from capture, through resolving its meaning, to retrieval practice generated on demand by a large language model, and around it an autonomy loop, a weekly self-regulative cycle of goal setting and reflection that keeps learner to engage with vocabulary loop turning. The five learners used the application 41 times for a combined 289 minutes, captured 179 words between them, and finished 27 of the 36 practice sessions they opened. Four reported higher self-regulation after the week and none lower. Usability averaged 80.5 on the System Usability Scale.

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1 Introduction

Language is the primary instrument through which people communicate, access knowledge, and participate in social and professional life. In an increasingly interconnected world, the ability to operate in more than one language has become a practical necessity for millions. A 2024 survey of over 26,000 EU citizens found that three out of five Europeans can hold a conversation in a language other than their mother tongue, and 86% believe everyone should speak at least one foreign language, and 69% more than one additional language [1].

People pursue second language learning for a wide range of reasons: career advancement, migration, education abroad, and cultural participation [2]. Although modern translation technologies can facilitate communication [3], fluency enables something beyond that: genuine participation, the ability to build relationships, and the confidence to navigate everyday life independently [4].

Learning a language as an adult is a significant undertaking. Unlike early childhood language development, which unfolds through years of immersive naturalistic exposure, adult learners must engage deliberately with the target language to make progress [5]. Among the many challenges of language learning, vocabulary stands out as the most immediate. Following a conversation or reading a text with confidence requires a sufficient stock of words to make the rest of it comprehensible [6]. This thesis focuses specifically on vocabulary acquisition: how learners encounter, capture, and retain new words, and how digital tools can better support that process.

This study is set among adults learning Danish in Denmark. Danish has acquisition difficulties of its own, an unusual sound structure among them [7], and those are not the subject of this thesis. Its subject is the circumstances these learners study under. They are enrolled in Danskuddannelse 3, the largest track of the publicly provided Danish programme, which is built for adults who are otherwise occupied: classes run outside working hours and commonly follow a full working day, contact time is thin, and the words arrive in class faster than they can be absorbed [8]. Consolidating them afterwards is left largely to the learner. That condition is not particular to Danish, and it is what this thesis designs for.

Language learners typically combine several strategies to learn new vocabulary: naturalistic exposure through media and everyday life, formal classes, and dedicated language learning applications. Beyond these, they rely on further tools for translation, recording, or lookup, each serving a different purpose in a different context. These tools rarely communicate with each other, so the work of bridging them falls to the learner [9]. Lacking a coherent system to capture and consolidate vocabulary, words encountered across these contexts seldom make it into deliberate study. A word heard in conversation, looked up on a phone, and never recorded is a missed opportunity for acquisition, since a single encounter rarely establishes a word and acquisition depends on meeting it repeatedly [6].

Recent advances in generative AI change what a vocabulary tool can be built around. An application could previously teach only words someone had written content for in advance, so supporting an arbitrary word meant paying an author for it, and content fixed ahead of the learner was the economical design [10]. A large language model breaks that coupling: a word the system has never seen can be given a translation, an example sentence and plausible distractors the moment a learner enters it [11]. A tool organised around the

words a learner actually meets, and not around a syllabus chosen in advance, therefore comes within reach of a small system. The capability settles nothing about how it should be used, though. It does not decide who chooses the words, when practice happens, or whether the learner comes back to any of it, and those are the terms this thesis takes up: how the possibility can be turned into a system designed around the real workflows of adult language learners, in a way that strengthens the learner's own effort instead of substituting for it.

This thesis answers that through design research, structured as a Double Diamond [12] and conducted as Research through Design, in which the built artifact is the instrument the work is answered with [13]. Which breakdown a system should target was not known at the outset, so the brief could not be settled from the literature and the problem had to be opened empirically before anything was designed against it. Two diamonds therefore run in sequence (Chapter 3). The first opens the problem and then closes it: a review of second language acquisition research establishes how vocabulary is learned, interviews with active Danish learners establish how that process unfolds in practice and where it breaks down, and the two converge into a final problem statement. The second opens the space of responses to that statement and narrows it to one: design constraints derived from the problem space findings, candidate concepts composed from an inventory of design mechanisms, one concept realised over three build cycles into a deployed application, and a seven-day evaluation in which learners used that application unsupervised in their own everyday context.

1.1 Initial problem statement

Adult language learners encounter new vocabulary across a wide range of everyday contexts, yet many struggle to develop a sustainable practice for learning and retaining the words they meet outside formal instruction. Provision is not what is missing. Digital tools for language learning are plentiful, and the difficulty lies in sustaining independent study with them once the structure of a class is removed. Which point in that practice actually fails is not yet established, and settling it is the work of the first diamond. The statement is therefore an initial one, broad enough to be investigated and too broad to be designed against, and it motivates two research questions:

- How do adult language learners currently manage vocabulary acquisition across their everyday contexts, and where does this process break down?
- How can a system be designed to reduce the friction of vocabulary acquisition and better support it?

The first is answered empirically. Adults learning Danish are interviewed about how they currently handle the words they meet, and what they describe is read against research on how vocabulary is learned, until the two converge on the one breakdown worth designing against. That convergence produces the **final problem statement** (Chapter 5), which replaces the statement above and becomes the brief the second question is answered against: a system is designed to it, built, and put in front of learners to use on their own.

1.2 Thesis structure

The thesis follows the two diamonds, and each chapter reports a part of them.

Chapter 2, Background, sets out the domain the work is situated in: vocabulary within second language acquisition, the digital tools available for it, what large language models change, and the Danish adult education context.

Chapter 3, Overall Methodology, states the research stance and the structure applied across the thesis, and which methods produce each phase.

Chapter 4, Problem Space, reports the first diamond, widening the problem through a literature review and user research, then narrowing it through synthesis.

Chapter 5, Final Problem Statement, states the problem the rest of the thesis is built against.

Chapter 6, Solution Space, reports the second diamond in two parts: the design work, from constraints through to candidate concepts, and the realisation and evaluation, from implementation through to the study with learners.

Chapter 7, Discussion, reads the thesis as a whole and what follows from it.

Chapter 8, Conclusion, states what the work established.

2 Background

The introduction argued that adult learners meet new words across scattered contexts, and that the tools available to them do not cohere into a single practice. Turning that claim into a design problem means setting out the domain it describes. This chapter does so in four steps: the research field vocabulary learning belongs to, so that design decisions answer to evidence rather than intuition; existing digital tools for vocabulary learning; what generative models have recently changed about what such a tool can hold; and the circumstances of the learners themselves.

2.1 Vocabulary within Second Language Acquisition

Second language acquisition (SLA) studies how people develop competence in a language beyond their first, and what conditions shape that development [5, 14]. Rather than treating a language as one undifferentiated object, the field organises work around the interacting subsystems competence is made of: phonology, morphology and syntax, pragmatics and discourse, and the lexicon, which concerns the words themselves and what it means to know one. These are distinct enough to be researched and taught separately, but not separable in use. Knowing a word includes knowing how it is pronounced and how it behaves grammatically; grammatical competence has nothing to operate on without words to fill it.

This thesis is located in the lexical subsystem, for two reasons. Word knowledge strongly conditions comprehension and production across the others [15]. And it is the one subsystem with no natural endpoint: a language's grammar is finite and can in principle be worked through, whereas the lexicon expands for as long as a learner keeps meeting the language. Vocabulary is therefore the area in which sustained personal effort matters most, and matters longest.

Vocabulary learning can be approached from two angles, and the research divides along them. The first asks what conditions produce durable word knowledge: how repeated retrieval, the spacing of practice, and sufficient comprehension of surrounding input support retention [6, 16]. And the second asks what learners actually do, producing taxonomies of the strategies by which they discover, record, and consolidate words, and of the metacognitive strategies by which they manage that work [17, 18].

A design project cannot proceed on the first angle alone. Findings about optimal conditions specify what a tool should do; they say nothing about what a learner will do on a weekday evening after a full day of work, or which practice they will abandon. A system built only on efficiency evidence risks being correct and unused.

2.2 Digital Support for Language Learning

Digital environments have changed the conditions under which vocabulary is met and retained. Learners now select their own material and encounter language never authored for teaching [2], picking up words from captioned video and streaming platforms [19, 20], short-form video [21] and text-based conversation [22]. Each raises the rate at which a learner meets unfamiliar words while leaving entirely to the learner what happens afterwards.

Applications built deliberately for language learning are the obvious candidate for closing that gap. Mobile-assisted language learning (MALL) is language study on mobile devices,

distinguished less by the technology than by its conditions of use: wherever the learner is, in short episodes, on a device that is always present [23, 24, 25]. That suits vocabulary, which decomposes into small units fitting the fragmented moments of a day [26, 27, 28].

Rosell-Aguilar sorts the market by whether an application was built for language learning at all, separating *apps designed for language learning* from *apps not designed for language learning but useful to language learners* [29]. The two groups most used for vocabulary study sit one on each side of that line: **whole-language applications** such as Duolingo and Babbel, which offer a complete course through varied exercises and grammatical explanation [30, 31], and **flashcard applications** such as Quizlet and Anki, built for study in any subject but used to rehearse learner-made sets of words [32, 33, 34]. Table 2.1 sets both groups out alongside a third and much smaller one.

Application	Origin of the words studied	How words are practised	Installs
Designed for language learning: whole-language applications			
Duolingo	Platform: a fixed curriculum sequence	Gamified exercises with repetition and immediate feedback	500M+
Babbel	Platform: a fixed curriculum, by subscription	Structured lessons with explicit instruction and review	50M+
Not designed for language learning: general flashcard applications			
Quizlet	Learner: typed in by hand, or a set copied from another user	Several study modes over learner-built sets	50M+
Anki	Learner: authored card by card, as text or image	Spaced repetition scheduling over learner-built decks	10M+
Designed for language learning: capture-oriented vocabulary tools			
Vocabuo	Learner: captured from browser text or video, words extracted automatically	Flashcards with generated examples, spaced repetition	10K+
SnapWords	Learner: captured as a photograph of an object, converted to a labelled word	Review of captured words, with no scheduling	5K+

Table 2.1: Commercial vocabulary applications characterised by where the words a learner studies originate, the dimension Rosell-Aguilar's taxonomy [29] does not record, though the group labels are its own. Installation brackets are those reported by the Google Play store [35].

The question this thesis puts to those applications is one the standard classification does not ask: where do the words a learner studies come from? The second column of Table 2.1 answers it: either the platform supplies the vocabulary, as Duolingo and Babbel do, so that words met outside the application remain outside its scope; or the learner types it in by hand, as Quizlet and Anki do. A third and much smaller group is built exactly for capturing self-encountered vocabulary: Vocabuo and SnapWords save a word at the moment it is met, instead of asking the learner to reconstruct it later.

A systematic review of fifty-one studies of MALL applications for vocabulary learning found the same prescribed-content model dominating [10], which is what leaves words met in class outside what those applications will practise. Systems built to close that gap between classroom and after-class practice do exist, and have been evaluated in research settings [36, 37]: enough to establish the bridge as a recognised design target, few enough to establish how rarely it is attempted.

2.3 Large Language Models and the Content Constraint

A large language model (LLM) is a neural network trained on very large text corpora to predict how text continues, built on the transformer architecture [38] and able to produce text to order from a natural-language prompt [11]. Language teaching was quick to take up the capability [39].

What this opens up shows at two levels. Output is produced one token at a time by sampling from a distribution over likely continuations [11, 40], so language is generated on request rather than retrieved from a store. For the interaction, an interface can accept whatever word a learner brings and return something studiable in the same moment, which moves the design question from which words a tool can cover to what it does with the words it is given. Sampling also means the same word can be practised differently on each return, which is what rehearsal needs and a fixed content set cannot give.

For the engineering, this replaces a content pipeline with an integration. The system stores learner data and asks for language when it needs it, so its own database holds no vocabulary content and the work moves into the prompt. What comes back is free text, so the caller defines the structure it expects and parses defensively. Steering runs through two levers: the prompt, itself a design surface with an established literature [41], and the sampling parameters a provider exposes, most commonly temperature, which narrows the distribution before each token is drawn [40]. Each generation is a remote request, so latency and failure are ordinary conditions to design for.

What it changes for a vocabulary application is where its content comes from, and Table 2.1 shows what the older arrangement produced. When every word a tool teaches has to be authored ahead of the learner, what it covers has to be decided in advance, which is much of why the prescribed-content model dominates [10]. A model lifts that requirement: a word the system has never seen can be explained beyond its direct equivalent, placed in context, and turned into practice material, at the moment the learner supplies it. That is what brings the third group in the table within reach of a small system, since words a learner chooses cannot be anticipated and so cannot be authored ahead of them.

The prospect is drawing active research, young enough that its questions are still open. What generated text can be relied on to do has been evaluated directly, including whether it can be produced at a specified proficiency level [42], which is what decides whether generated material can be matched to where a learner actually is. Vocabulary learning with an AI chatbot has been studied directly [43], and chatbot-assisted vocabulary learning has been examined for how it relates to learners' study behaviour, motivation and performance [44]. Work on generative AI in foreign language learning more broadly reports that learners who come to lean on such assistance can do worse despite otherwise sound study behaviour [45].

That work points at a design question and not only a technical one. A model can supply everything a learner-collected word needs to become studiable, and what the evidence keeps returning to is what it cannot supply: the learner's own decisions about which words matter and whether to come back to them. An application that begins where a learner meets a word is now buildable in a way it recently was not. What such a system should ask of that learner in return is the open question.

2.4 Adult Language Education in Denmark

The setting in which the learners this thesis addresses study is a classroom, and it is also where their vocabulary load is heaviest, since the words are set by the material and not

chosen by the learner.

The empirical work was carried out with adults learning Danish in Denmark, in a publicly provided programme that 66,253 learners attended in 2024 [8, 46]. It draws on Danskuddannelse 3, the largest of the three parallel tracks at 46% of all learners [8].

The track is delivered as six modules taken in sequence, each ending in a test that has to be passed before the next begins [47]. Progression is therefore by demonstrated level and not by time enrolled, which makes a learner's module the most informative single statement of where they are, and this thesis identifies participants by module throughout. Across the six, the programme runs from roughly level A2 to level B2 [47]: an early module indicates a learner who can handle familiar everyday exchanges, a middle one a learner who can manage most situations arising in daily life, and a late one a learner who can follow and produce connected discourse on unfamiliar subjects. The later modules lead to Prøve i Dansk 3, the examination the track is built towards [46].

Two features of that programme bear on how vocabulary is studied. It is built for learners who are otherwise occupied, with attendance required to be possible outside normal working hours, so classes commonly follow a full working day. And contact time is thin: an average of 117 teaching lessons across 2024, the fewest of the three tracks, against which registered absence ran at 28.4% [8].

The learner these conditions describe is committed, externally scheduled, under time pressure, meeting a heavy load of unfamiliar words, and left to consolidate it largely alone. What that learner lacks is not content but a way of carrying the words they self-encountered into deliberate study, which makes the opportunity a *learner-side* one, working within the conditions of the class rather than seeking to alter them.

How that learner in fact manages vocabulary, and where the process breaks down, is the subject of Chapter 4.

3 Overall Methodology

This thesis is a design-research project: it combines empirical, theory-grounded investigation of a real-world learning problem with the design, engineering, and field evaluation of a working software system. This chapter sets out the overall research methodology of the thesis. Section 3.1 situates the thesis within Research through Design and states what adopting it commits the work to: design decisions count as findings, the built system is the instrument through which the second research question is answered, and the problem space is free to reshape the brief that system is built against. Section 3.2 then introduces the Double Diamond as the structure applied across the whole thesis, and describes each of its four phases in turn: what each phase broadens or narrows, which methods produce that movement, and where it is reported. The methods are introduced there at the level needed to follow the overall process, and their detailed application is documented in the chapter in which each one is used.

3.1 Research Through Design

Research through Design (RtD) is a methodology in which building an artifact generates new knowledge, rather than only applying knowledge established beforehand [13]. Design decisions, such as which constraints to impose, which mechanisms to include, which concept to pursue, are part of the inquiry, and the finished artifact is the instrument through which the research question is answered.

Both research questions of the initial problem statement (Section 1.1) call for this approach. The first, how learners currently manage vocabulary acquisition and where that process breaks down, is answered empirically, and its consequence for the methodology is one of ordering: which breakdowns a system should target was unknown at the outset, so the design brief could not be settled in advance. RtD accommodates this, leaving what the problem space finds free to reshape what the solution space builds. The second, how a system can be designed to reduce friction and better support long-term retention, is answered by the design itself, since friction and retention are properties of sustained behaviour and observable only once learners use a working system under real conditions. Building and deploying the artifact is therefore part of the method rather than only its outcome. Because the first question has to converge before the second can be scoped, the thesis runs as two sequential diamonds.

3.2 Double Diamond as Overall Structure

The Double Diamond is a model of the design process built from four phases, Discover, Define, Develop, and Deliver, which alternate between widening and narrowing the work [12]. Discover and Define form the first diamond, the problem space (Chapter 4); Develop and Deliver form the second, the solution space (Chapter 6). Figure 3.1 sets out the whole structure, and each phase is described below in turn.

Discover. The first diamond opens from the **initial problem statement** (Section 1.1) and broadens the understanding of the problem through a domain literature review of second language vocabulary acquisition (Section 4.1), which establishes how that acquisition, and vocabulary learning strategies specifically, are understood to work. It is followed by user research with four active Danish language learners enrolled in Danskuddannelse 3, conducted as semi-structured interviews (Section 4.2). The interviews are analysed using framework analysis [48]: coded against the same structure used to conduct them, then

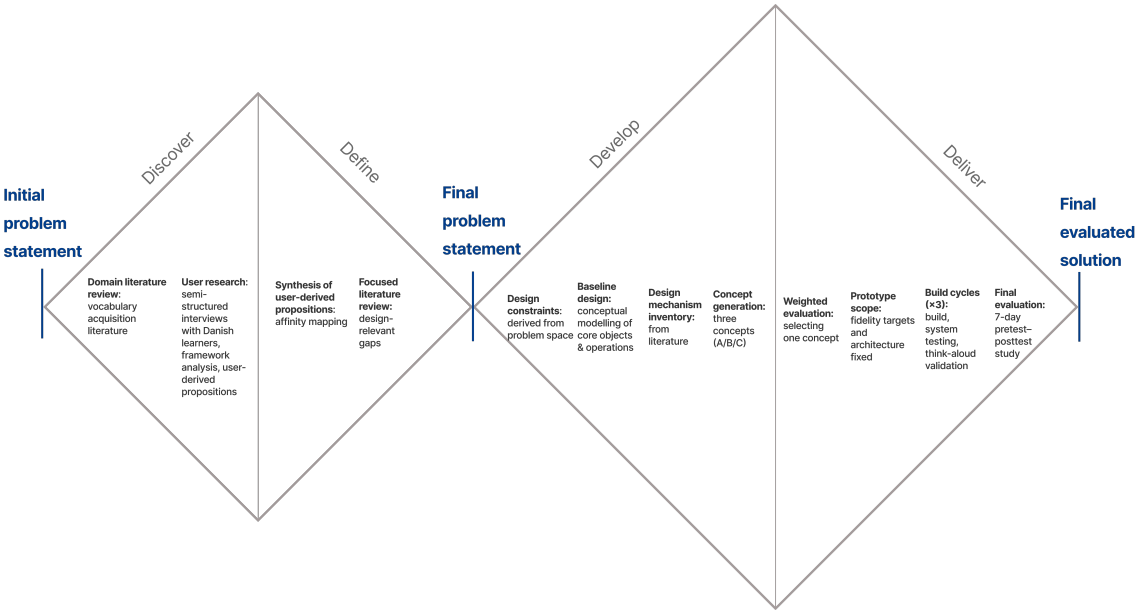


Figure 3.1: The Double Diamond structure of the thesis, applied as two sequential diamonds: a problem space diamond (Discover, Define) producing the final problem statement, and a solution space diamond (Develop, Deliver) producing the results of the solution space.

charted into a participant-by-theme matrix that allows comparison across participants, from which user-derived propositions are extracted.

Define. This phase narrows domain review and interview findings into a single statement of the problem. The user-derived propositions carried out of the framework analysis are regrouped by conceptual similarity through affinity diagramming [49], deliberately without reference to the interview structure that produced them, surfacing the specific points at which learners' vocabulary practice breaks down (Section 4.3). The breakdowns exposed by this synthesis raise design-relevant questions that the domain review was not scoped to answer, and a focused literature review is conducted to close those gaps (Section 4.4). The two converge into the **final problem statement** (Chapter 5), which reframes the initial broad concern with vocabulary acquisition into a single design problem, and is the brief the second diamond is built against.

Develop. The second diamond takes the final problem statement as its brief and broadens the space of possible responses to it. Design constraints are first derived from the problem space findings and grounded in established models of vocabulary learning and self-regulated learning (Section 6.1). What those constraints already settle is fixed as a baseline system, through conceptual modelling of the core objects and operations it needs (Section 6.3), so that what remains open can be explored on a stable base. A focused literature review then yields an inventory of design mechanisms addressing that open part (Subsection 6.4.2), and the inventory is composed into three candidate concepts, each a coherent design direction rather than a checklist of isolated features (Subsection 6.4.4).

Deliver. This phase narrows the three concepts down to one built and evaluated system. The concepts are compared using the weighted objectives method [50], and one is selected (Subsection 6.4.5). How finished that concept had to be is settled before implementation starts, as a set of fidelity targets across McCurdy et al.'s five dimensions [51] and the architecture that delivers them (Section 6.5). The concept is then realised over three build cycles (Section 6.6), each pairing a round of implementation with two evaluations, on the software-engineering distinction between verification and validation

[52]: black-box system testing [53] checks internally that the round behaves as intended, and user validation puts that round's design question to learners, through think-aloud usability testing [54]. What the cycles produce, and what is fixed as the final system, is a deployed web application (Section 6.7). That system is then evaluated with learners using a single-group pretest–posttest design [55]. Participants complete a baseline questionnaire, use the application independently over a 7-day period while their usage is logged automatically, and then repeat it alongside a standardised usability instrument covering their experience of the system [56] (Section 6.8). Section 6.7 and Section 6.8 together present the output of the second diamond, the **final evaluated solution**.

3.2.1 Summary

The work yields two outputs, each closing one diamond. The final problem statement (Chapter 5) closes the problem space and serves as the brief for everything after it. The final system (Section 6.7) and its evaluation (Section 6.8) close the solution space: the system is the artifact through which the final problem statement is solved, and the evaluation is what tests it.

4 Problem Space

This chapter reports the first diamond. Its starting point is the initial problem statement set out in Section 1.1, which frames vocabulary acquisition broadly, and its end point is a single problem the solution space is built against.

Discover widens that problem in two directions: a domain review of what research establishes about vocabulary learning (Section 4.1), and user research with four adult Danish learners establishing what they actually do and where it breaks down (Section 4.2). Define then converges both, first by synthesising the propositions the interviews produced (Section 4.3), then through a focused review of the questions that synthesis raises (Section 4.4). The result is stated as the final problem statement in Chapter 5, the brief for Chapter 6.

4.1 Domain Literature Review: Vocabulary Acquisition

This section reviews what second language acquisition research establishes about vocabulary learning. Chapter 2 located the thesis in the lexical subsystem and identified the two literatures that bear on it: one asking what conditions produce durable word knowledge, and one asking what learners actually do when they meet an unfamiliar word. Both are reviewed here.

The review is organised around a single question: what happens between the moment a learner meets an unfamiliar word and the point at which that word can be said to be known. Both endpoints are easy to state, what separates them is not, and it is where a design intervention would have to act. Figure 4.1 sets the question out.

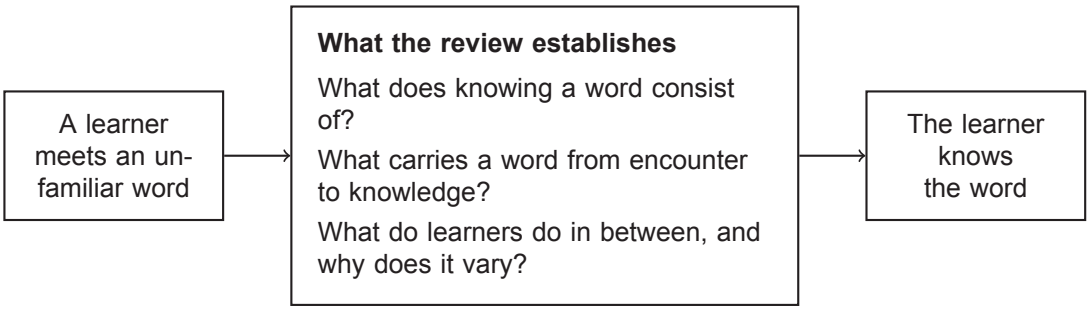


Figure 4.1: The question the domain review is organised around. Both endpoints are given: a learner meets an unfamiliar word, and at some later point knows it. What lies between them is what the review sets out to establish.

The literature was gathered through Google Scholar, in a sequence driven by that question rather than by a fixed protocol. The first searches concerned what it means to know a word, returning the mental lexicon literature and the multidimensional accounts of word knowledge. Those accounts raise how such knowledge is acquired, which led to the distinction between incidental and intentional learning and to the conditions each one depends on: lexical coverage for the first, retrieval practice and its distribution over time for the second. Following the same citation trails to what learners actually do led to the vocabulary learning strategy taxonomies and the studies applying them to particular populations.

The review proceeds in three steps. Subsection 4.1.1 establishes what it means to know a word, which determines what a system must supply when a learner resolves one. Sub-

section 4.1.2 sets out how the lexicon develops, and why deliberate study is required alongside exposure. Subsection 4.1.3 turns from conditions to behaviour, reviewing how vocabulary learning strategies have been classified and what governs a learner's use of them. The section closes by stating what the review settles and what it leaves open, which is what the user research in Subsection 4.2.1 is designed to establish.

4.1.1 What it means to know a word

The vocabulary a learner holds in a given language is referred to in cognitive linguistics as the *mental lexicon*: the internal, organised store of word knowledge a speaker draws on when comprehending and producing language [57]. Unlike a dictionary, which lists words as discrete entries, it is an interconnected network in which words are linked through meaning, sound, grammatical behaviour, and contextual association [58].

The question is therefore not how many words a learner knows but what knowing one consists of. Thornbury argues that word knowledge is multidimensional, encompassing at minimum the word's form, its spoken and written representation, and its meaning, including denotation, connotation, and the contexts in which it is appropriately used [59]. Knowing a word is consequently not a binary state reached at the moment of lookup: a learner may recognise a form without being able to deploy it, and a translation equivalent supplies only one of the dimensions the account requires. This matters for design, because it separates resolving what a word means from acquiring the knowledge needed to use it, and the tools learners have to hand commonly conflate the two.

4.1.2 How the lexicon develops

Lexical development refers to the process by which a learner's mental lexicon grows over time, encompassing both the expansion of vocabulary size and the deepening of knowledge about individual words [60]. SLA research identifies two primary mechanisms through which this development occurs, incidental learning and intentional learning, both of which are considered essential for robust lexical growth [61].

Incidental vocabulary learning occurs as a by-product of meaning-focused activity such as reading, listening, or conversation, without a deliberate intention to memorise specific words [6]. Words are acquired through repeated encounters in context, building form–meaning connections without directed study. This depends on the learner comprehending the surrounding input, which is a matter of lexical coverage, the proportion of words in a text or utterance that a learner already recognises [62]. Numerous studies point to 95–98% as optimal for both comprehension and incidental acquisition [6, 63, 64]. Below that, the density of unknown words exceeds the learner's capacity to infer meaning from context, so incidental learning rests on a foundation of vocabulary already held.

In practice, however, incidental learning alone is rarely sufficient for systematic lexical development. Reaching the 95% coverage threshold required for basic comprehension demands knowledge of at least 4,000 word families, a level most early and intermediate learners have not yet achieved [65, 64]. As a result, large portions of authentic input remain incomprehensible and yield little incidental acquisition.

That is the argument for intentional vocabulary learning. By deliberately studying new words, learners raise their lexical coverage in a targeted way [6]. The two mechanisms are not competing but mutually reinforcing: deliberate study raises coverage, which makes more input comprehensible, which in turn creates more opportunity for incidental acquisition [66, 6].

A further complication is lexical attrition, the loss or degradation of previously acquired vocabulary [67]. Without sufficient rehearsal and use, previously acquired words can be

forgotten entirely, meaning that the mental lexicon is a dynamic state requiring active maintenance rather than a permanent achievement [68].

This thesis focuses on intentional vocabulary learning as the mechanism through which learners can expand their lexicon deliberately and counteract attrition. It takes several forms, from dictionary lookup and flashcard review to contextual production and mnemonic techniques [6, 69, 70]. Among them retrieval practice matters most, because it requires the learner to recall a word's form, meaning, or use from memory rather than only re-read it.

The testing-effect literature supports this. Testing improves long-term retention compared with restudy, and retrieval is itself a mechanism of learning rather than only a measure of it [16, 71]. In vocabulary learning this means that exercises requiring learners to recognise, recall, or produce words are learning events and not only assessments. Retrieval works better still when repeated over time: distributed practice research shows that spacing produces stronger long-term retention than massed practice, which makes spaced repetition a practical way to organise repeated retrieval [72, 73, 74]. Retrieval practice is not the only useful vocabulary learning strategy, but it connects intentional study to long-term retention most directly, and is therefore the foundation the design work builds on.

4.1.3 Vocabulary learning strategies

The literature reviewed so far establishes what has to happen for a word to be learned. What learners actually do belongs to a second literature, and it bears on this thesis more directly, because any system that is built takes a position on it whether or not the position is deliberate. Every vocabulary tool embodies a strategy: whatever it makes easy becomes the strategy its users adopt, so an application offering recognition-based flashcards has selected rote consolidation on the learner's behalf. The artifact participates in the learner's strategy rather than merely supporting it [9], so the strategies learners already employ have to be understood before a system intervenes in them.

Vocabulary learning strategies (VLS) are the techniques learners employ to acquire new words, discover their meanings, and retain them [75, 76]. Classifying them is meant to allow systematic comparison of what learners do and of how effective different approaches are. The classifications serve the first aim better than the second, in two respects that both bear on this thesis.

The first is that the field has produced several classifications organised on incompatible principles. Schmitt's, the most widely adopted in vocabulary research, sorts fifty-eight strategies by their place in an episode, into discovery strategies used on first meeting an unknown word and consolidation strategies used to retain it [17]. Oxford organises by psychological function instead, across language learning as a whole rather than by position in a vocabulary episode [77]. Ma, holding a two-category division too coarse to capture the sequential character of acquisition, organises by memory process into four stages running from discovering a word to consolidating it through use [78]. Each is defensible on its own terms, but they cut the same behaviour along different axes and the field works with all three.

The second follows from the first. Empirical studies apply a single taxonomy, or an extension of one, to a particular learner population and report which strategies it uses [79, 80, 81, 76]. This has produced descriptive breadth, but the results stay indexed to the populations they came from, describing what those learners report doing rather than what a given learner would gain most from doing.

The reason lies deeper than sampling: effectiveness is a property of the fit between a strat-

egy and its circumstances rather than of the strategy alone. Gu argues that the choice, use, and effectiveness of a strategy depend jointly on the learner, the task, and the learning context [18]. Synthesising forty-five empirical studies, Boonkongsaen groups the recurring influences into learner-individual factors, such as motivation and prior language learning experience, and social or situational factors, such as class level and opportunities for interaction [82]. A strategy that suits a learner with daily opportunities to use the language may prove unworkable for one whose contact with it is confined to a weekly class, and may be abandoned by that same learner when the task shifts from an examination to a conversation.

The consequence is specific. An account of the most efficient way to learn vocabulary, with learner, task, and context held together, lies outside what the strategy literature offers, and the contingency above suggests any such account would hold only for the population it was derived from. Proceeding on laboratory effectiveness alone would risk a system that is efficient in principle and abandoned in practice. This thesis therefore answers the open question empirically: which strategies this population employs, under their actual conditions, and at which points the process fails them. The same contingency carries a methodological consequence, since the framework for that investigation has to account for learner, task, and context alongside strategy, which motivates the choice of framework in Subsection 4.2.1.

4.1.4 Summary of the domain review

The review opened by asking what happens between meeting an unfamiliar word and knowing it. The literature's answer is that nothing happens on its own, and that what separates meeting from knowing is neither a single event nor a fixed set of them, but something individual to the learner. Usually the word has to be resolved along more than one dimension, its written and spoken form as well as its meaning, connotations, and the contexts in which it is used. It then has to be brought back deliberately and repeatedly after the encounter, and maintained afterwards against loss. Which strategies serve a learner best at each of those points varies widely, and no classification of them is fine-grained enough to hold for all.

Each of those points becomes a requirement on the design. Support at the moment of encounter has to supply more than a translation equivalent, since a translation covers one dimension of several. Deliberate study has to be built into the system rather than left to exposure, because at the proficiency levels this thesis addresses coverage is too low for incidental acquisition to carry the load, and retrieval practice distributed over time is the best-evidenced form that study can take. Because words are lost without use, the system has to support maintenance rather than one-off acquisition.

What none of this settles is whether any of it happens for a given learner. The literature states what has to be true for a word to be retained, and what options exist for making it true. Whether those conditions are met, and which option a particular learner will sustain, depends on circumstances no review can supply. The following section therefore turns from what the literature establishes to how the adult Danish learners studied here handle vocabulary in practice, and where that handling breaks down.

4.2 User research

The domain review closed on a question the literature cannot settle from the outside: which strategies this population employs under its actual conditions, and at which points

the process fails them. That question is empirical, and this section reports the investigation designed to answer it, conducted as semi-structured interviews with four adults enrolled in Danskuddannelse 3.

An investigation of this kind needs an analytical framework before it needs an interview guide. Vocabulary learning is not a single observable behaviour but a set of loosely connected episodes spread across settings, tools, and weeks, so what the interviews could see depended on what the framework directed them to ask.

The section proceeds in three steps. Subsection 4.2.1 sets out the criteria a framework had to meet, compares the candidates against them, and adopts Gu's tetrahedral model as the primary organising structure with distributed cognition as a second lens. Subsection 4.2.2 describes how the two shaped participant selection, the interview structure, and the analysis. Subsection 4.2.3 reports what the interviews established.

4.2.1 Choice of analytical framework

Answering that question required a framework capable of structuring inquiry into a complex, situated human activity. Subsection 4.1.3 established what an account of this kind requires: strategy use is contingent on the learner, the task, and the context, so a framework that records strategies alone cannot explain why the same strategy succeeds for one learner and fails for another. Two criteria follow. The framework must structure inquiry across person, task, and context alongside strategy, and it must be specific enough to vocabulary learning to direct attention to the points within a learning episode at which the process can break down. Table 4.1 presents the frameworks considered against these criteria.

Framework	Origin	What it addresses
Activity Theory [83]	HCI & cognitive science	How people pursue goals through interactions with tools and within social and organizational contexts
Situated Action [84]	HCI & cognitive science	How action emerges in context; people adapt their behavior based on the specific situation, available resources, and unfolding circumstances rather than following pre-defined plans
Distributed Cognition [9]	HCI & Cognitive Science	Cognitive work as distributed across people, artifacts, and the environment rather than confined to an individual mind
VLS taxonomy [17]	SLA	Classification of vocabulary learning strategies by cognitive type into discovery and consolidation categories
Tetrahedral model [18]	SLA	Vocabulary learning as a problem-solving activity shaped by four interacting dimensions: person, task, context, and strategy

Table 4.1: Analytical frameworks considered for structuring the user interviews.

Activity Theory is among the most widely adopted frameworks in HCI for investigating complex human behaviour. It treats activity as hierarchically structured around goals, mediated by tools, and shaped by social and cultural conditions [83]. It meets the first criterion, since goals, tools, and social conditions cover much of what varies across learners.

It does not meet the second: it says nothing about the internal structure of a vocabulary learning episode, which would have to be imported from elsewhere.

Situated Action offers a contrasting perspective. Rather than treating behaviour as goal-driven, it understands action as an improvised response to the situation as it unfolds, shaped by the resources at hand rather than by pre-formed plans [84]. This lens suits the moment of an unexpected word encounter. The learners studied here, however, are already committed to engaging with unknown words, and what the investigation needs to follow is what becomes of that intention over time, which a goal-oriented framework represents better.

Distributed Cognition holds that cognitive work is not confined to an individual mind but is distributed across people, artifacts, and the environment [9]. This bears directly on vocabulary learning, where dictionaries, notebooks, and flashcard applications participate in the cognitive process rather than merely supporting it. Like the two frameworks above it supplies no domain structure, so it fails the second criterion as a primary framework. The property it isolates, whether the work at a given stage is carried by the learner alone or by something outside them, is captured by none of the others.

Within SLA, the investigation could instead classify what learners do directly, using an established **VLS taxonomy** such as the one described in Subsection 4.1.3 [17]. This meets the second criterion and would return a well-organised inventory of participant behaviour. It fails the first, for the reason that section set out: a strategy-only lens records the choice without the person, task, and context that account for it.

The **tetrahedral model** meets both criteria. Its four dimensions cover person, task, and context alongside strategy, and it is specific to vocabulary learning [18], combining the goal-directed structure of Activity Theory with the domain-specific concerns of VLS research.

This study therefore adopts the tetrahedral model as the primary organising framework for the interviews, complemented by Distributed Cognition as a second analytical lens. The tetrahedral model structures what participants are asked about; Distributed Cognition directs attention to how the cognitive work at each stage is supported or impeded by the tools and people around them. The model is set out in full below, and what the second lens contributed to the interview guide is described in Subsection 4.2.2.

The adopted framework: the tetrahedral model

The tetrahedral model treats vocabulary learning as a problem-solving activity in which the choice and effectiveness of learning strategies depend jointly on the learner, the task they are trying to accomplish, and the context in which learning takes place [18]. Each dimension is described below.

- **Person** refers to the individual characteristics a learner brings to the encounter, including age, motivation, and prior linguistic background.
- **Task** is the objective a learner is trying to reach, for example, remembering the meaning of a word or preparing for a language exam.
- **Context** is the environment that defines what strategies are appropriate or available, including the classroom, available tools, and social setting.
- **Strategy** is a sequence of deliberate cognitive operations a learner employs to achieve the task at hand, ranging from guessing meaning from context to rehearsal and active use [18].

Alongside this four-dimensional view, Gu describes vocabulary learning as a set of related sub-tasks that unfold when a learner meets an unknown word [18]. These are illustrative

of the cognitive operations that may occur across a learning episode rather than a formal taxonomy. Three of them bear on this investigation.

Meaning resolution is the process by which a learner resolves the meaning of an unknown word, either by inferring it from surrounding context or by consulting an external source [6].

Recording is the act of externalising a newly encountered word into an artifact such as a notebook, a phone note, or an annotation, so that it can be revisited later [18].

Retention is the process of consolidating a word into long-term memory through repeated and sufficiently deep engagement [18]. The strategies learners employ at this stage vary considerably, ranging from shallow techniques such as rote repetition [85] to deeper encoding approaches such as spaced repetition [86].

None of the three occurs entirely within the learner's mind. Each involves a tool, another person, or the environment, which is what makes distributed cognition useful as a complementary lens.

4.2.2 Methods

Semi-structured interviews were conducted with learners attending formal Danish language education. The format follows from what is being investigated: a vocabulary practice is not observable in a single sitting but reconstructed from what a learner reports doing across weeks, and a semi-structured guide keeps the dimensions of the tetrahedral model covered while leaving room to follow an account the guide did not anticipate. The population was chosen because learners in formal education already have an established practice shaped by their classes, which gives the interview something concrete to ask about.

Participants

Four participants took part, all of them enrolled in Danskuddannelse 3, the track described in Section 2.4, and all of them attending classes for at least one year. They were aged 23–27 and were either master's students or young professionals, and were recruited through personal networks and Danish language school contacts.

Prior to the interview, all participants signed a written consent form (see Appendix B) agreeing to participation and audio recording for transcription purposes. All data was anonymised prior to analysis, and audio recordings were stored securely and will be deleted upon completion of the thesis.

Interview Structure

The interview was organised into four blocks. Three of them follow the dimensions of the tetrahedral model: person, context, and tasks and strategies. Task and strategy were combined into a single block, since strategies are inseparable from the tasks that motivate them, a learner's choice of strategy being a response to the goal they are pursuing at that moment. Recording a new word is a task, but whether a learner writes it in a notebook, types it into their phone, or annotates the margin of a text is a strategic choice shaped by the context of that encounter. A fourth block, reflection, was added so that participants could assess their own practice rather than only describe it.

Each interview lasted approximately 20–30 minutes.

Block 1: Person. This block established the participant's background as a language learner: their motivation for attending Danish classes, duration of study, prior language learning experience, and the time they invest in vocabulary learning outside the classroom.

Block 2: Context. This block explored the environments in which participants encounter Danish, in class, in conversation, and in everyday life, and what resources each of those environments makes available to them.

Block 3: Tasks and Strategies. This block traced what happens when a participant meets an unknown word, from the moment of encounter through to whatever becomes of the word afterwards.

Block 4: Reflection. This block invited participants to reflect on their vocabulary learning as a whole: what they found frustrating, what they wished worked differently, and where their current strategies felt insufficient.

Distributed cognition formed no block of its own but ran across the others. Whenever a participant described doing something, they were asked what carried the work besides themselves, which application, notebook, or device and which people, and then what their experience of it was: what helped, what was effortful, and whether they had abandoned a tool previously and why.

The four blocks served as a starting point for the conversation rather than a rigid script, and their order and emphasis varied across participants depending on the direction the interview took. Questions were phrased in everyday language, and participants were encouraged to describe their experience in their own terms.

Data analysis

Interviews were analysed using framework analysis [48], which codes data against a structure fixed in advance rather than one built up from the transcripts. The tetrahedral model supplied that structure, so the interviews were analysed against the same dimensions used to conduct them. Interviews were first transcribed and reviewed to establish familiarity with the data, and each transcript was then read against the four blocks, which structured the initial coding pass.

The coding frame is hybrid in the sense that its two levels came from different places. The top level was applied deductively: every segment of a transcript was assigned to person, context, tasks and strategies, or reflection. Nothing below that level was set beforehand, and the sub-themes within each block were derived inductively from what participants described. The person block resolved into learning background and time invested outside class, the context block into where participants encounter Danish and where they use it, and the tasks and strategies block into the stages a word passes through from encounter onwards. The tasks reported in Subsection 4.2.3 were derived in this way rather than taken from an existing VLS taxonomy, for the reason given in Subsection 4.2.1: an imported taxonomy would have fixed in advance what the investigation set out to find. The model therefore supplies the skeleton and the transcripts supply what hangs on it.

Coded data was organised into a framework matrix in which rows represent participants and columns the identified themes, allowing systematic cross-participant comparison (Appendix C). The matrix was then interpreted for patterns, similarities, and differences across participants.

Following each block, user-derived propositions were extracted to summarise the patterns observed. Propositions are concise, generalisable statements derived directly from empirical observation, intended to carry a finding beyond the individual account it came from into a claim about the user population [87]. They are the unit of analysis carried forward into the affinity diagramming and synthesis.

4.2.3 Findings

These findings report the results of the semi-structured interviews. The full framework matrix is presented in Appendix C.

The section follows the three analytical blocks of person, context, and tasks and strategies, and closes with a distributed cognition analysis of how the cognitive work is supported or impeded by the tools and social resources available to participants. Each part ends with user-derived propositions [87], presented as **UP X**, which are synthesised at the end of the section.

Person

The four participants are summarised in Table 4.2. The spread in their module levels reflects different lengths of study rather than differences in motivation or commitment.

Participant	Occupation	Origin	Other languages	Time studying Danish	Danish level
P1	Engineer	Poland	Polish, English	3 years	Module 2
P2	MSc student	Argentina	Spanish, Catalan, English	1 year	Module 3
P3	MSc student	Germany	German, English	2 years	Module 5
P4	MSc student	Poland	Polish, English	1 year	Module 3

Table 4.2: Overview of study participants

All four shared a broadly integrative motivation: they are internationals in Denmark and treat the language as a means of participating in Danish society rather than a professional requirement. As one participant put it:

> *"In work conditions, everyone speaks English, but just in normal settings, everyday life usage, knowing Danish is really useful."*

Attitudes towards learning differed despite the shared motivation. Some participants described structured, test-driven habits from prior schooling that their current classes do not impose.

> *"I was just lazy to learn new words on my own. I need someone harsher to give me like, next week you have to do this, otherwise you get a bad grade."*

Among the participants who relied on external structure such as mandatory tests or teacher-assigned tasks, a further pattern emerged: they were also the ones more likely to describe themselves as lazy, attributing limited progress to personal failings rather than to the absence of supportive tools or environments. Both, P1 and P4, share a Polish educational background. No causal claim can be made from two participants, but the pattern is consistent with research suggesting that educational culture shapes learners' expectations of how vocabulary learning should be organised and supported [88].

Three participants, P1, P3, and P4, use or wish to use consolidation strategies such as spaced repetition. P2 instead defines their learning by exposure and practice in real-world situations, treating the classroom as a complement to it. The contrast tracks prior language learning experience rather than current proficiency.

UP 1. *Learners with prior experience of test-driven educational systems expect external accountability and struggle to self-regulate deliberate study in its absence.*

UP 2. *Positive prior experience with specific tool types, particularly progress-tracking flashcard apps, shapes learners' expectations of what effective vocabulary learning should feel like and what tools should provide.*

UP 3. *Learners orient toward either exposure-driven or consolidation-driven strategies based on prior language learning experience, and this orientation shapes which learning tasks they invest in and which they neglect.*

Context

For most participants, the Danish language class was the primary environment in which they encountered and engaged with Danish. Outside the classroom, exposure was limited and largely opportunistic: interactions at shops, ambient exposure in everyday life, and reading encountered across a range of sources, from Danish news and physical books to official letters arriving through e-Boks. The setting of an encounter shapes what resources are available at the stages that follow, since a word met while reading on a device is within reach of a dictionary or capture tool in a way that a word heard in passing conversation, or read on paper, is not.

Participants described their classes differently. P1 drew a sharp contrast between two: their previous class was content-heavy but provided no practice, which they blamed for the vocabulary attrition they experienced during a two-year break, while their current class is conversation-based but introduces no new content.

The volume of new vocabulary divided them as well. Two found the amount sufficient, one felt there was not enough, and P4 described it as often overwhelming:

> *"In class there are just so many new words every time, I write some down but by the next lesson half of them are gone."*

Despite this variation, all four agreed that classroom exposure alone was insufficient to move words into long-term knowledge.

Outside the classroom, the social environment shaped how much Danish participants encountered and could practise. P1 and P4 were surrounded by English speakers, with workplaces and social circles providing no regular opportunity to use Danish productively. P1 identified this as the single biggest barrier to fluency:

> *"You don't get any practice outside of the class if you don't really force yourself, because people around you, they don't speak Danish to you."*

A recurring frustration was the tendency of Danish speakers to switch to English the moment a learner hesitated or made an error, withdrawing an opportunity for spoken practice just as it began:

> *"I start speaking Danish and maybe I hesitate or I pronounce something wrong and they switch to English."*

P2 and P3 had more ambient Danish available to them. P2 actively initiated Danish conversations in everyday situations, supported by a native-speaking landlady and regular contact with shopkeepers and coworkers, and described real-world exposure as their primary mechanism for acquiring new vocabulary. P3 had occasional conversations outside class, less deliberate but still more than P1 and P4.

UP 4. *Danish language classes vary considerably in structure and content across teachers and modules, ranging from content-heavy instruction with no practice to conversation-based classes with no new vocabulary input.*

UP 5. *A single classroom encounter with a word, without subsequent recording, rehearsal, or use, is broadly insufficient for retention.*

UP 6. *Opportunities for spoken practice outside class depend on availability of Danish-speaking contacts.*

UP 7. *Opportunities for spoken practice outside class depend on willingness to initiate Danish in everyday situations.*

Tasks and Strategies

As described in Subsubsection 4.2.1, tasks and strategies are interrelated: tasks represent what the learner is trying to achieve, and strategies are how they attempt to achieve it. Analysis of interview transcripts using framework analysis identified six recurring tasks that participants engage in as part of vocabulary acquisition. Tasks are presented descriptively, grounded in participant behaviour, and broadly consistent with the sub-task structure described by the tetrahedral model [18].

Task	Description
Resource seeking (outside classroom)	Deliberately seeking out Danish language input outside formal instruction, for example through reading, media, or social interaction, to increase exposure and encounter new vocabulary in context.
Selective engagement	Deciding which unknown words are worth attending to, based on criteria such as frequency, or relevance, rather than engaging with every unknown word encountered.
Meaning resolution	Resolving the meaning of an unknown word, either by inferring it from surrounding context or by consulting an external source such as a dictionary or translation tool.
Recording	Externalising a newly encountered word into an artifact such as a notebook, document, or app so that it can be revisited later.
Retention	Consolidating a word into long-term memory through repeated and sufficiently deep engagement, ranging from rote repetition to spaced retrieval practice.
Learning scheduling	Planning and regulating when and how vocabulary study takes place outside class, including decisions about how much time to invest and which words to prioritise.

Table 4.3: Vocabulary learning tasks identified through user research.

These tasks are not sequential stages but independent sub-goals that learners pursue in different combinations, orders, and frequencies depending on the personal and contextual factors described above. The effort invested in a task also varies with external pressure. P3 acknowledged that flashcard review works well when applied, but described the behaviour as reactive, activated by an approaching exam rather than sustained as a habit.

Table 4.4 summarises each participant's strategies across the six tasks, colour coded by the participant's own perception of their strategy use.

A pattern runs across the matrix. P2 reports the most functional set of strategies, with only meaning resolution presenting a mixed experience. P3 is similar in most tasks, handling meaning resolution well, though recording and retention remain contingent on external factors such as exam pressure. P1 and P4 show clear pain points across multiple tasks: both lack any active retention or learning scheduling strategy, and neither records vocabulary consistently. Across all four, recording and retention are the most prevalent pain

Task	P1	P2	P3	P4
Resource seeking (outside classroom)	Opportunistic reading, prior Duolingo use	Daily Danish exposure, proactive reading	Leisure reading	Duolingo only
Selective engagement	Notices by frequency	Notices by part of speech	Notices by frequency	No strategy
Meaning resolution	Google Translate; no context	Context inference; digital backup	DeepL; Kindle dictionary	Glosbe; sentence-level lookup
Recording	Notebook in class only	Grammatical notebook; inconsistent	Word document; not revisited	No strategy
Retention	No strategy; prior flashcard use	Interest-driven encoding, by practise	Flashcards; exam-driven only	No strategy; prior Quizlet use
Learning scheduling	No strategy	Opportunistic; no schedule	Exam motiveted	No strategy, wish for external structure

Table 4.4: The strategy each participant employs for each of the six vocabulary learning tasks derived by framework analysis. Rows are tasks, columns are participants, and each cell states the strategy that participant reported using for that task. Colour reflects the participant's own perception of it: red where a strategy is absent or consistently fails, yellow where it exists but is partial, inconsistent, or contingent on external factors, and green where the participant reported a functional approach and identified no pain point.

points, while meaning resolution is the most consistently supported task, even if the quality of that support varies.

UP 8. *Recording and retention, are the most prevalent pain points across participants.*

UP 9. *Meaning resolution is the most consistently supported task.*

The following subsections examine each stage in turn, drawing on participant accounts to explain the patterns visible in the table.

Resource seeking (outside classroom)

All four engage in resource seeking outside class to some degree, consistent with the integrative motivation identified in the person block: they do not limit their exposure to class but seek out Danish in everyday contexts. P2 is the most active case, describing real-life conversation and ambient exposure as their primary mechanism for acquiring new vocabulary, with class serving as a complementary resource rather than a central one.

UP 10. *All participants engage in some form of resource seeking outside class, consistent with integrative motivation.*

Selective engagement

Participants described broadly similar approaches to selective engagement, though the mechanisms varied. P1 and P3 attended to words that appeared repeatedly across a

text, treating frequency as a signal of importance. As P1 put it:

> *"When I see it occur multiple times across the text, I'm like, oh, maybe I should know that word. But if it's some random word I've never seen before, and I can get it out of context, I don't really pay attention."*

P2 used grammatical category as a guide, focusing on nouns and adverbs because these carry the core meaning of a sentence, which makes the overall sense easier to follow without knowing every word. P4 reported no specific strategy, engaging with unknown words as they arose when the meaning mattered, without a deliberate filter.

Selective engagement carries no pain point in Table 4.4 because it was neither a source of reported friction nor a stage participants reflected on critically. All four had arrived at workable implicit approaches without deliberate effort, and none expressed dissatisfaction with how they manage it.

UP 11. *Selective engagement is handled implicitly by all participants through self-developed filtering strategies, and none identified it as a source of friction.*

Meaning resolution

Participants used broadly similar digital tools for meaning resolution, but their strategies and the points at which friction arose differed considerably.

P1 relied on a basic digital translator, which provided direct word-level equivalents but lacked the usage context needed to understand how a word functions in practice:

> *"If I got just the word and the Google translation, I wouldn't know how to use it."*

The other three had independently arrived at more contextually rich tools. P2 was unwilling to interrupt their reading flow for individual lookups, preferring to infer meaning from context and read on:

> *"It's annoying to just stop for looking words. I just keep reading and try to understand it as a whole."*

P3 addressed the same friction differently, switching from printed texts to a Kindle, whose built-in dictionary allows lookup without leaving the reading environment. Instead of picking up a phone, opening a translation app, and returning to the text, the lookup happens in the same interface, which removes the disruption. P4 translated at sentence level rather than word by word, to preserve contextual meaning.

Meaning resolution was not a stage at which participants reported major breakdowns, but two recurring needs emerge: access to contextual meaning rather than direct word-level translation, and low-friction lookup that does not interrupt reading flow.

UP 12. *Learners consistently need contextual meaning rather than direct word-level translation*

UP 13. *Learners prefer low-friction lookup that does not interrupt reading flow.*

Recording

Three of the four showed some form of recording behaviour, but none applied it consistently, and P4 had no recording practice of any kind. All four understood why recording matters, though for different reasons. P1 treated it as a prerequisite for later flashcard review, and wanted the manual entry barrier removed:

> *"If there was some automated way to just add it to like a flashcard app, that would be great because then I can just use it in my free time."*

P2 maintained a notebook organised by grammatical category, which served a dual purpose: an encoding strategy in the moment, and a structured reference for later use:

> *"In my notebooks I have a list, for example, of verbs, and I try to classify them by time, or mode, so that I can reuse them later."*

The habit was applied inconsistently, particularly outside class. P3 recorded a few words per class and intended to return to them, though in practice this rarely happened beyond the session itself. For P4 the absence of recording was not a matter of effort but of perceived purpose: without an external structure assigning which words to learn and testing retention of them, recording felt unmotivated.

A further problem was common to all four, the disconnect between where words are encountered and where recording takes place. Recording was almost exclusively a classroom behaviour, and words met outside class, in conversation, while reading, or in everyday life, were rarely captured in any form.

UP 14. *Recording is almost exclusively a classroom behaviour. Words encountered outside class are rarely captured.*

Retention

As established above, the absence of reliable recording left most participants with little material to return to. The breakdown at recording therefore propagated directly into retention: without a captured record of encountered words, systematic rehearsal was not possible in the first place.

P1 recalled a school-era flashcard application that required typed input in both directions, and credited the active typing rather than passive recognition for the retention effect:

> *"Every week we had tests on new words in high school, and I was obsessed with my flashcard app. I would go through it whenever I was on the train or bus."*

P4 described a similar experience with a flashcard application that tracked sessions and gave a sense of measurable improvement over time. P3 used flashcards as a retention strategy, but only in the days immediately before an exam, and described the strategy as reactive rather than habitual:

> *"I think flashcards work well for me, if I really use them and if I don't just write them and then never look at them again."*

P2 was the only participant whose retention functioned without deliberate rehearsal. Daily conversations provided enough re-encounter to consolidate words over time, and organising recorded words by grammatical category functioned as elaborative encoding, requiring active analysis of each word rather than passive copying, which SLA research identifies as deeper and more effective than rote repetition [85].

Retention was therefore a consistently broken task. Where strategies existed they were reactive rather than habitual, and for two of the four participants words encountered in class or in everyday life had no reliable path to long-term knowledge, relying instead on the chance of re-encounter or on memory alone.

UP 15. *Where retention strategies exist, they are reactive rather than habitual, activated by external pressure such as exam deadlines rather than sustained independently.*

Learning scheduling

Approaches to learning scheduling vary considerably. P2 values exposure and real-world use over systematic study and, while aware the approach lacks structure, does not identify this as a problem. P3 recognises that their engagement is exam-driven but frames it as a matter of priority rather than an absence of strategy. P1 and P4 have no scheduling strategy and actively wish for one, describing a desire for externally imposed structure to compensate for the absence of self-directed study habits. As P4 put it:

> *"In my dream world, I would rather have it more structuralized. For instance, teacher gives us every week 20 new vocabulary and every week we do a test of 20 new words."*

UP 16. *Participants who lack self-directed scheduling strategies do not simply need reminders but externally imposed accountability structures.*

Distribution of cognitive work

To examine how the cognitive work of vocabulary acquisition is distributed beyond the individual learner, Table 4.5 maps the social resources and material artifacts each participant draws on across the six tasks identified in Table 4.3. For each task, the table shows whether the cognitive work is carried by the learner alone, delegated to a material or digital artifact, supported by social interaction, or absent entirely.

Task	P1	P2	P3	P4
Resource seeking	**Danish news, online content**	**landlady, shopkeepers media**	**Kindle, Danish texts**	**Duolingo**
Selective engagement	*Internal*	*Internal*	*Internal*	*Internal*
Meaning resolution	**Google Translate teacher**	**Google Translate, ordbog.dk teacher**	**DeepL, Kindle dictionary**	**Glosbe, AI screenshots**
Recording	**notebook (class only)**	**grammatical notebook**	**Word document**	None
Retention	None	**notebook as encoding daily conversations**	**GoodNotes flashcards**	None
Learning scheduling	None	*Internal*	None	None

Table 4.5: Distribution of cognitive work across social resources and material artifacts for each task. **Blue text** indicates reliance on a material or digital tool. **Purple text** indicates reliance on other people. *Internal* indicates the task is handled entirely by the learner's own cognition. None indicates the absence of any strategy.

Comparing Table 4.4 with Table 4.5 relates the distribution of cognitive work to the presence of pain points. P2 shows the most distributed work, across both social resources and artifacts, and has the fewest pain points. P1 and P4 show the most concentrated work, relying almost entirely on their own unaided cognition at the stages where breakdowns occur. Where a strategy is absent in the first table, the second consistently shows no artifact or social resource carrying the work. The pain points are therefore not simply a matter of motivation or effort but reflect a genuine absence of external support in the distributed system available to these learners.

The task-level picture is consistent with this. **Meaning resolution** is the most externally supported stage, the one task where every participant uses at least one digital artifact. **Selective engagement** is internal for all four, the only stage where no external resource, social or material, takes part. **Recording**, **retention**, and **learning scheduling** show the largest gaps: P4 has no external support at any of the three and P1 none at retention or scheduling, so for these participants the workflow terminates after meaning resolution, with words looked up in the moment and nothing carrying them forward.

UP 17. *Meaning resolution is the most consistently externally supported task: every participant uses at least one digital artifact to resolve word meaning.*

UP 18. *Recording, retention, and learning scheduling are the least externally supported tasks. Where support is absent, the workflow terminates after meaning resolution.*

UP 19. *Learners with more distributed cognitive work across artifacts and social resources report fewer pain points than those relying on unaided cognition alone.*

4.2.4 Summary of user research

The user research consisted of semi-structured interviews with four adults enrolled in Danskuddannelse 3, structured and analysed against the four blocks of the tetrahedral model, with distributed cognition asking at each stage what carried the cognitive work besides the learner. Framework analysis produced a participant-by-theme matrix, from which the six vocabulary learning tasks reported above were derived, and each block yielded a set of user-derived propositions.

Across the four blocks the interviews describe a practice that holds up to the moment of encounter and gives way after it. All four participants seek out Danish beyond the classroom, filter which unknown words are worth attention without deliberate effort, and resolve meaning using tools they found for themselves. What follows is where it fails: recording is inconsistent and almost exclusively a classroom behaviour, retention strategies appear only reactively under exam pressure, and scheduling is largely absent. How far a participant gets tracks their prior educational experience rather than their proficiency, with those accustomed to test-driven schooling expecting an external structure their Danish classes do not supply. The distributed cognition analysis accounts for where the practice fails, since the stages that break down are exactly those at which no artifact and no other person carries any of the work.

Each proposition above is tied to the block it came from, and those blocks were set before the interviews rather than drawn out of them. That grouping now works against the analysis, because one breakdown runs through all of them and looks like something different in each. Under person it looks like a personal failing, in participants who call themselves lazy for not studying alone. Under context it looks like circumstance, in having nobody to speak Danish with. Under tasks and strategies it looks like a missing tool, in having nowhere to put a word once it has been looked up. Taken one block at a time these read

as three problems. Taken together they are one, and recovering it means detaching the propositions from the blocks that produced them and regrouping them by what they share.

4.3 Synthesis of User-Derived Propositions

To converge the problem space, the user-derived propositions were extracted from the findings above and treated as independent, atomised units. They were then grouped by conceptual similarity, without reference to the interview structure that produced them, following the affinity diagramming method [49]. The process produced the five thematic clusters shown in Figure 4.2. Each is taken in turn below: three are set aside as outside the scope of this design intervention, and two are carried forward as design targets.

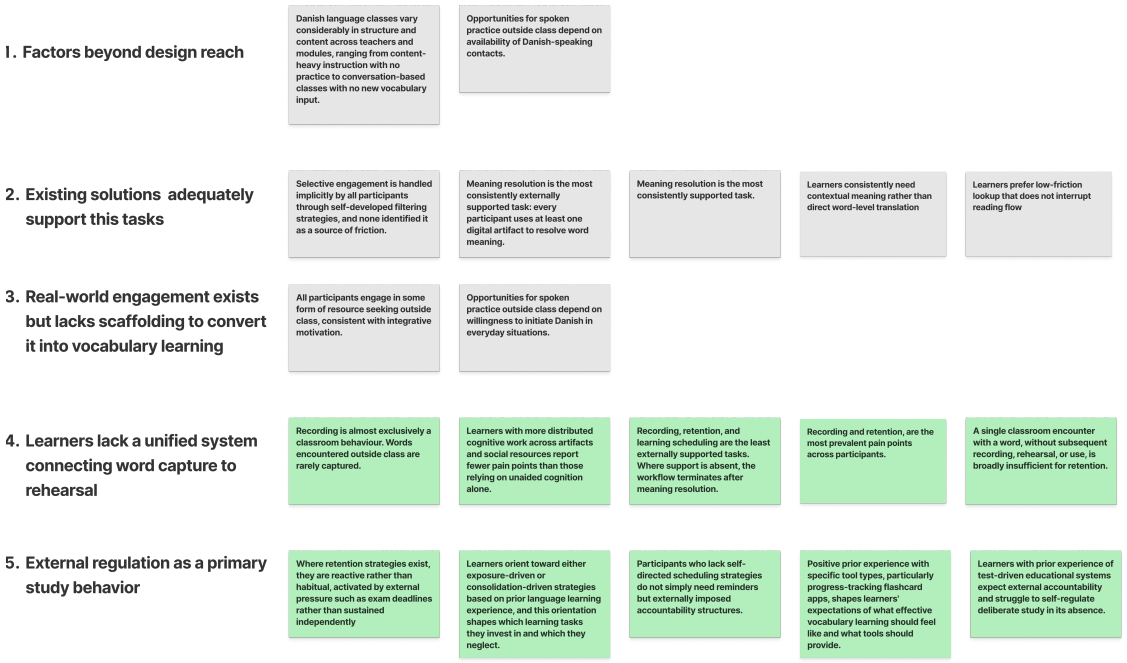


Figure 4.2: Affinity diagram of user-derived propositions grouped by conceptual similarity. Clusters with a grey background identify areas outside the scope of this design intervention. Clusters with a green background represent design-relevant findings carried forward into the focused literature review.

The first, *Factors beyond design reach*, corresponds to **UP 4** and **UP 6**. These propositions describe conditions shaped by the structure of Danish language education and the social environment a learner inhabits, neither of which a vocabulary tool can directly influence. While both factors shape the overall learning experience [18, 82], addressing them falls outside the scope of a tool-level intervention and they are not carried forward as design targets.

The second, *Existing solutions adequately support the tasks*, corresponds to **UP 11**, **UP 17**, **UP 9**, **UP 12**, and **UP 13**. Participants reported no friction at the stage of selective engagement, having developed implicit filtering strategies without deliberate effort. Meaning resolution is similarly well supported: all four participants had independently arrived at functional digital tools for word lookup, including Google Translate, DeepL, Glosbe, and the Kindle built-in dictionary. Designing a new solution for these tasks risks replacing

what already works. However, integration of meaning resolution into a unified workflow remains a relevant design consideration.

The third cluster, *Real-world engagement exists but lacks scaffolding to convert it into vocabulary learning*, corresponds to **UP 10** and **UP 7**. All four participants demonstrated integrative motivation and actively sought Danish beyond the classroom, whether through reading, ambient exposure, or everyday conversation. However, words encountered in these contexts were rarely captured or carried forward into deliberate study. While this cluster presents a design opportunity, scaffolding real-world engagement depends on first solving the more fundamental breakdown identified in the clusters below.

The fourth cluster, *Learners lack a unified system connecting word capture to rehearsal*, corresponds to **UP 14**, **UP 18**, **UP 19**, **UP 5**, and **UP 8**. Learners do not lack encounters with new words; they lack a coherent personal system for keeping those words in a form that supports later retrieval and rehearsal. Research on learner-selected vocabulary suggests that owning and organising self-encountered words, rather than studying prescribed lists, produces stronger motivation and retention [89]. This cluster is identified as a primary design target.

The fifth cluster, *External regulation as a primary study behavior*, corresponds to **UP 15**, **UP 3**, **UP 16**, **UP 2**, and **UP 1**. For most participants, sustained vocabulary study outside class depended on external pressure such as exam deadlines or teacher-assigned tasks rather than independently maintained habit. Deliberate rehearsal occurred reactively, triggered by upcoming exams or teacher expectations, and collapsed in their absence. Participants frame it as a personal failing, describing themselves as lazy or unmotivated rather than pointing to any failure of their tools. The pattern suggests that the barrier is the absence of a structure that would prompt and sustain the study rather than an absence of will. Whether and how tool design can substitute for or scaffold that missing structure in mobile learning contexts is examined in Section 4.4.

4.3.1 Summary of the synthesis

The affinity diagramming converged the user research into two primary design opportunities. The first is the absence of a unified system connecting personal word capture to rehearsal: words encountered in everyday contexts are lost before learners can engage with them deliberately. The second is the dependence on external control: deliberate study happens under imposed structure and stops without it.

Section 4.4 examines what existing research and tools have addressed in this space, and where the gap remains that this thesis sets out to fill.

4.4 Focused Literature Review: Design-Relevant Gaps

This review is the last stage of the problem space, connecting the findings of the empirical study back to the literature so that the problem statement can be settled. Its three themes come directly from the affinity clusters. The first, self-encountered vocabulary, follows from participants understanding recording as meaningful but having no consistent structure for it. The second, MALL application effectiveness and learner preferences, follows from their using existing tools for meaning resolution while none supported the workflow as a whole. The third, learner autonomy in MALL, follows from sustained study depending on external pressure rather than independent habit. Two of the three concern mobile-assisted language learning (MALL), which Section 2.2 introduced as language study on a

device that is always present, in short episodes and wherever the learner happens to be. Together the themes triangulate the gap between what existing tools provide and what these participants needed.

Each theme was searched separately in Google Scholar, starting from the wording of the cluster that raised it and following citation trails out from the studies that matched. Work was retained where it bore on what a tool can or cannot do for the behaviour the cluster describes, rather than on vocabulary learning in general.

4.4.1 Self-Encountered Vocabulary

Participants were aware that capturing encountered words was a prerequisite for later review, and several wished for a system that would support it (UP 14). This points to a need for a collection of self-encountered vocabulary, meaning words the learner meets through their own activity rather than words assigned to them, held in a form that supports later retrieval and rehearsal.

The Vocabulary Self-Collection Strategy (VSS) formalises the principle [89]. In VSS, learners identify words from their own reading and experience, nominate them for study, and take responsibility for explaining why each word matters to them [89, 90]. Learner-selected groups consistently outperform control groups on vocabulary size and retention, with the difference attributed to ownership over the words studied [91, 92], and VSS instruction also improves learner perception and engagement with the learning process [93]. Combining learner selection with a personalised recording artifact, such as a Personal Vocabulary Note in which students document self-chosen words, increases vocabulary gains and autonomy further [94].

That evidence, however, comes almost entirely from classroom settings, where a teacher mediates the nomination and provides the scaffolding. The tools that could carry the strategy outside the classroom do not offer it: the systematic review discussed in Section 2.2 finds MALL applications delivering fixed word sets rather than supporting learner-driven collection [10]. Strong pedagogical evidence and thin technical implementation is the gap this theme identifies.

4.4.2 Effectiveness of MALL apps for vocabulary learning, and learner preferences

Dedicated vocabulary learning applications produce measurable gains in word knowledge compared with traditional instruction [10], though much of the evidence concentrates on a small number of widely adopted tools. Duolingo is the most studied content-based platform, delivering a fixed curriculum sequence built on retrieval practice, repetition, and feedback [95]. It improves discrete language skills including vocabulary and grammar, and promotes learner autonomy and motivation [96, 10, 97, 95]. The vocabulary a learner engages with is nonetheless determined by the platform, so words met in everyday life remain outside its scope.

Quizlet is the most studied of the general purpose tools learners repurpose for vocabulary study, a flashcard application not built for language learning [29] that has been shown to produce significant gains in vocabulary acquisition [98, 99, 96].

Preferences between the two are not uniform and appear to reflect educational culture and prior learning experience. Korean learners have been found to favour structured, repetition-based tools such as Quizlet, while Hong Kong learners preferred more individualised tools such as Duolingo [96]. No single design therefore fits all learner populations, which is consistent with the participants of this study differing in their orientation toward consolidation-driven and exposure-driven strategies (UP 3).

These findings raise a further question. Most participants did not lack awareness of useful tools; they lacked the regulation to use them consistently outside external pressure (UP 15, UP 16). What research says about supporting that regulation is examined next.

4.4.3 Learner autonomy in MALL

As described in Section 4.3, vocabulary study outside class depended on external pressure rather than independently maintained habit. Research on learner autonomy addresses exactly this, and in MALL it has largely been pursued through self-regulated learning (SRL), the cognitive and metacognitive processes through which learners plan, monitor, and evaluate their own study [100]. The lens has been applied to understand why learners engage with mobile tools inconsistently and what design can do to support more sustained independent use.

SRL is consistently identified as one of the strongest predictors of sustained vocabulary outcomes in MALL contexts [101, 102]. Yet many learners remain passive consumers of app content rather than active self-regulating users, accessing apps reactively and for short periods rather than as part of a sustained independent study practice [103].

Tools that embed explicit self-regulation mechanisms produce measurably better outcomes than those that do not. A quasi-experimental study of a vocabulary learning app with built-in goal-setting and progress monitoring found significant improvements in both vocabulary mastery and motivation compared to a control group, with the self-regulation mechanism producing consistent gains regardless of learner gender or prior ability [101]. A larger scale study of a mobile app with an explicit self-regulation scheme bridging classroom and after-class vocabulary learning found not only vocabulary gains but an evolution in learner behaviour, with autonomy developing from basic self-control toward more sophisticated strategies including goal-setting and self-reflection over time [36].

Research applying self-determination theory in a self-directed learning application points the same way, finding that applications satisfying the basic psychological needs of autonomy, competence, and relatedness sustained better motivation and engagement than those that did not [104].

Where AI has been examined in this connection, the design-relevant result is a boundary rather than an endorsement. Scaffolding a learner's own regulatory behaviour works and substituting for it does not: metacognitive strategies fully mediate the relationship between autonomy and vocabulary performance [44], while high dependence on generative AI is negatively associated with language skill outcomes even where mobile self-regulated learning is otherwise present [45]. That line, between scaffolding learner effort and replacing it, is a constraint on the design rather than a feature to choose.

Taken together, this body of work establishes that autonomy in vocabulary acquisition is a capacity tool design can actively support or undermine. What the evidence does not show is a tool that does the supporting for words of the learner's own choosing, since the applications shown to work embed regulation into content the platform itself supplies.

4.4.4 Summary of focused literature review

The three themes converge on the same gap, as Figure 4.3 sets out. The evidence for learner-selected vocabulary is strong, and the tools that could carry the strategy outside the classroom do not support it. The tools that are well evidenced work on content they supply themselves, which leaves words met in everyday life outside their scope. And sustaining either without external pressure depends on regulatory behaviour that most learners do not bring and most tools do not scaffold.

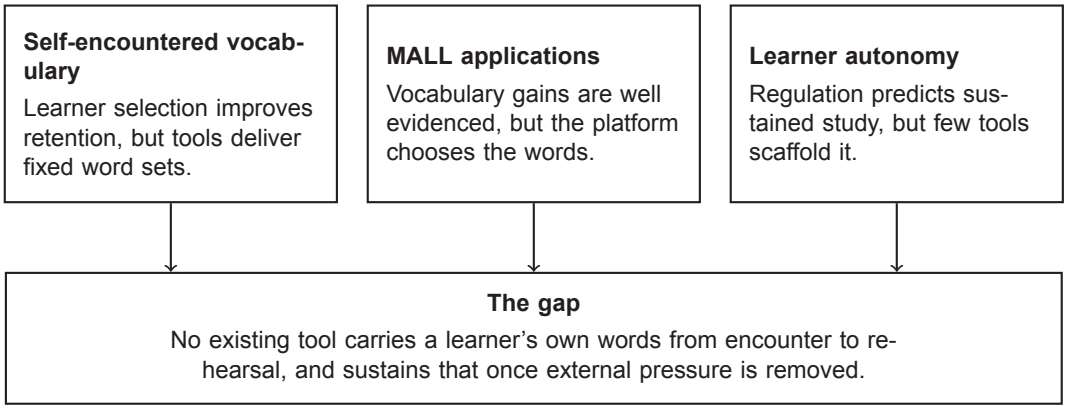


Figure 4.3: The three themes of the focused review. Each finds strong evidence for one part of what these learners need and an absence of it in the tools available to them. Together they identify a single gap.

What follows from this is a single design target rather than three. A system for self-encountered vocabulary is only viable if the learner keeps returning to it, and returning to it is exactly what the participants of this study could not do unaided. The two halves of the problem are therefore one: carrying words from encounter into rehearsal, and sustaining that once external pressure is removed.

4.5 Problem Space Conclusion

This chapter traversed the first diamond. Discover opened from the initial problem statement and widened it in two directions, a domain review of what second language acquisition research establishes about vocabulary learning (Section 4.1), and user research with four adults enrolled in Danskuddannelse 3 (Section 4.2). Then, Define converged user research findings. The user-derived propositions carried out of the interviews were regrouped by affinity into five clusters, three of which fall outside what a vocabulary tool can influence (Section 4.3), and a focused literature review took the questions the remaining two raised (Section 4.4).

What the diamond produced is a design target. These learners meet words they care about and resolve their meaning without difficulty, but nothing carries those words into deliberate study, and the study that does happen depends on a structure their classes no longer supply. The literature confirms both halves and offers no tool that closes them together. Chapter 5 states this as the final problem statement, defines the two terms it turns on, and sets out how the thesis approaches it.

5 Final Problem Statement

The problem space converged on a single breakdown. The adults it studied, enrolled in Danskuddannelse 3 (Section 2.4), understand the words they meet. What is missing is anything that carries a word past that moment, or that sustains the returning to it once the pressure of a class is removed.

The final problem statement is therefore:

> **How can a system be designed to support learner autonomy in retaining self-encountered vocabulary?**

5.1 Breaking down the problem statement

Two terms carry the weight of the statement, *retaining self-encountered vocabulary* and *supporting learner autonomy*, and each bounds the design problem in a different way.

5.1.1 Retaining self-encountered vocabulary

This thesis uses *self-encountered vocabulary* to describe words that learners meet through their own activity, rather than words assigned by a teacher, textbook, curriculum, or application. In the problem space, these encounters occurred in Danish classes, conversations, reading, digital media, and everyday situations. What unifies them is not the setting, but the mode of encounter: the word is noticed by the learner, and the learner decides whether it is personally worth keeping. This combines the incidental encounter emphasised in second language acquisition research [69] with the learner-driven selection emphasised in vocabulary self-collection research [89, 90].

Retaining such vocabulary means more than briefly understanding a word at the moment of lookup. It is treated here as three connected stages, and these are the terms the rest of the thesis uses: **capture**, getting the word out of the encounter and into a form that outlasts it; **resolution**, establishing what it means beyond a direct translation (Subsection 4.1.1); and **rehearsal**, bringing it back deliberately and repeatedly over time (Subsection 4.1.2).

This carries two consequences for the design. It settles the form of the artifact: because capture has to happen where the word is met, and those encounters are spread across the settings listed above, support has to be carriable into them. Mobility is not what solves the problem, it is what makes capture possible at all. It also scopes the content model, since the system should not prescribe which words a learner studies but hold the words that originate in the learner's own encounters and carry them through to rehearsal.

5.1.2 Supporting learner autonomy

Subsection 4.4.3 established that autonomy is a capacity tool design can support or undermine. What it left open is what the term commits this thesis to, which this section settles.

Learner autonomy is broadly understood as the capacity to take control of one's own learning. It includes the ability to make decisions about what to study, sustain learning activity without constant external pressure, and reflect on progress in order to adjust future learning [105, 106]. While definitions differ across research traditions, they share a central concern: responsibility for learning shifts from external agents, such as teachers, exams, and institutional schedules, toward the learner [107].

This does not mean that an autonomous learner studies alone or without tools. This thesis follows Little's distinction between autonomy and self-instruction [105]: autonomy concerns who directs the learning, not whether support is present. A learner can therefore be autonomous while using a system, provided the system supports the learner's own goals and decisions rather than replacing them.

The traditions that study autonomy divide the capacity differently, and the distinction decides what a system can be held to. *Self-directed learning* concerns learner control over the external conditions of study, what to learn, when, and with which resources [108]. *Self-determination theory* concerns its motivational basis, the difference between externally pressured activity and activity driven by interest, value, or perceived competence [109, 110]. *Self-regulated learning* describes the process through which autonomy is enacted: learners plan, monitor, evaluate, and adjust their own learning [111, 112].

This thesis therefore operationalises learner autonomy through self-regulated learning (SRL). The reason is pragmatic: autonomy is a capacity, whereas SRL names observable activities through which that capacity can be supported and evaluated. A system cannot give a learner autonomy, but it can scaffold the activities through which autonomy is practised [112, 113].

The design problem is therefore not to automate vocabulary learning on behalf of the learner. It is to design a system in which the learner and the artifact together can become less dependent on external structures. The learner remains the source of direction, while the system carries part of the organisational work needed to keep self-encountered words moving from capture to rehearsal [9].

5.2 Approach

The question of the final problem statement is answered by building a system and evaluating it with learners. The two terms broken down above are what that system is held to: it must carry self-encountered words from capture through to rehearsal, and it must support the learner in keeping that going without the external structures their classes supply. Chapter 6 takes this as its brief. It opens by turning what the problem space already settles into design constraints, and ends with the system built against them and what learners did with it.

6 Solution Space

This chapter reports the second diamond of Figure 3.1. Its starting point is the final problem statement (Chapter 5), and its end point is a built system and evidence of what learners did with it.

Develop widens the space of possible responses. Constraints are derived from the problem space (Section 6.1), and what they settle is the structure of the system itself (Section 6.2) together with a baseline for the part of it those constraints already close (Section 6.3). What they leave open is explored through a review of self-regulated learning (Subsection 6.4.2) and composed into three candidate concepts rather than a checklist of isolated features (Subsection 6.4.4).

Deliver narrows the three to one. A single concept is selected (Subsection 6.4.5), its scope is fixed before implementation begins (Section 6.5), and it is built over three cycles (Section 6.6). What the cycles produce is the final system (Section 6.7), which is then evaluated with learners over seven days of independent use (Section 6.8). Section 6.9 states what the diamond produced, the final evaluated solution.

6.1 Design constraints

Five design decisions follow directly from the problem space findings and require no further deliberation. They are treated as constraints that bound the solution space rather than as choices the design process has to justify.

DC	Constraint	Rationale
DC1	Carriable to the point of encounter	Words are met across everyday contexts outside the classroom, and a desktop tool would not be available where they are met (UP 14, UP 4).
DC2	Capture of self-encountered words	Capture is the entry point of the process and the stage at which it currently breaks (UP 14, UP 18).
DC3	Resolution without leaving the system	Switching tools at the point of resolution was a source of friction that prevented words from being captured at all (UP 13).
DC4	Rehearsal through retrieval practice	Retrieval practice is a fundamental mechanism of intentional vocabulary learning [114], and most participants identified it as their most effective strategy (UP 2).
DC5	Regulation across the whole process	No participant had a self-directed strategy for sustaining engagement, and regulation came from external structure such as teachers and tests, or was absent (UP 16). The system must therefore carry the continuity of the process, prompting and scheduling engagement, under the learner's agenda.

Table 6.1: Design constraints derived from the problem space. Each is traced to the user-derived propositions supporting it, and the first four follow the capture, resolution, rehearsal sequence in order.

Each is settled by the problem space and none is revisited later. What they do not settle is how they fit together, which the next section takes up.

6.2 The resulting dual-loop structure

The five constraints are not independent requirements. Read together they describe a structure, and this section derives it, checks each half against the model it corresponds to, and states which half the design work still has to explore.

The first four constraints in Table 6.1 are interdependent: they trace the stages a self-encountered word or phrase passes through, capture at the point of encounter **DC2**, **DC1**, resolution **DC3**, and rehearsal toward retention **DC4**. This capture–resolution–rehearsal sequence constitutes the **vocabulary learning loop** of the system: the recurring cycle a newly encountered word passes through, from encounter to retention. The loop corresponds closely to Ma's four-stage model of vocabulary learning introduced in Subsection 4.1.3, in which learners discover a new word, obtain its meaning, map the word's meaning to its form, and consolidate knowledge through use [78]. Table 6.2 summarises the correspondence: each stage of the vocabulary learning loop operationalises one of Ma's stages, with form–meaning mapping distributed across resolution and rehearsal rather than assigned a stage of its own.

Vocabulary learning loop stage	**Ma's stage**	**Correspondence**
Capture	Discovering a new word	Capture is the behavioural trace of discovery: the learner notices an unknown word in context and records it at the point of encounter.
Resolution	Obtaining its meaning; mapping meaning with form (initial)	Resolution operationalises obtaining meaning through lookup and contextual explanation; the contextual explanation supports the initial form–meaning mapping.
Rehearsal	Consolidating knowledge through use; mapping meaning with form (strengthened)	Retrieval practice operationalises consolidation and strengthens the form–meaning mapping; productive use lies outside the system's scope.

Table 6.2: The system's vocabulary learning loop and its correspondence to Ma's [78] four-stage model of vocabulary learning. Ma's form–meaning mapping stage is not assigned a loop stage of its own but is distributed across resolution and rehearsal.

The final design constraint **DC5** concerns whether and when the cycle runs at all. At its core, the system supports the vocabulary learning loop, in which the effort of moving a word through capture, resolution, and rehearsal is consolidated into one continuous process, while the learner supplies its content: the self-encountered words themselves. Around it sits a second loop, the **autonomy loop**, comprising the regulation of learning: deciding what to study, when to return to it, and how to keep the process moving, the function participants previously obtained from external structure or lacked entirely.

The two loops differ in their design status: the vocabulary learning loop is closed by the constraints above, derived primarily from user research and confirmed by theory [78], while the autonomy loop is bounded but not settled. To develop it further, this thesis draws on Zimmerman's cyclical model of self-regulated learning [112], which is used in the literature both as an organising framework for autonomy in MALL [115] and as a basis for designing learning systems that support learner autonomy [25, 36].

Self-regulated learning is the operationalisation of autonomy adopted in Subsection 5.1.2,

and each of Zimmerman's three phases corresponds to one component of the autonomy defined there, as summarised in Table 6.3. The choice of Zimmerman's model at this stage is pragmatic rather than exhaustively justified: learner autonomy emerged as the central concern of the thesis later than the vocabulary learning process itself, once the problem space findings made its absence apparent.

Autonomy loop stage	Description	Aspect of autonomy exercised
Forethought	Learners set goals and plan their study, selecting strategies and forming expectations before learning begins	Making independent decisions about one's own learning
Performance	Learners apply the chosen strategies while monitoring their own engagement, attention, and effort	Sustaining learning activity without external direction
Self-reflection	Learners evaluate outcomes against their goals, attribute success or failure, and adjust goals and strategies for the next cycle	Reflecting critically on progress

Table 6.3: The system's autonomy loop, based on Zimmerman's three stages of self regulated learning [112]. Each stage marks an occasion at which one aspect of the learner autonomy defined in Section 5.1.2 is exercised.

Figure 6.1 summarises the resulting system process as two nested cycles. The inner vocabulary learning loop corresponds to the constraints and Ma's model described above; the outer autonomy loop corresponds to Zimmerman's three phases just introduced. The figure shows the loop structure only, and not the distribution of control between the learner and the system.

This dual-loop structure is the first design decision the thesis makes rather than inherits. Everything preceding it was derived, from learners or from the literature; the separation into two loops is a commitment about how the system is organised, and under Research through Design such commitments are themselves findings (Section 3.1). It forms the backbone of the design that follows.

6.2.1 Summary of the dual-loop structure

This dual-loop structure is the first design decision the thesis makes rather than inherits. Everything preceding it was derived, from learners or from the literature; the separation into two loops is a commitment about how the system is organised, and under Research through Design such commitments are themselves findings (Section 3.1).

It also divides the design work that remains. The vocabulary learning loop is closed by the constraints, so it can be specified directly and held fixed. The autonomy loop is not, so it has to be designed, and designing it is the open question of this chapter. The next two sections take them in that order.

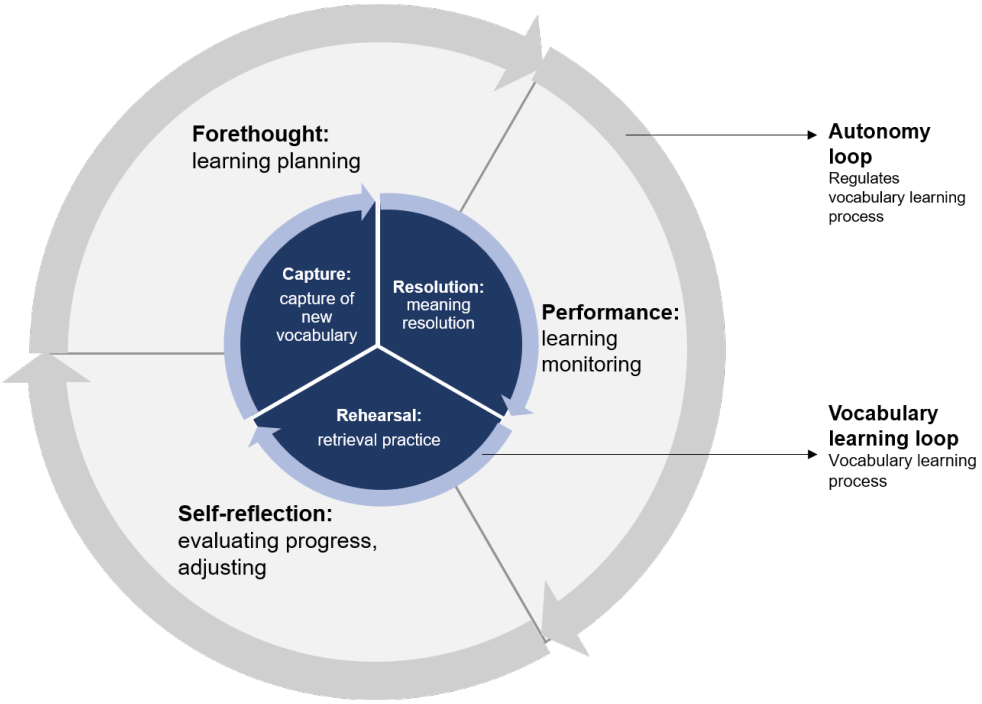


Figure 6.1: The dual-loop structure of the vocabulary learning system. The inner vocabulary learning loop, capture, resolution, and rehearsal, follows Ma's model [78] and the constraints derived from the problem space (Table 6.1). The outer autonomy loop, forethought, performance, and self-reflection, follows Zimmerman's self-regulated learning cycle [112] and addresses the regulation of learning that the problem space found largely absent (Table 4.4). The outer loop governs the inner one.

6.3 Baseline design of the vocabulary learning loop

This section fixes the inner loop of the dual-loop structure as a specification. It places the baseline among existing applications, sets out a conceptual model naming the objects the system holds and the operations performed on them, and maps those operations onto the capture, resolution, and rehearsal stages established in Section 6.2. The reasoning behind each decision follows, and the section closes with what the baseline settles.

The baseline specifies the minimum vocabulary learning system on which autonomy-supporting functionality can later be integrated: a learner enters self-encountered words, receives support in resolving their meaning, and practises them through a sequence of retrieval exercises. This follows the standard engineering notion of a *baseline*, a specification fixed and agreed upon so that later work proceeds through controlled additions to it rather than through renewed redesign [116]. It acts as a neutral reference design, showing how vocabulary learning happens without explicit self-regulation support.

Holding the vocabulary learning loop fixed is also a prototyping decision. A prototype manifests the design dimensions under investigation and holds the remaining ones at conventional values [117, 118], which means deciding what is under investigation before anything is built. That decision was already made by the constraints, and the vocabulary learning loop is the half they left conventional.

Conceptually, the baseline system is a personal vocabulary companion. Table 2.1 grouped existing applications by where the words a learner studies originate, and the baseline sits

in the third group there, alongside Vocabuo and SnapWords, which already capture self-encountered words and provide retrieval practice over them [34].

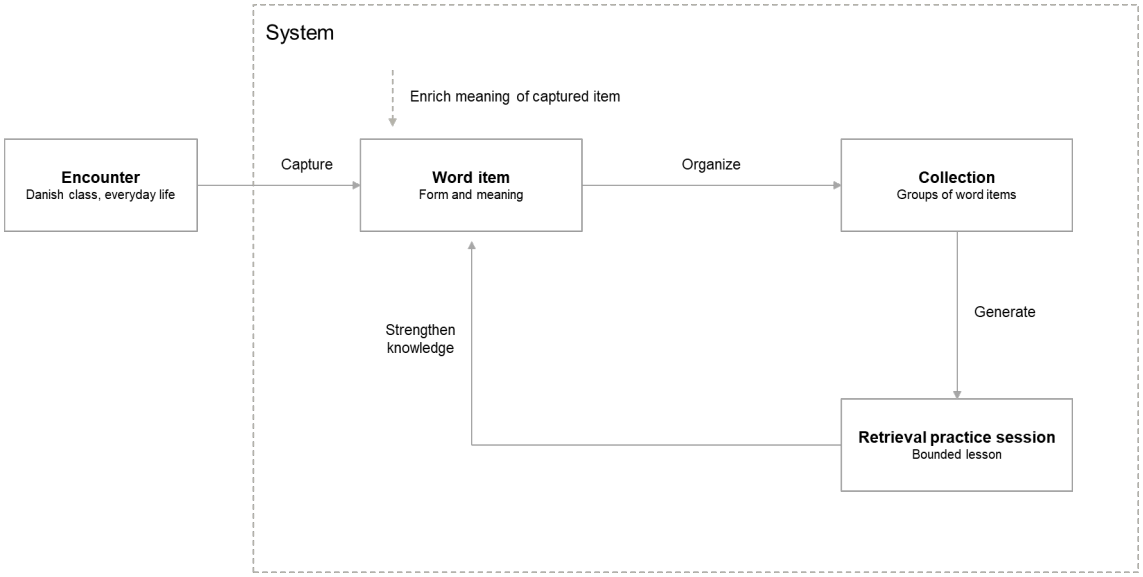


Figure 6.2: Conceptual model of the vocabulary learning loop. Boxes are the objects the learner and system operate on: an Encounter outside the system boundary, and the Word item, Collection, and Retrieval practice session inside the system. Solid arrows are the operations connecting these objects: Capture turns an Encounter into a Word item, Organise assigns a Word item to a Collection, Generate creates a Retrieval practice session from a Collection, and Strengthen knowledge returns the result of practice to the Word item. Enrich is shown as a dashed arrow because it modifies the Word item in place rather than creating a connection between two separate objects.

Following the notion of a conceptual model [119], the system exposes three core objects, the Word item, the Collection (a set of Word items), and the Retrieval practice session, and a small set of operations on them (see Figure 6.2). Object names are capitalised throughout to distinguish them from the loop stages they serve. Capture creates a Word item, the captured item is enriched with meaning, Word items are organised into Collections, and Retrieval practice sessions are generated from those Collections. The practice session then strengthens the learner's knowledge of the Word item. The distribution of actions between learner and system is described in Table 6.4.

The decisions behind each of these operations are set out below.

6.3.1 Design rationale

The baseline system follows the vocabulary learning loop established in Section 6.1: capture, resolution, and rehearsal. It is kept minimal because its purpose is to provide a stable vocabulary-learning core before autonomy-supporting mechanisms are added. The design therefore includes only the functions needed to move a self-encountered word from encounter to later practice.

Capture is limited to manual text entry and clipboard input because these directly cover the problem-space cases of recording words from class, conversation, reading, and digital text. More advanced modalities, such as camera input or speech recognition, may reduce friction but are not necessary for testing the core learning loop.

Resolution happens inside the same system to avoid the tool-switching friction identified in the problem space. Each Word item is enriched with translation, pronunciation, part of

speech, and an example sentence because vocabulary knowledge involves form, meaning, and use rather than a translation pair alone [69, 70].

Vocabulary learning loop stage	Operation in the system	Learner action	System action
Capture	Capture	Types the item manually or pastes it from the clipboard.	Creates a word item and keeps the captured form available for resolution.
Resolution	Enrich	Approves the system's input or edits it.	Provides English translation, pronunciation, part of speech, and an example sentence.
Rehearsal	Organise	Adds the item to an existing Collection, creates a new Collection, or skips assignment.	Stores the item in the selected Collection; skipped items are placed in an unassigned Collection.
	Generate	Selects a collection to practice.	Generates a short practice session from the selected collection.
	Strengthen knowledge	Completes exercises within a collection of words.	Presents exercises and gives immediate task-level feedback.

Table 6.4: Swimlane representation of the vocabulary learning loop operations, showing the distribution of actions between learner and system before self-regulated learning functionality is introduced.

Collections are introduced as the middle layer between separate word items and bounded practice sessions. A practice session needs a limited set of words in order to function as a short mobile learning activity, since mobile microlearning is based on focused units that can be completed in brief use moments [27]. Collections therefore provide the selection basis from which a retrieval practice session can be generated. They are learner-made so that this bounded set can still reflect self-encountered vocabulary and learner-selected relevance [89, 90]. Organise therefore belongs to rehearsal rather than to capture or resolution, since what it does is prepare the set a practice session is drawn from, even though the learner performs it while adding the word.

Practice is therefore collection-based and bounded as a short retrieval practice session. This follows the learner-made set model common in flashcard tools [34, 32, 33]. The exercise sequence moves from recognition toward production, reflecting the distinction between receptive and productive vocabulary knowledge and the value of retrieval practice for retention [69, 74].

6.3.2 Summary of the baseline design

The baseline is now a specification: three objects, five operations, and a stated division of work between learner and system. This is the shape the vocabulary learning loop keeps, and the design work that follows adds to it rather than reworking it. What it does not do is regulate whether any of it happens, and that is what the next section designs.

6.4 Design of the autonomy loop

This section develops the autonomy loop, the half of the system the constraints bounded but did not settle. It proceeds in four steps: the objectives a concept has to meet, the literature review that supplies the design mechanisms, the concepts composed from those mechanisms, and the evaluation that selects one to build.

The objectives are fixed first (Subsection 6.4.1). They derive from the final problem statement, the problem space findings, and the baseline system, and are weighted against one another. They are set before any concept exists and applied at the end, when the concepts are scored, so that the concepts are judged against the problem the thesis set itself.

A review of self-regulated learning then supplies the design mechanisms (Subsection 6.4.2). Zimmerman's three phases are read as design activities, and the mechanisms collected for each are gathered into an inventory of what a system can do to support forethought, performance, and self-reflection (Subsection 6.4.3).

The mechanisms depend on one another, so they are composed into three concepts, each taking a different view of what regulating vocabulary learning mainly consists of (Subsection 6.4.4). Each concept is then scored against the weighted objectives, and one is selected and carried forward into implementation (Subsection 6.4.5).

6.4.1 Weighted objectives for concept selection

Three concepts are generated later in this section, and only one is built. The objectives set out here are what that choice is made on.

The comparison uses the weighted objectives method [50], in which design objectives are ranked by importance, assigned weights accordingly, and each candidate scored against them. The objectives follow from the final problem statement, the problem space findings, and the baseline system, so that a concept scoring well is a concept that answers the question the problem space produced.

The objectives are as follows.

Objective A: Support the SRL cycle across the whole vocabulary loop. The concept should create a coherent forethought–performance–reflection cycle across the vocabulary loop, from adding and organising words to practising, evaluating, and returning to them.

Objective B: Support transfer from Danish class and everyday encounters into practice. The concept should help learners turn vocabulary encountered in Danish class, work, reading, conversation, or other daily situations into material for later self-directed practice.

Objective C: Develop autonomy as a skill. The concept should help learners practise planning, monitoring, reflection, and adjustment over time without replacing the learner as the source of goals and final decisions.

Objective D: Keep mobile interaction burden low. The concept should support fast capture and short practice with low overhead.

Objective E: Be feasible to implement as a web-app prototype. The concept should be realistic to implement and demonstrate within the planned prototype scope.

Table 6.5 presents the weight of each objective, with the highest weight given to Objective A, support the SRL cycle across the whole vocabulary loop. The method itself is docu-

mented in Appendix D, and the pairwise comparison matrix and ranking behind these weights in Appendix E.

Objective	Weight
Objective A	0.30
Objective C	0.25
Objective B	0.20
Objective D	0.15
Objective E	0.10

Table 6.5: Weighted objectives used for evaluating the candidate concepts. The full pairwise comparison and weighting calculation are provided in Appendix E.

The objectives are now fixed and are not revisited until the concepts are scored.

6.4.2 Design Literature Review: Self-Regulated Learning

The literature was gathered through Google Scholar. What it is asked for is what a system can do to support the learner's regulation of vocabulary learning and learning in general. Work was retained where it stated something a system can do, so that what it describes could be carried into the mechanism inventory follows (Table 6.7); work establishing that self-regulated learning matters, without naming what a design should provide, was not.

Zimmerman's cyclical model is used because it breaks self-regulated learning into three phases that can be addressed by design: *forethought*, where the learner sets goals and prepares action; *performance*, where the learner acts, monitors, and controls learning; and *self-reflection*, where outcomes are evaluated and the next cycle is adjusted [112, 120]. This fits the definition of autonomy used in this thesis because it describes how learners take control by deciding, sustaining, reflecting, and adjusting learning activity.

MALL and learning analytics research supports translating SRL into system features where the system makes activity visible and usable for regulation [25], and vocabulary studies using SRL mechanisms report benefits for performance, motivation, or self-regulated behaviour [101, 36]. AI-supported feedback widens the same design space by treating explanations and recommendations as scaffolds for monitoring and adjustment [44, 45].

Zimmerman's subprocesses as design activities

Zimmerman's phases contain subprocesses that can be read as design activities. Some, such as goal setting, monitoring, self-recording, and self-evaluation, can be directly scaffolded through interface structures. Others, such as self-efficacy, task value, and attribution, are not features in themselves, but can be influenced by how feedback, progress, and choice are framed.

Forethought	Performance	Self-reflection
Goal setting; strategic planning; task value; self-efficacy; goal orientation	Task strategies; time management; help-seeking; monitoring; self-recording	Self-evaluation; causal attribution; self-satisfaction; adaptive inference; next-cycle adjustment

Table 6.6: Compact overview of Zimmerman's SRL phases and the activities used to organise the design-mechanism review.

The overview in Table 6.6 organises the three sections that follow: forethought mecha-

nisms support planning and motivation before practice, performance mechanisms support action and monitoring during it, and self-reflection mechanisms support evaluation and adjustment after it [120, 113].

Supporting forethought
Forethought concerns what the learner is trying to do and why it is worth doing, and its most direct design mechanisms are goal setting and strategic planning. Specific, proximal goals are more actionable than vague intentions because they give the learner a standard for later evaluation [121]: in the baseline system, a target collection, a number of words, or a number of practice sessions. Strategic planning extends the goal by linking it to when and how practice will happen, as in MALL designs that connect classroom learning with practice outside class [36].

Forethought also includes motivational beliefs. These cannot be inserted as a single feature, but design can support them by making goals attainable, progress visible, and content personally meaningful [120]. This is why learner-selected vocabulary matters: it supports task value and ownership [89, 90]. Competitive mechanisms such as badges or leaderboards may support accountability, but they also risk shifting attention from mastery to external reward or comparison [109, 122].

Supporting performance
Performance concerns sustaining action and observing what happens while learning takes place. The baseline system already contains one task strategy: retrieval practice, which supports retention more strongly than passive review [114, 74]. SRL mechanisms can extend this through session choices, spaced scheduling, prompts, help-seeking, monitoring, and self-recording.

Prompts and reminders are relevant because the problem space showed difficulty returning to vocabulary work. Mobile vocabulary studies indicate that reminders can increase study contact, but they are weak when disconnected from a broader self-regulation strategy and may become disruptive if poorly timed [26, 28]. AI feedback is relevant where learners produce language and need help judging correctness or use; studies of AI-assisted vocabulary learning and explainable feedback suggest that feedback can support monitoring and adjustment when it helps learners understand what to do next [44, 123]. Finally, digital systems can automatically record captures, saves, practice attempts, errors, and completed sessions, creating the evidence needed for monitoring and later reflection [25].

Supporting self-reflection
Self-reflection concerns how the learner interprets outcomes and adjusts the next cycle. Dashboards, session summaries, reflection prompts, and open learner models can support self-evaluation by making progress and difficulty visible [124, 25, 125]. However, visibility alone is insufficient: learners also need support in interpreting what the trace means and how it should change the next goal, plan, or practice focus [126, 127].

Causal attribution and self-reaction matter here because feedback can either support or undermine autonomy. Missed practice or weak words should be framed as adjustable information about the plan, not as personal failure, and the same caution about external reward applies to points, streaks, and shared challenges as it does to the competitive mechanisms noted above [109, 122]. Reflection mechanisms should therefore lead to adaptive inference: what the learner should do differently in the next cycle.

6.4.3 Mechanism inventory and integration opportunities

ID	SRL activity	Design mechanism
M1	Goal setting (F)	Learner-defined proximal goals, such as words per week, sessions per week, or target collections [**locke2002goal**, 101].
M2	Strategic planning (F)	Study plan that links a goal to collection choice, session frequency, and preferred practice strategy [36].
M3	Self-efficacy (F)	Competence-oriented progression with attainable challenges, visible mastery gains, and early success moments [120].
M4	Task value and interest (F)	Use self-encountered words as study content, preserving relevance and ownership [89, 90].
M5	Goal orientation (F)	Mastery framing of progress rather than competitive ranking [109].
M6	Task strategies (P)	Retrieval exercise progression from recognition to recall and production [69, 74].
M7	Time management (P)	Adaptive spaced scheduling that decides when words should return for practice [114].
M8	Prompts and reminders (P)	Notifications tied to the learner's own plan rather than generic engagement pressure [26, 28, 25].
M9	Help-seeking (P)	AI feedback and explanation during difficult tasks, especially sentence production [44, 123, 45].
M10	Metacognitive monitoring (P)	In-session progress indicators and word-level status shown during practice [101].
M11	Self-recording (P)	Automatic logging of captures, saves, reviews, errors, and completed sessions [25].
M12	Self-evaluation (R)	Dashboards and summaries showing progress against goals and collection status [124, 25].
M13	Open learner model (R)	Expose the system's estimate of word knowledge so the learner can inspect it [125].
M14	Causal attribution (R)	Frame lapses as adjustable planning problems rather than personal failure [120].
M15	Adaptive inference (R)	Post-session reflection that feeds into the next goal, schedule, or collection choice [126, 127, 36].
M16	Social comparison and relatedness (R)	Optional friend leaderboard or shared challenge that supports accountability, while avoiding making competition the primary goal [109, 122].

Table 6.7: Inventory of SRL design mechanisms identified from Zimmerman's subprocesses. Activity labels indicate the primary SRL phase: (F) forethought, (P) performance, and (R) self-reflection.

The inventory shows that SRL support cannot be designed as a set of independent features. The mechanisms depend on one another because the cycle only becomes meaningful when forethought, performance, and self-reflection are connected: goals require monitoring and later evaluation, reminders require a learner-owned plan, and reflection requires a prior goal or strategy against which progress can be interpreted. Selecting isolated mechanisms would therefore risk fragmented support, while implementing all sixteen would make the mobile system too complex and shift attention away from the baseline system.

The inventory is therefore a set of options rather than a selection. The next section composes the mechanisms in Table 6.7 into three candidate concepts.

6.4.4 Concept generation

The mechanisms are grouped into three coherent directions, each organised around a different interpretation of what regulating vocabulary learning mainly consists of. The three interpretations mirror Zimmerman's phases: one concept is led by forethought, one by performance, and one by self-reflection. Each keeps the baseline system unchanged and adds mechanisms from Table 6.7 around it, so the concepts differ in what they regulate rather than in the vocabulary learning loop they regulate.

Concept A: Goal-regulation assistant

This concept treats explicit learner goals as the main organising element of the autonomy-supporting layer. In the baseline system, the learner can add words, organise them into collections, and start practice, but the system does not help them define what they are trying to achieve across time. Concept A addresses this by making the learner set a concrete target for a collection, for example practising a class-related collection during the week or learning a selected number of words. The app's role is then to make practice accountable to that goal: it helps the learner see which collection the goal concerns, start practice in relation to it, and notice whether practice is moving the goal forward.

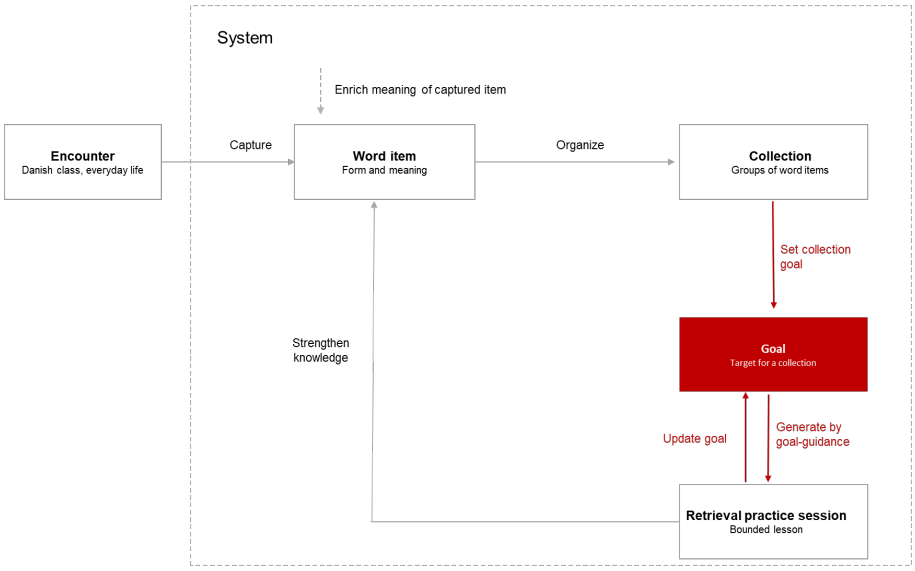


Figure 6.3: Conceptual model of Concept A, the goal-regulation assistant. Grey elements show the baseline system: Encounter, Word item, Collection, and Practice session. The added Goal object links a Collection to goal-guided Selection and receives progress updates from Retrieval practice.

As shown in Figure 6.3, selection becomes goal-guided because the learner starts practice in relation to a Collection goal, and Retrieval practice updates progress toward that Goal. The main regulatory loop is therefore not a new vocabulary-learning loop, but an added goal loop around the existing Collection and Practice session. The learner remains responsible for setting and revising the Goal, while the system makes the relationship between goal, collection, and practice visible.

The key elements distinguishing Concept A from the other concepts are:

- **Dominant SRL phase: forethought.** Concept A is organised around the learner's goal setting before and between practice sessions, rather than around adaptive scheduling during practice or reflection after practice.
- **Supporting performance mechanism: goal-guided Selection.** The performance phase is still present because the Goal changes how the learner enters practice: Selection is framed around a goal-linked Collection.

- **Supporting reflection mechanism: goal progress.** Self-reflection is present in a lightweight form because Retrieval practice updates progress toward the Goal, giving the learner a basis for judging whether the Goal should be kept or revised.
- **Key design mechanisms:** goal setting (M1), self-efficacy through attainable goals and visible gains (M3), mastery-oriented goal framing (M5), lightweight progress visibility (M12), and goal revision after practice (M15). Prompts and reminders (M8) are not among them: what carries the learner back to practice here is the Goal they set, not a message the system sends.

Concept B: Consistency coach

This concept makes the baseline system's vocabulary loop easier to sustain over time. In the baseline system, the learner can capture a word, organise it into a collection, and practise a collection, but the system does not help the learner keep returning to these steps. Concept B addresses this by adding an Activity record that stores capture and practice history. The purpose is not to define a learning target, as in Concept A, but to make the learner's recent activity visible to the system so it can support regular return to the loop.

As shown in Figure 6.4, the learner still practises through Selection from a Collection, and the system still generates a bounded Practice session. What changes is the support around the loop: captures, organisation activity, and completed practice can update the Activity record, while the Activity record can prompt the learner back to relevant Word items, Collections, or practice through Selection. The concept therefore supports consistency by helping the learner keep the baseline system's workflow active, rather than by introducing a new goal object or a reflection-heavy review step.

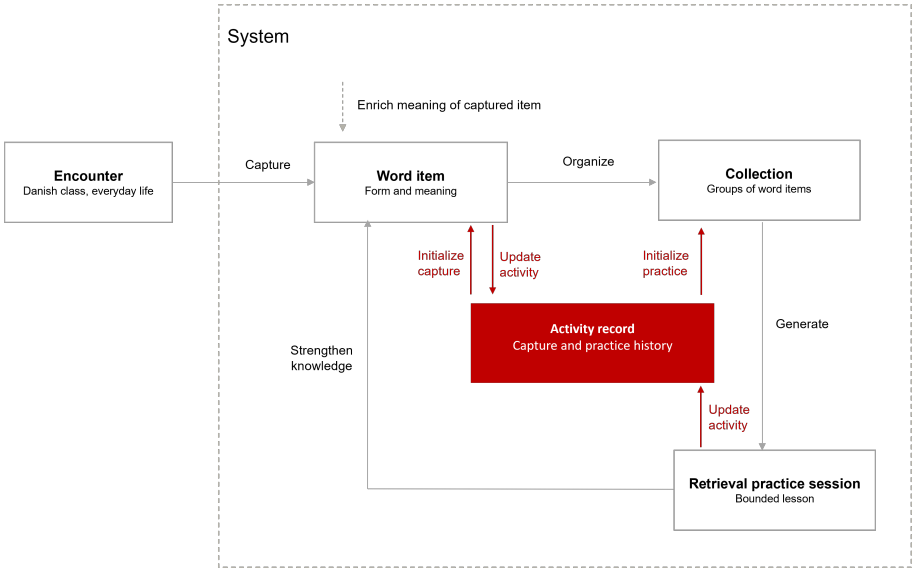


Figure 6.4: Conceptual model of Concept B, the consistency coach. Grey elements show the baseline system: Encounter, Word item, Collection, and Practice session. The added Activity record stores capture and practice history across the baseline loop, such as added Word items and practised exercises, and uses this history to prompt return to existing workflow objects, such as Word items and Collections.

The key elements distinguishing Concept B from the other concepts are:

- **Dominant SRL phase: performance.** Concept B is organised around sustaining action over time, especially returning to capture, organisation, and practice.
- **Supporting forethought mechanism: lightweight intention.** Forethought is present as an intention to keep the vocabulary loop active, such as returning to recently cap-

tured Word items or practising a Collection that has not been used recently.

- **Supporting reflection mechanism: activity visibility.** Self-reflection is present in evidence of recent learning activity, such as how many Word items were added and how many exercises were practised, but it does not introduce a full post-session reflection process.
- **Key design mechanisms:** prompts and reminders (M8), metacognitive monitoring (M10), self-recording (M11), dashboard of activity (M12), and self-efficacy through small attainable returns to practice (M3).

Concept C: Reflective focus coach

This concept makes reflection the main driver of the next learning action. In the baseline system, the learner can complete Retrieval practice and receive feedback on individual tasks, but the system does not preserve difficult outcomes as a separate focus for later practice. As a result, the learner may know that some answers were wrong during a session, but this information is not turned into a visible object that can guide the next session. Concept C addresses this by adding a Weak-word set: a set of Word items that have shown difficulty during practice and therefore need further attention.

As shown in Figure 6.5, a practice session is generated both from a learner-selected Collection and the Weak-word set, allowing practice to focus on Word items that were previously difficult. This keeps Collection-based practice available while adding a reflection-to-action loop: Retrieval practice identifies difficulty, the Weak-word set makes that difficulty inspectable, and the learner can decide whether it should shape the next Practice session.

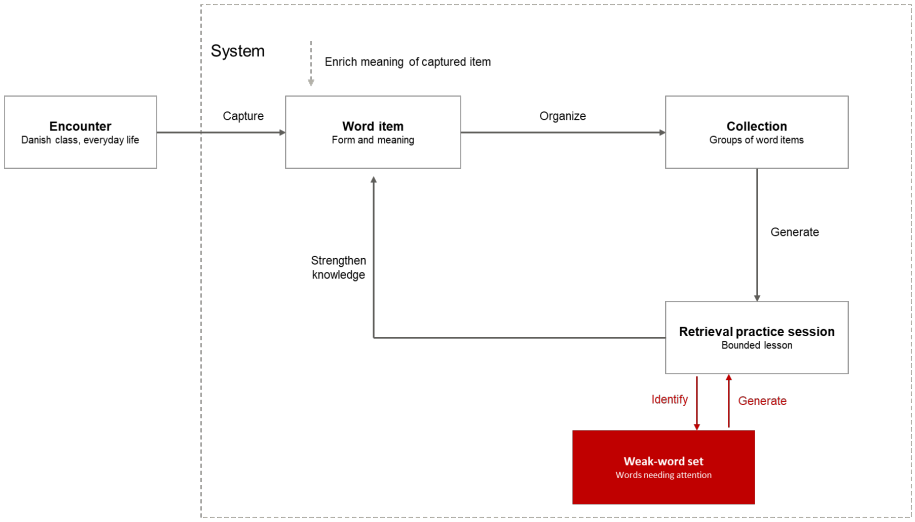


Figure 6.5: Conceptual model of Concept C, the reflective focus coach. Grey elements show the baseline system: Encounter, Word item, Collection, and Practice session. The added Weak-word set is generated from difficult outcomes in Retrieval practice and can become an alternative basis for Selection into a new Practice session. The model shows how self-reflection feeds into the next cycle by turning weak Word items into an actionable practice focus; it does not describe every supporting mechanism.

The key elements distinguishing Concept C from the other concepts are:

- **Dominant SRL phase: self-reflection.** Concept C is organised around interpreting practice outcomes after a session and using them to decide what should happen next.
- **Supporting forethought mechanism: next focus.** Forethought is present because the Weak-word set gives the learner a concrete basis for choosing the next practice focus, rather than choosing only from regular Collections or starting a ses-

sion without evidence from previous practice.

- **Supporting performance mechanism: focused practice.** The performance phase is still present because the Weak-word set can be used to generate a new Practice session, but the purpose of that session is shaped by reflection on previous difficulty.
- **Key design mechanisms:** self-recording of practice outcomes (M11), self-evaluation through post-session difficulty summaries (M12), an open learner model through visible weak Word items (M13), causal attribution that frames weak words as adjustable learning targets rather than failure (M14), adaptive inference into the next practice focus (M15), and on-demand help-seeking or AI explanation for difficult items (M9).

The three concepts share the baseline system as their core and differ only in the SRL mechanisms integrated around it. The next section scores them against the weighted objectives and selects one.

6.4.5 Evaluation of alternative concepts

Each concept is scored against each objective on a five-point scale from 1 (*inadequate*) to 5 (*excellent*), described in Appendix D, and each score is multiplied by the corresponding objective weight listed in Table 6.5.

The evaluation gives the highest score to Concept A, the goal-regulation assistant, with a weighted score of 4.20. Its main strength is that it develops learner autonomy through explicit goal setting and revision, while remaining straightforward to implement as a web-app prototype, since goal objects, progress counts, and goal-linked practice entry need nothing the platform cannot deliver. Its main limitation is the additional interaction burden created by asking the learner to define and maintain a goal.

Concept	Weighted score
Concept A: Goal-regulation assistant	4.20
Concept B: Consistency coach	4.00
Concept C: Reflective focus coach	4.00

Table 6.8: Summary of weighted concept evaluation results.

Concepts B and C both receive a weighted score of 4.00, but for different reasons. Concept B supports the whole vocabulary loop well by prompting return to capture and practice, but its dependence on reminders makes it less feasible to demonstrate in a web-app prototype and introduces the reminder trade-off identified in the literature: reminders can increase return to study, but their effectiveness depends on timing and relevance, and poorly fitted reminders can become disruptive [28]. Concept C is highly feasible and has low interaction burden, but its SRL support is concentrated around practice and weak words rather than the full loop from encounter to later practice. The detailed score calculations and reasoning are provided in Appendix F.

On this basis, **Concept A** is selected for further development.

6.4.6 Summary of the autonomy loop design

The autonomy loop is now designed as far as concept work can take it. The weighted objectives fixed what a concept had to achieve, the review of self-regulated learning supplied the mechanisms available for achieving it, and the three concepts showed that the same inventory supports more than one coherent system. The scores separated them by

how much of the vocabulary loop each one regulates, which is what Objective A weighted most heavily.

The conceptual model now covers both halves of the dual-loop structure: the vocabulary learning loop held fixed as a baseline, and a Goal object around it that the learner sets and revises (Figure 6.3). What remains is to carry that model into an executable prototype. The next section fixes the scope of that prototype and its technical realisation.

6.5 Prototype Scope and Technical Realisation

This section establishes what kind of prototype Concept A required and how it was technically realised, before the build cycles that produced it are presented (Section 6.6). The word *prototype* is used here in a specific sense: the artifact is a software prototype, an executable, deployed system, rather than a mock-up, a scenario, or a design representation of any other kind [118]. That choice follows directly from how the concept is to be validated, and the fidelity targets and architecture set out below are settled against it. Both are fixed here, before the first build cycle, so that the cycles are a record of reaching them rather than of deciding them.

6.5.1 Prototype scope

The validation approach adopted in this project is to test Concept A with its intended users as working software: learners use the system themselves, on their own devices and in their own everyday context, over sustained independent use rather than in isolated supervised sessions. What the evaluation requires in turn determines what had to be built. Independent, sustained use rules out a local demonstrator and a click-through mock-up alike; the artifact had to be a persistent, multiuser system holding each learner's own accumulating data and reachable from their own phone.

How finished such a system needs to be varies by aspect. McCurdy et al. decompose a prototype's fidelity into five dimensions that can be planned independently rather than as one overall low-to-high score: level of visual refinement, breadth of functionality, depth of functionality, richness of interactivity, and richness of the data model [51]. The value of the decomposition here is that it lets fidelity be planned as a set of targets fixed before building, rather than judged after the fact. Effort is allocated across the five dimensions, and the allocation follows from the final problem statement, since the autonomy loop is what is under investigation while the vocabulary learning loop is the substrate it operates on. Figure 6.6 states the target profile; the arguments for each level follow.

- **Breadth of functionality (target 5/5).** Every operation in Concept A's model must be implemented (Figure 6.3). A learner working alone cannot be routed around a missing one, and a partial model would validate a different concept.
- **Richness of interactivity (target 4/5).** Interactions must run end-to-end against live services, with nothing operated by the researcher, or observed self-direction could not be attributed to the artifact. Capped at 4 because full interactive richness would depend on platform capabilities beyond what a prototype of this scope can be built and deployed with; the deployment that follows from this is set out below.
- **Richness of the data model (target 5/5).** Accounts, word items, collections, exercise logs, and goal commitments must all persist and accumulate across days, with nothing seeded or simulated, since independent use over time is what the validation observes.
- **Visual refinement (target 4/5).** Not under investigation in its own right, so polish is not the aim; but visual quality shapes engagement and perceived usability [128], and neglecting it would confound how learners experience the autonomy mechanisms.

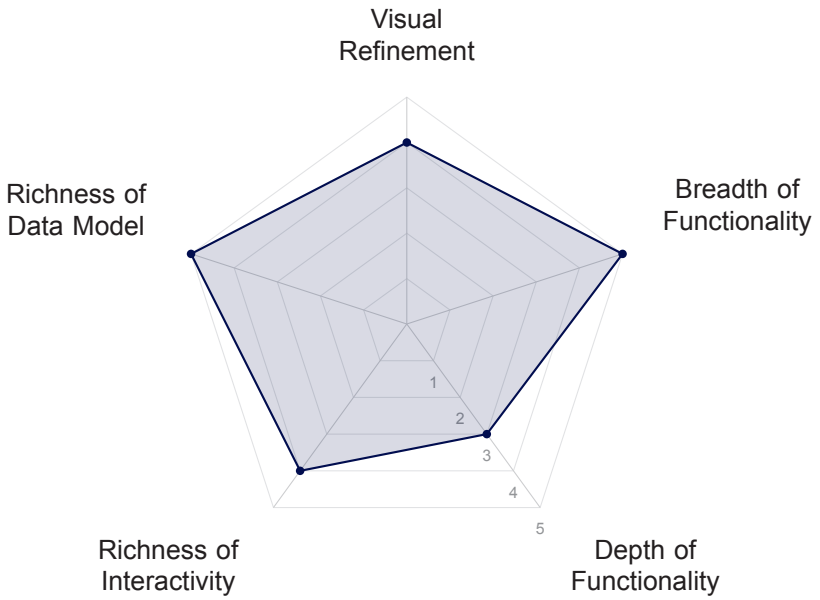


Figure 6.6: Target fidelity profile set before the first build cycle, on McCurdy et al.'s five dimensions [51]. Levels run from 1, indicative, to 5, complete within the scope the validation demands, so completeness is judged relative to the loops the system supports rather than to a commercial product. The arguments for each level follow below.

- **Depth of functionality (target 3/5).** Targeted low by choice. Hardening each function exhaustively would demand testing effort beyond what is feasible, and translation quality or unusual word forms bear nothing on who directs the learning. Functions need only behave correctly in ordinary use.

6.5.2 Technical realisation

The profile above narrows the range of workable deployments considerably. A web application was preferred over a native mobile app for the same practical reason that motivated a mobile-carriable artifact in the first place (Subsection 5.1.2): it reaches a learner's own phone without going through app-store distribution, and it stays within what a prototype of this scope could realistically deliver. The system was therefore implemented as a single-repository Flask [129] web application backed by a PostgreSQL [130] database, both deployed on Render [131], a managed cloud hosting platform, and served to the learner's browser as a single-page application. It is delivered as a progressive web app, so a learner can install it to the home screen of their own phone and open it as they would any other application, without it passing through an app store. The aim was a working software prototype adequate for validation with real learners: all learners shared a single deployment and database, each with their own account and their own persistent data.

Two of the operations in Table 6.4, *Enrich* and *Generate*, rely on external services rather than on logic written from scratch. Enrich supplies the English translation, part of speech, and example sentence a learner then approves or edits, along with pronunciation audio: translation comes from machine translation, through the Google Cloud Translation API [132], and audio from Cloud Text-to-Speech [133], while the free-text elements, the part of speech and the example sentence, are produced through the OpenAI chat-completion API [134] using the `gpt-5.4-nano` model. The same API drives Generate, which composes a short practice session, in the form of write, multiple-choice, and cloze exercises, from the collection the learner selects, and serves as the translation fallback should Cloud Translation be unreachable. Figure 6.7 summarises how these pieces fit together.

The `gpt-5.4-nano` model was chosen to fit the shape of the requests. The operations ask for short and structured output, a part of speech, one example sentence, a fixed number of exercise items, rather than open-ended prose. Response time matters as well, and a request is issued for every word captured and every practice session composed. A small, fast model meets both conditions, so the smallest one adequate to the task was the appropriate choice.

That choice also reduces the two levers described in Section 2.3 to one. `gpt-5.4-nano` exposes no sampling parameters, so temperature and its equivalents were unavailable and all steering was done through prompt design: the structure the backend expects is stated in the prompt, and the returned text is parsed defensively, since that structure is a convention the model has been asked to follow and not a guarantee.

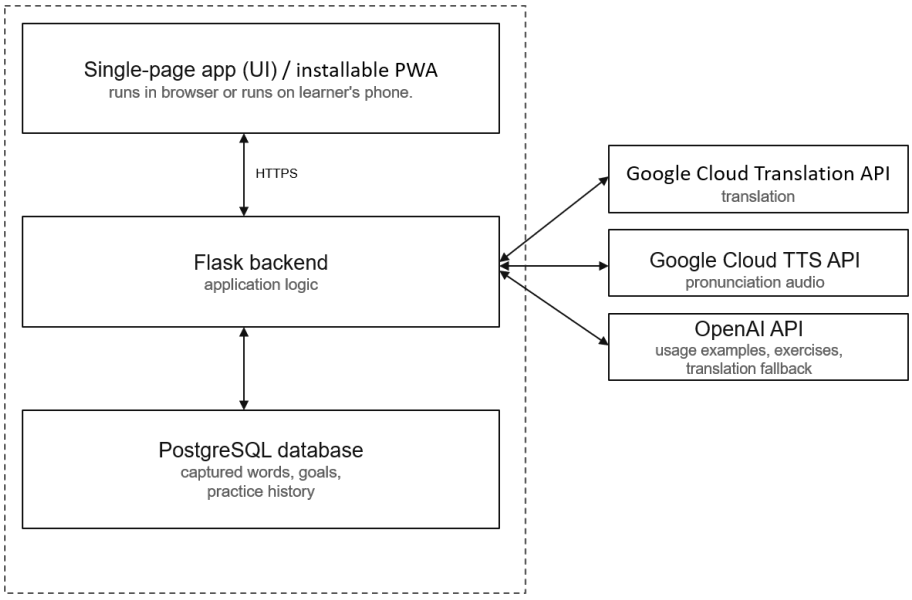


Figure 6.7: Architecture of the prototype, settled before the first build cycle and unchanged across all three. A single-page application on the learner's device, installable as a progressive web app, communicates over HTTPS with a Flask [129] backend and PostgreSQL [130] database deployed on Render [131]. The dashed boundary marks the system itself, holding the autonomy loop and all learner data; the language content presented to the learner is produced by external services [132, 133, 134] beyond that boundary and consumed without verification.

6.5.3 Summary of the prototype and technical realisation

The prototype is now specified on both counts the build needs: how finished it has to be, and what it runs on. Breadth of functionality and richness of the data model are targeted at 5 because a learner working alone cannot be routed around a missing operation, and independent use over time cannot be observed on seeded data. Depth of functionality is targeted at 3 because ordinary correctness is all the autonomy loop requires of the vocabulary learning loop beneath it.

The architecture serves the same end: an installable progressive web app over a Flask backend and PostgreSQL database, with the language content drawn from external services outside the system boundary. Both the targets and the architecture are fixed before the first build cycle, so next section reports how they were reached.

6.6 Prototype Build Cycles

The previous section fixed what the prototype had to be: an executable system meeting the fidelity targets of Figure 6.6, built on the architecture of Figure 6.7. What it could not fix is how well Concept A's mechanisms (Subsection 6.4.4) actually serve a learner once implemented, since that is not something the targets or the architecture decide, and not something further reading could settle either. Reaching the targets was therefore not a single build but a sequence of build cycles, in the design-science sense of a construct-and-evaluate loop in which each pass produces a better-informed artifact than the last [135, 136]. Within each cycle the work moved from a low-fidelity sketch, used to fix tab structure and operation placement before any code was written [137], to an executable build, implemented with the assistance of agentic coding tools under the design decisions made at that stage.

Each cycle follows the same three steps, which rest on the software-engineering distinction between *verification*, are we building the system right, and *validation*, are we building the right system [52]. A **build** step changes the prototype; a **system testing** step verifies internally that the change behaves as intended; and a **user validation** step establishes externally, with real participants under a documented protocol, whether the change resolves the design question it was meant to answer. System testing therefore carries no design question of its own. It asks whether the build works, not whether the design is right, and only the build and the user validation are framed around a question.

Verification was carried out manually and black-box, judging each component by its externally observable behaviour rather than its internals [53], so that effort could be concentrated on validation with learners, where the design questions actually lie. The same procedure ran at the end of every round, before that round was handed to participants. The application was run with real credentials and no stubs for the OpenAI, Google Cloud Translation, or Text-to-Speech APIs; every user-facing flow was walked through in the actual interface, issuing real requests to the Flask backend; server-side effects were cross-checked against the database row by row, to confirm persistence rather than a correct-looking screen; and raw responses were inspected for malformed payloads, particularly LLM output, which is free text parsed defensively. Any defect found was fixed and re-tested, and only a clean pass allowed a round to proceed to user validation. The emphasis suits the architecture: a single-server monolith whose main risk is integration failure between Flask, the database, and three external APIs, rather than faults in isolated units of logic.

Three rounds were carried out, and the design question shifted at each one. Round 1 built the full stack, from the PostgreSQL schema and the Flask endpoints through the external-service integrations to the single-page interface, and realised Concept A in its most direct form, a goal attached to a collection, asking whether that alone is enough to scaffold forethought and hold practice accountable to a learner-set target. It was not: goals were set easily and then forgotten. Round 2 therefore rebuilt the goal mechanism itself, replacing the per-collection goal with a recurring weekly goal that resurfaces commitments, adjusts its own targets, and reports outcomes as accomplishments rather than counts, with the interface reorganised around it: a dedicated goals tab, an achievements page in place of the progress count, and capture moved out of its own tab.

Round 3 closed the remaining gaps, resolving the visual and interaction defects that Round 2's participants exposed and adding the usage-tracking infrastructure the system needed to be observed in real-world use. Table 6.9 sets out each round by architectural layer, so that what changed and what was checked can be read against Figure 6.7; the

full protocols, participants, and findings are reported in Appendix G.

Round	Built (by layer)	Verified	Design question	Answer
Round 1 *Concept A implemented*	**Database:** the full schema, from accounts to completed exercises. **Backend:** accounts, capture, enrichment, goals, progress, and the assembly of sessions of eighteen exercises from the chosen collection. **External:** translation with a language-model fallback, pronunciation audio, and the wording of examples and exercises. **Interface:** tabs for Capture, Practice, and Progress, with Concept A as a goal on each collection (M1, M5, M12, M15), installable to the phone.	Rows inspected directly for persistence and ownership; progress figures checked against recorded activity; services called live, under odd input and with translation made unavailable; every flow walked through by hand, including offline.	Is a goal attached to a collection, on its own, enough to scaffold forethought and make practice accountable to a learner-set target? *Think-aloud, 3 participants.*	**No.** Goals were set easily, but their usefulness wasn't confirmed. Progress read as a count, not an accomplishment; sessions felt too long; participants wanted goals in one place and questioned the separate capture tab.
Round 2 *Goal mechanism redesigned*	**Database:** extended for weekly goals, sessions, and mastery. **Backend:** the per-collection goal replaced by a weekly cycle ending in a review the learner must complete (M1, M8, M14), four goal categories, self-adjusting targets (M15), re-earnable mastery (M12, M13), and sessions cut to twelve exercises, six words with two each. **External:** both exercises generated afresh each session and tappable to save (M4, M6); a spelling check while typing. **Interface:** tabs reorganised into Current Goals, Practice, and Achievements (M5, M16), capture folded into practice, sessions resumable.	Week turns, target adjustment, and mastery exercised by setting week dates by hand; session composition and counts checked against records; generated alternatives read for plausibility; walkthrough repeated, with continued use surfacing a misleading chart scale, figures stuck at zero, missing exercises, and corrupted input.	Does restructuring the autonomy loop around a recurring weekly goals, ones that resurfaces commitments and frames outcomes as accomplishments, resolve the Round 1 breakdowns? *Think-aloud, 2 participants.*	**Yes.** The Round 1 breakdowns did not recur and reception was positive. Two further findings concerned finish rather than the mechanism: inconsistent colours, and a few interface defects.
Round 3 *Tracking added, interface refined*	**Database:** logins and app openings recorded separately, so returning without a fresh login still counts. **Backend:** a script drawing logins, openings, and sessions into a summary of use. **Interface:** colours revised into a consistent palette and the Round 2 defects repaired. No new mechanisms; external services unchanged.	Events checked for one entry per login and one per opening; summary figures matched against the underlying records; walkthrough repeated, including the steps behind each defect, to confirm none recurred.	Does the resulting system support learner autonomy in retaining self-encountered vocabulary? *Sustained independent use in learners' own context (Section 6.8).*	Reported in Section 6.9.

Table 6.9: The three build cycles, each pairing a build, described by architectural layer and verified through the black-box procedure above, with a design question put to participants. Design mechanisms (Subsection 6.4.4) are referenced where realised; more details in Appendix G.

6.6.1 Summary of the build cycles

Three rounds reach the fidelity targets, and what moved between them was the goal mechanism rather than the architecture, which is unchanged from Figure 6.7. Round 1 established that Concept A in its most direct form does not hold: a goal attached to a collection is set once and then forgotten. Round 2 answered that by making the goal recur, resurface commitments, and report outcomes as accomplishments, and its participants did not reproduce the Round 1 breakdowns. Round 3 added no mechanism at all, resolving finish and adding the tracking the final evaluation depends on.

Round 1's answer is the one that matters for the rest of the chapter. The system carried into evaluation is not Concept A as selected but Concept A as revised, and under Research through Design that revision is itself a finding rather than a correction to be hidden (Section 3.1). Section 6.7 describes what the three rounds produced.

6.7 The Final System

The system is a vocabulary application for adult learners of Danish, used on the learner's own phone. It is deployed online (here), with username `tester` and password `1234`, and the source code is available on GitHub (here). This section describes the system first as a whole, through the two loops it is built around, and then through selected elements of the implementation.

The system maps onto the dual-loop structure proposed in Section 6.2: a vocabulary learning loop nested inside an autonomy loop that governs it. The **vocabulary learning loop** moves a word through three stages: **Capture**, where the learner types or pastes in a word met in everyday life, in class, in conversation, or in writing; **Resolution**, where the learner is shown a proposed translation, pronunciation, and example of the word in use, which they confirm or edit before it is saved into a collection they have created and named; and **Rehearsal**, where the learner opens a practice session built from a collection they choose. Over this vocabulary work sits the **autonomy loop**, with its own three stages: **Goal setting**, where the learner commits to a set of targets at the start of each week; **Goal monitoring**, where progress accrues against those targets as the week goes on; and **Goal reviewing**, where the outcome is shown at the week's end and must be acknowledged before the next week's targets can be set. The autonomy loop is cyclical: once reviewed, the week resets and Goal setting opens again, with the cycle's period fixed at one week. Figure 6.9 sets both loops out screen by screen, one screen per stage. Both sit behind the same three tabs at the foot of the screen, so committing to a goal or reading a review is not a separate destination from the vocabulary work itself; it is reached the same way Capture and Rehearsal are.

6.7.1 The two loops and how they connect

The two loops did not carry equal design weight. The vocabulary loop was closed early, fixed by the constraints established in Section 6.3 and confirmed by theory; the autonomy loop was left open, and it is what the build cycles iterated on. Concept A originally attached a goal to a single collection; Round 1 showed that such a goal was set easily and then forgotten (Section 6.6), and the loop was rebuilt around a recurring weekly cycle instead.

The two loops are coupled in both directions (Figure 6.8). A committed goal reaches across the whole vocabulary cycle rather than any single operation of it: adding words bears on Capture, practising and mastering bear on Rehearsal. In the other direction, the goal counters read the learner's recorded activity, so progress is a record of what was actually done rather than something the learner reports. Closing the cycle, the week's

review feeds forward into the next, both by requiring the learner to acknowledge the outcome before new goals can be set and by adjusting what the system recommends.

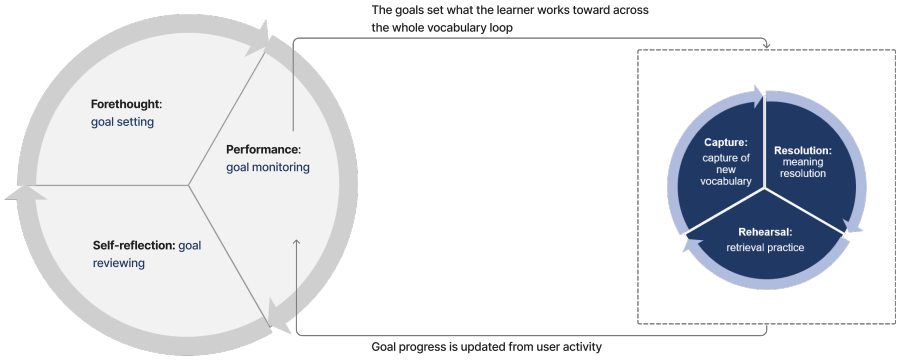


Figure 6.8: How the two loops couple: goals set in the autonomy loop reach into vocabulary work, and the resulting activity is what updates goal progress.

6.7.2 Implementation

The architecture is as set out in Figure 6.7 and unchanged since the first build cycle: a single-page application on the learner's phone, installable as a progressive web app, over a Flask backend and PostgreSQL database deployed on Render, with translation, pronunciation, and generated text drawn from external services. What follows is not a complete account of the implementation but a set of examples, organised by the two loops in turn.

Vocabulary learning loop

Capture and **Resolution** are where a word enters the system (see Figure 6.9). The learner types the word or pastes it from the clipboard; while typing, a live spelling check raises a suggested correction unprompted, which matters because a misspelled entry would otherwise be resolved against the wrong word entirely. Resolution then proposes a translation, an example sentence, and pronunciation audio, drawing the translation from a dedicated translation service with the language model as fallback, and the part of speech and example from the language model directly. The learner approves or edits the proposal, and only the confirmed version is saved, so nothing enters a collection the learner has not seen and accepted. Collections are set up before Capture, not after: a word is only saved once a collection exists to receive it, with no unassigned catch-all as in earlier rounds.

A practice session, in the **Rehearsal** stage (see Figure 6.9), is a session of twelve exercises covering six words from the chosen collection. Each word has two exercises, multiple choice and gap fill in a sentence (see Figure 6.11). The system selects the words as follows: candidates are drawn at random, but a word whose most recent session was answered correctly is set aside and another drawn in its place, biasing the six words of a session toward what the learner has not yet answered correctly (Figure 6.10). Starting the session triggers generation, and all of it completes before the first question appears. A multiple-choice item takes one call to the language model; a gap-fill takes two, one for the Danish sentence and one for its translation. Six words therefore issue eighteen requests, sent concurrently and not one after another, and the exercise queue is assembled in the browser from what returns, pairing each word with its two exercises. This is where the response time that governed the choice of model (Section 6.5) is spent. Listing 6.1 gives the multiple-choice prompt as an example. Its distractors are not matched on surface

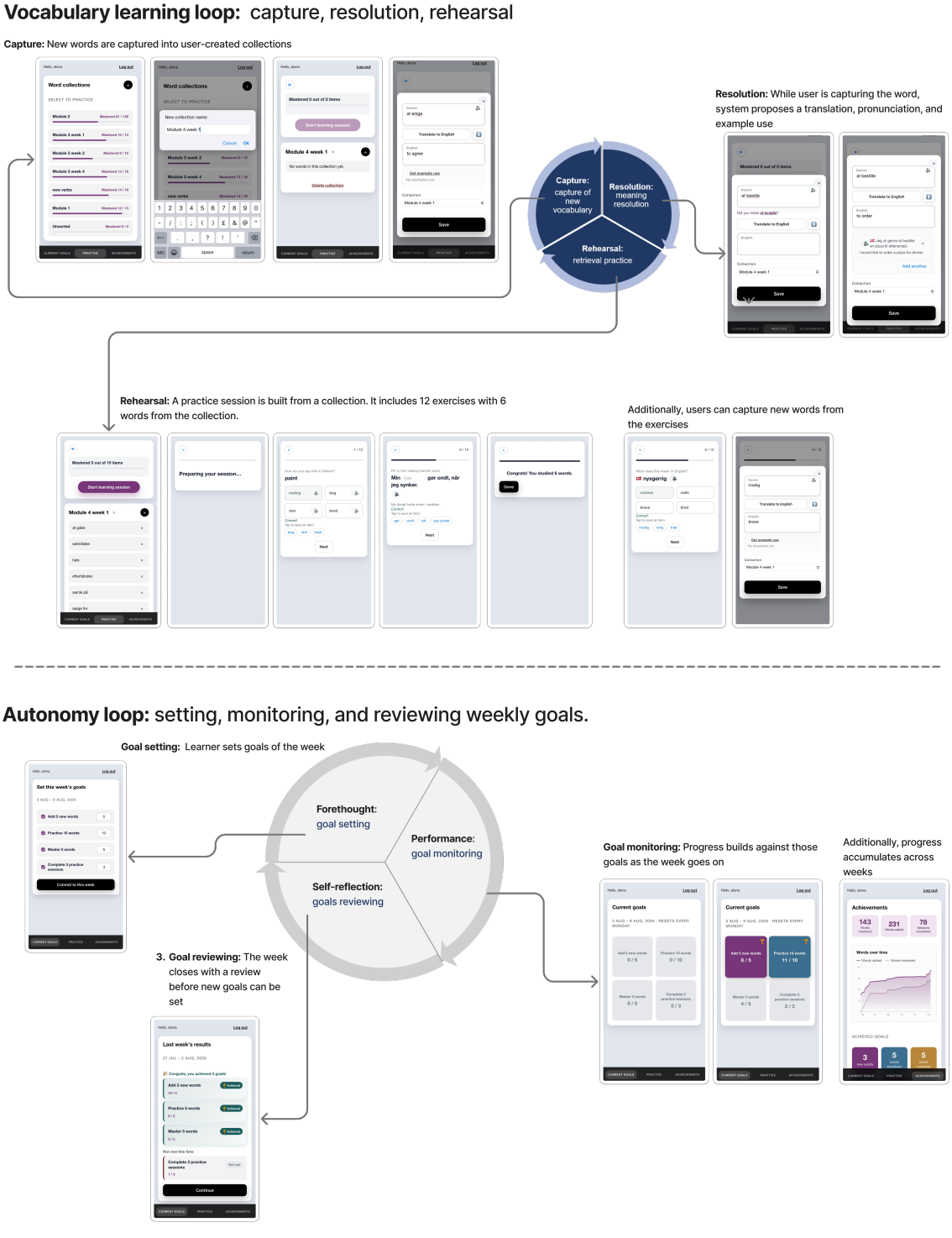


Figure 6.9: The final system: the vocabulary learning loop above, the autonomy loop below. Screens are grouped by the stage they address.

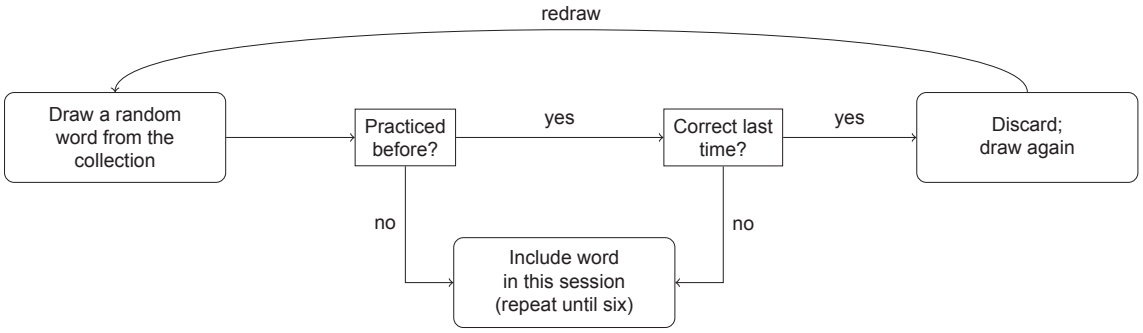


Figure 6.10: How the six words for a practice session are chosen: a correctly-answered word is set aside and another drawn in its place.

form alone: each must carry a meaning distinct from the target rather than a synonym or related term, hold a similar language level, and avoid an obvious thematic association with it, so a wrong option stays plausible without being a near-miss.

```
System:
You are a careful language tutor helping English speakers learn
Danish words. Given an English target word/phrase and its Danish
translation, produce JSON describing the target's part of speech
and three fresh distractor words in English that share the same
part of speech. Each distractor must have a distinct meaning (not
a synonym or semantically related term), be of similar language
level and naturalness, and not share a clear thematic or emotional
association with the target. If the target is a multi-word phrase,
each distractor should contain the same number of words (+/-1) and
feel like a natural expression of similar length.

User:
TARGET_ENGLISH: {target}
TARGET_DANISH: {translation}

Return JSON with the shape:
{"part_of_speech": "...", "distractors": [{"text": "...",
"translation": "...", "note": "..."}]}
- Provide exactly three distractors.
- Distractor texts must be different from the target and from
  each other.
- Distractors must be natural English words/phrases a learner
  might encounter.
- Keep the translation concise Danish and the note within 12
  words.
- Respond with JSON only.
```

Listing 6.1: Example prompt sent to the language model for multiple-choice distractor generation.

Either exercise can also be tapped to capture a word from it as a new item in its own right, and not only the target: a distractor option, or another word inside the gap-fill sentence, can be saved just as the target can, which turns Rehearsal into another route into Capture rather than only a use of what Capture has already produced. A session left unfinished can be resumed rather than restarted, so an interruption does not cost the exercises already completed.

Every answer is logged with its attempt score and the key of the session it belongs to. An item answered wrongly replays once before the session ends, reusing the content already generated for it and issuing no further requests. That log is also what the selection above reads, so each session is scored by what the learner did in the ones before it.

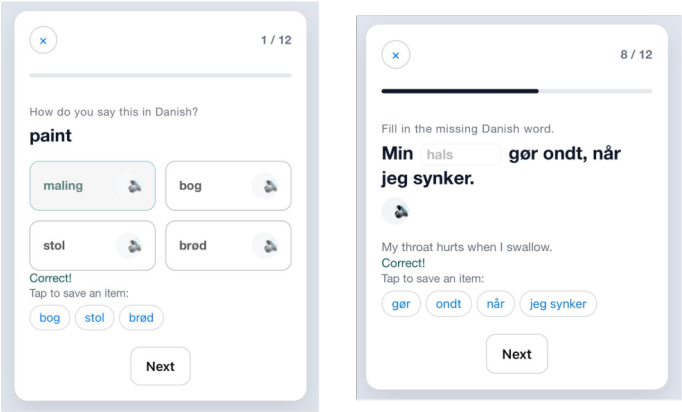


Figure 6.11: The two exercise types built for every word drawn into a practice session: multiple-choice (left) and gap-fill (right).

Autonomy loop

The weekly autonomy cycle moves through its three stages in strict order (Figure 6.12, Figure 6.9). **Goal setting** opens the week: the learner commits to a target for each of the four goal categories, and nothing else in the cycle proceeds until they have. **Goal monitoring** then runs for as long as the week lasts: rather than a stored running counter, each category's progress is computed fresh, on every load, from the learner's own recorded activity in Capture, Rehearsal, and mastery, so what is shown is always current rather than a figure that could drift out of sync with it. **Goal reviewing** closes the week: each goal's outcome is shown as met or not met, and must be acknowledged before a new week's targets can be set. That review gate is what makes self-reflection part of the loop: a learner cannot pass an unfavourable week by without first seeing what happened.

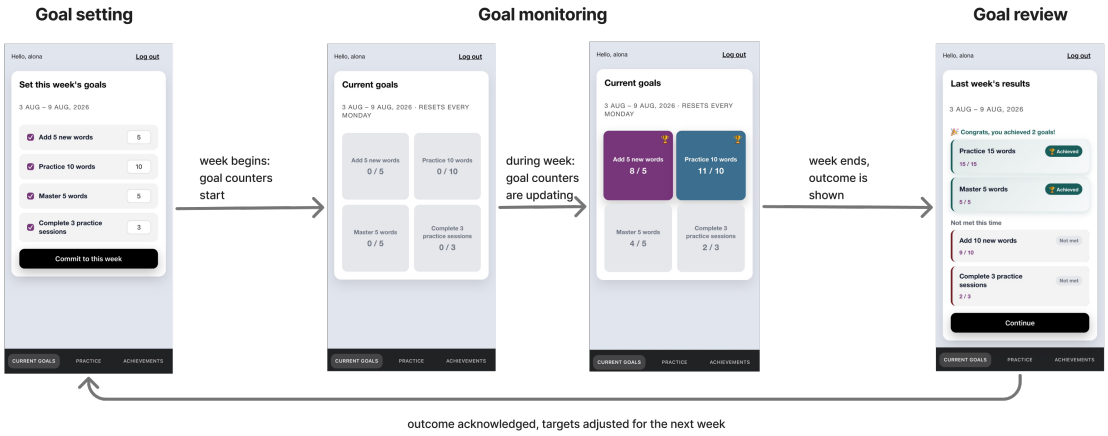


Figure 6.12: The weekly autonomy cycle, screen by screen: targets are committed at the start of the week, progress accrues against them while it runs, and the outcome must be acknowledged at the week's end before the next week's targets can be set.

The recommended target then carries forward asymmetrically: a met goal raises the next week's recommendation by five, while a missed goal leaves it unchanged rather than lowering it, so a missed week is not read as grounds to ask for less. The learner can still set a target above or below the recommendation, since the commitment remains theirs; what the system adjusts is only the starting point it offers.

The four categories of goals (see Figure 6.12) divide the vocabulary loop between them. **Add words** targets Capture; **practice words**, **complete sessions**, and **master words**

target Rehearsal, counting exercises worked through, returns across the week, and words answered without error. The split follows the distinction between process and outcome goals, with three of the four specifying what the learner will do and only mastery specifying what will result [138].

6.7.3 Summary of the final system

The system realises the dual-loop structure as working software: a vocabulary learning loop of Capture, Resolution, and Rehearsal, and a weekly autonomy loop of goal setting, monitoring, and reviewing that governs it. The two are coupled in both directions, since a committed goal reaches across the whole vocabulary cycle while the goal counters read the activity that cycle records, and the week's review gates the targets of the week after it.

Measured against the fidelity profile fixed before the first build cycle (Figure 6.6), the system reaches both of its full targets: every operation in Concept A's model is implemented, and all learner data persists and accumulates across days rather than being seeded. Richness of interactivity lands where the profile predicted, since the system runs live but remains browser-bound, without push notifications or any reminder beyond what the interface shows when opened. Depth of functionality was held at a working level by choice, correct in ordinary use but not hardened against unusual word forms, since translation quality bears nothing on who directs the learning. The delivered system therefore matches the profile it was planned against, on the dimension held low as much as on the two held full.

6.8 Final Evaluation

The final evaluation puts the system to learners using it on their own. Over seven days participants used the application independently, in their own everyday context, with no supervision, prompting, or researcher involvement once the study began. Self-regulation was measured before and after that period using the same instrument, and what participants did in between was recorded by the application itself, so that any reported change could be read against actual use. The method is set out first, then the results, reported per participant across sources that can be checked against one another.

As established in Subsection 5.1.2, self-regulation serves here as the operationalisation of learner autonomy (Table 6.3), so what the questionnaire measures is autonomy rather than a separate construct. Two namings follow from that and are held apart throughout this section: everything derived from the questionnaire is named for self-regulation, since that is what the published instrument measures, while *autonomy loop* continues to name the outer loop of the design.

6.8.1 Method

A single-group pretest–posttest design was adopted [55], chosen for the exploratory nature of the study, the small anticipated sample, and the short intervention, which together made a controlled design impractical. Three sources were combined: a pre-treatment questionnaire measuring baseline self-regulation, usage logs recorded automatically during the intervention, and a post-treatment questionnaire repeating the same items alongside an evaluation of the application.

Participants and Recruitment

Participants were recruited through posts in Danish-learning Facebook groups, and all of them were attending Danskuddannelse 3 classes at the time of the study. The recruitment

post stated what the application is and what it asks of a learner, that words met in everyday life are collected in it and practised against goals the learner sets for the week, so everyone who volunteered knew what kind of tool they were committing a week to before the study began. Participation was voluntary and could be withdrawn at any time.

Instruments
Following instruments were used in the evaluation.

Self-regulation scale The eleven items of the self-regulation subscale of the Language Learner Autonomy Scale [139], rated 1 to 5 and reproduced in Appendix H. They are called the *self-regulation items* below, and their mean the *self-regulation composite*. All eleven are taken together as one scale, for reasons set out under Data Analysis. The original wording was left unaltered, with a contextual clarification added beneath ambiguous items to anchor them to Danish vocabulary. The post-treatment questionnaire repeated the items under a heading noting they were the same as before and asking participants to answer as they applied now.

Usability. The System Usability Scale, ten items yielding a score from 0 to 100 [56], administered post-treatment only. Its purpose is diagnostic: published norms place the mean at around 68 [140], so a score establishes whether the application was serviceable enough for the intervention to have been a fair test of the design.

Open questions. Three, all optional. Two after the usability items ask what made the application easy or hard to use, and whether anything was particularly useful or frustrating. One after the self-regulation items asks whether using the application changed anything about how the participant approached their Danish vocabulary, noting explicitly that a negative answer is equally useful. That last wording is deliberate: asking whether the application improved a participant's capacity to regulate their learning would invite a claim seven days of light use cannot support.

Usage logs. App openings, active days, time spent, words added and where they were added from, weekly targets committed and adjusted, and practice sessions opened and completed, all per participant and recorded automatically. These allow a genuine null effect to be distinguished from non-use.

Procedure

1. **Pre-treatment survey.** Participants gave consent, created their personal code, and completed the eleven self-regulation items.
2. **Registration and onboarding.** Participants registered in the application using the same code and received a written user guide covering how to capture a word, build a practice session, and set and review weekly goals. A short call was offered on request and taken up by one participant.
3. **Seven-day intervention.** Participants used the application independently, with no prompting or further support. Usage was recorded automatically. Registration fell in the middle of a calendar week, so the seven days spanned one week boundary and contained one full turn of the autonomy loop: targets proposed at registration and committed as proposed or adjusted, progress accruing over the remainder of that week, and one review at the boundary, where each target is shown as met or not met and the next week's targets are proposed in turn. No learner had a second full week under committed targets within the study window.
4. **Post-treatment survey.** Participants completed the SUS, two open questions on using the application, the same eleven self-regulation items, and a final open question.

Data Analysis
Pre- and post-treatment self-regulation scores were compared within subjects at composite level and reported as the direction and size of each participant's change. SUS was scored by the standard procedure and read against published norms. Usage logs were summarised descriptively and used as context for the questionnaire results. Open responses were coded inductively into short descriptive groupings and reported with counts and quotations, reproduced as written with obvious typing errors silently corrected. Given the sample size, results are reported per participant rather than in aggregate, and interpretive weight rests on agreement between sources that could disagree: what the logs record, which items moved, and what participants wrote in their own words.

The eleven self-regulation items are scored as one scale. The instrument separates items about identifying needs and setting goals from items about planning and monitoring, but with five pairs, choosing after seeing the data which grouping to present is precisely the freedom a study this small should not take. A single composite is therefore the outcome measure, and the question of where any change sits is put to the item level, where all eleven items are reported rather than selected.

6.8.2 Results
Five participants completed the study, and results are reported per participant throughout. The sections are ordered so that each constrains the next: where participants started, what they did over the week, whether the application was serviceable enough for that use to count as a fair test, whether self-regulation moved, where in the instrument it moved, and what participants said had changed. Any one of these is thin on its own; what carries weight is where they agree.

Participation and Pretest Scores
Seven participants completed the pre-treatment questionnaire. Five went on to use the application over the seven days and complete the post-treatment questionnaire; two did not continue. All five are included in the analysis, including those whose use was light, since how much a voluntary tool is used is itself a finding about the design. Table 6.10 sets out the pretest self-regulation composite of all seven, on the original 1 to 5 scale. The instrument carries no published cut-offs, so these values are interpretable relative to one another rather than against an external standard.

	Danish level	Pretest self-regulation	Completed the study
P1	Module 3	2.27	no
P2	Module 3	2.82	yes
P3	Module 3	3.09	yes
P4	Module 4	3.27	yes
P5	Module 1	3.45	yes
P6	Module 5	3.55	no
P7	Module 3	4.09	yes

Table 6.10: The seven participants, ordered by their pretest self-regulation composite, the mean of the eleven self-regulation items on the original 1 to 5 Likert scale.

Three features bear on how the intervention should be read. Composites run from 2.27 to 4.09 with a median of 3.27, so participants who already report directing their learning confidently have little room to move. Module level does not account for the spread: the four learners in Module 3 hold the lowest composite, the highest, and the whole range between them.

Application Use
Use varied considerably (Table 6.11), from three days and fourteen minutes at one end to seven days and roughly two hours at the other.

	Presence			Capture		Rehearsal	Autonomy loop
	Days used	App opens	Time (min)	Words added	From exercise	Sessions done/opened	Targets adjusted
P3	7	17	119	64	12	13/15	2
P7	6	7	53	36	0	4/7	3
P5	6	10	78	42	13	4/7	1
P2	4	4	25	15	6	3/4	0
P4	3	3	14	22	0	3/3	0
Total	—	41	289	179	31	27/36	6

Table 6.11: Recorded application use over the seven-day intervention, ordered by days used. Time is rounded to the nearest minute.

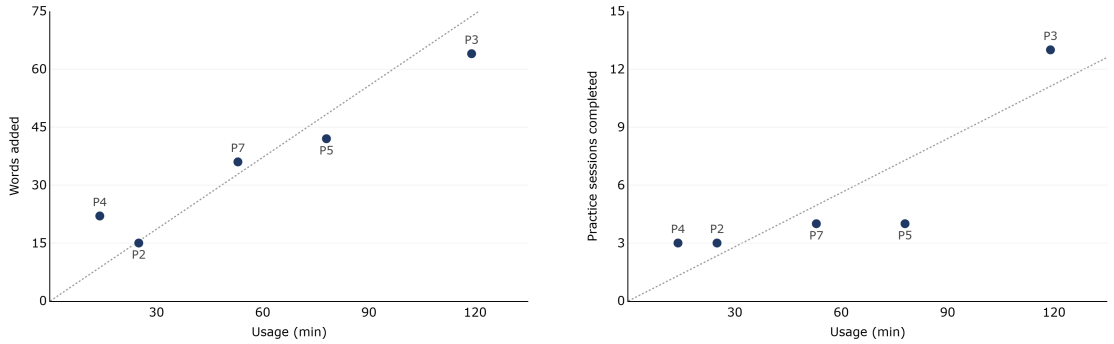
An app visit lasted 7.0 minutes on average, and no participant averaged less than 4.7 or more than 7.8, so the application was used in short repeated visits whoever made them. What separates fourteen minutes of total use from 119 is how often a learner came back, and that is where they part: P3 opened the application 2.4 times on each day it was used, P5 1.7, and the other three about once. Heavy use was built out of more days and more openings within a day, not longer stays.

The adjustments in the final column fall at the second goal setting, after the review at the week boundary had shown which of the first week's targets were met. Three of the five adjusted there, P7 three targets, P3 two and P5 one, so three learners passed through the review and committed the following week's targets with figures of their own.

Two further observations concern what learners did with the words themselves. Across the group 179 words were captured and 27 of 36 practice sessions completed, a rate of 75 %, and the nine unfinished sessions are not spread evenly: seven of them belong to P5 and P7, who each left three of the seven they opened, while P3 finished 13 of 15 and P4 all three. Round 1 participants had found sessions of eighteen exercises too long (Section 6.6), and the shortened twelve-exercise session was carried to the end three times in four, enough to call it workable in independent use but not comfortable for everyone. Second, the route from Rehearsal back into Capture was taken: three of the five saved words met inside an exercise rather than captured directly, 31 words in all, and for P2 that was 6 of 15, almost half of everything added.

Capture and Rehearsal do not draw on time in the application in the same way (Figure 6.13). Words added track the minutes closely (Figure 6.13a): four of the five captured between 0.54 and 0.68 words a minute, so the group rate of 0.62 describes them individually and not only in aggregate, and only P4 departs from it, adding 22 words in fourteen minutes. Completed sessions do not (Figure 6.13b). P3 accounts for 13 of the 27, and the remaining four completed three or four each whether they spent fourteen minutes in the application or seventy-eight: P5's 78 minutes produced the same four sessions as P7's 53 and one more than P4's 14. Part of that flatness is abandonment rather than absence, since P5 and P7 each opened seven sessions and left three, but P2 and P4 opened few in the first place.

A practice session issues eighteen requests to the language model before its first question appears (Section 6.7), and they are issued when a session is opened and not when it



(a) Words added

(b) Practice sessions completed

Figure 6.13: Time recorded in the application against what came of it, one marker per participant.

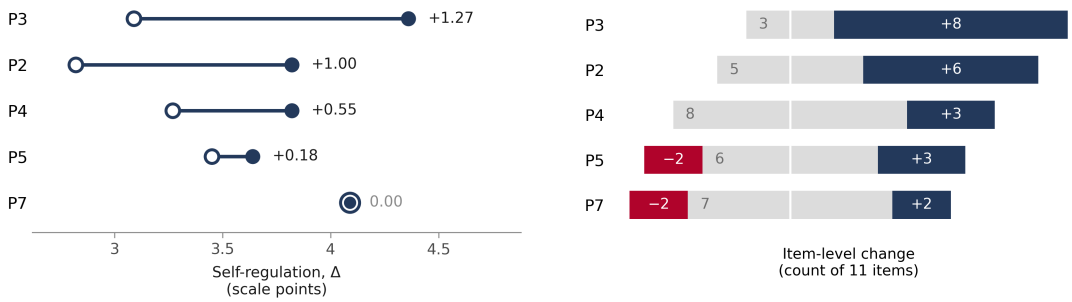
is completed, so all 36 opened sessions carry the full count: 648 requests across the week, of which the nine sessions left unfinished account for 162 generated and never answered. Generations that failed and retried (Section 6.6) are not counted, so the true figure is somewhat higher.

Usability

SUS scores run from 57.5 to 92.5, with a mean of 80.5 and a median of 85, one per participant on the right-hand axis of Figure 6.15. Four of the five sit above the published mean of about 68 [140], the lowest of those by almost ten points; P7 returned 57.5 and is the one below it. The two lightest users, P4 with three days and P2 with four, returned 85 and 77.5, so neither case of thin use can be attributed to an application the participant found unusable. P7, conversely, returned the lowest score while using the application on six of the seven days, and the open answers locate that dissatisfaction not in operating the interface but in remembering to open it at all (Subsubsection 6.8.2).

Self-Regulation Before and After

Figure 6.14 shows each participant's self-regulation composite before and after the seven days, and how many of the eleven self-regulation items moved. The composite rose for four participants and was unchanged for one, a median change of $+0.55$ scale points, and no participant scored lower after the week than before. With five pairs that is a statement about how many participants moved and in which direction, not about how far.



(a) Self-regulation composite

(b) Self-regulation items moved

Figure 6.14: Self-regulation change per participant, both panels ordered by pretest composite. In (a) the open marker is the composite before the intervention and the filled marker after; the dashed rule is the scale midpoint. In (b) the same participants' eleven items, split into those that rose (navy, left), did not move (grey) and fell (red, right), which distinguishes a composite that held still because nothing moved from one that held still because changes cancelled.

Two qualifications attach to that result. The unchanged composite belongs to P7, who

began at 4.09, the highest pretest composite in the study: seven of P7's eleven item responses did not move, and the two that rose were offset by two that fell. More seriously, the four who did move are exactly the four who began lowest, which is the pattern regression toward the mean produces as well as the pattern a working intervention produces [55], and a single-group design cannot separate them.

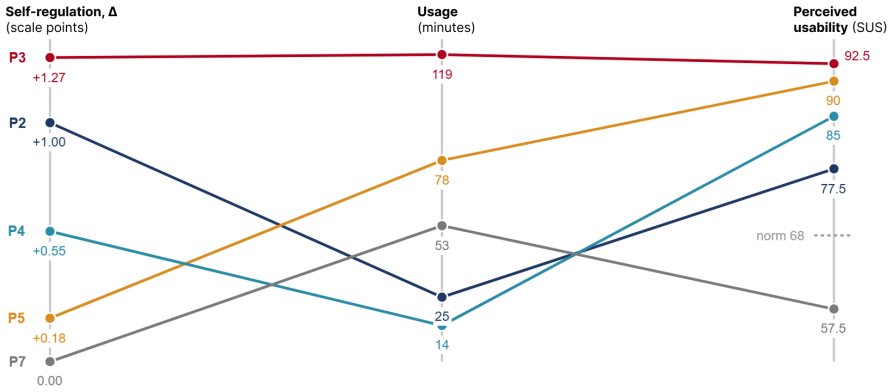


Figure 6.15: The three outcome measures as parallel coordinates, one line per participant. The axes are not on a common scale: vertical position gives the ordering within an axis and the printed number the value, so the slope between two axes carries direction only, not magnitude.

Participants do not hold the same position across the three measures (Figure 6.15). Only P3 sits near the top of all three, with the largest change and the most time in the application at 119 minutes. P2 has the second largest change on 25 minutes across four days, while P5 spent 78 minutes and moved $+0.18$. Among these five, how far the composite moved does not follow how much the application was used.

Where the Change Sits

All eleven self-regulation items are reported, ordered by mean change (Table 6.12). Ten item means rose, one fell, and the mean across items is $+0.6$. The rises are not of equal size, and where the largest of them sit decides whether the movement is concentrated on what the design addresses or spread across the subscale.

Item	P2	P3	P4	P5	P7	**Mean**
SRN1	+2	+1	+3	0	0	**+1.2**
SRN2	0	+2	+2	-1	0	**+0.6**
SRN3	0	+2	+1	-1	0	**+0.4**
SRN4	0	0	0	0	+1	**+0.2**
SRN5	+1	+1	0	+2	0	**+0.8**
SRP1	0	0	0	0	-1	**-0.2**
SRP2	+2	0	0	0	0	**+0.4**
SRP3	0	+3	0	+1	0	**+0.8**
SRP5	+2	+1	0	+1	0	**+0.8**
SRP6	+3	+1	0	0	-1	**+0.6**
SRP7	+1	+3	0	0	+1	**+1.0**
Mean across items						**+0.6**

Table 6.12: Change on each of the eleven self-regulation items, in scale points per participant. Items are worded in full in Table H.1; a rise is navy, a fall red and an unchanged response grey, as in Figure 6.14b.

Across five participants and eleven self-regulation items there are 55 responses, of which 22 rose, 29 did not move and 4 fell, and three participants recorded no decline on any item. Movement acting narrowly on the items the design addresses would leave declines

elsewhere, and these data do not show that, so the item level cannot rule out a general tendency to answer more affirmatively the second time.

The two largest item means are setting one's own learning goals (SRN1, $+1.2$) and developing daily or weekly learning plans (SRP7, $+1.0$), which is what committing to a week's targets and repeating that commitment ask, with evaluating whether learning goals have been achieved (SRN5, $+0.8$) among the three tied behind them.

What Participants Reported

All five participants answered all three open questions. Every substantive statement was coded as raised positively or as friction, and assigned either to a stage of the vocabulary learning loop or to the autonomy loop. Table 6.13 gives the representative points on each side.

	Valued or reported as changed	Raised as friction
P2	Recognised a need to devote more time to learning; the pronunciation audio	Forgot the application was there; wanted more exercise types
P3	A place to store vocabulary, allowing a more active approach than before	Capture too slow to use during class; purpose of the Achievements page unclear
P4	Being able to sort words into collections; a more structured practice than before, having previously used nothing to organise vocabulary	Classification does not record part of speech
P5	Liked the idea	Translations were sometimes inaccurate
P7	Helped in getting used to newly encountered words	Near-synonyms hard to tell apart, nothing prompting a return to the application

Table 6.13: Representative points from the three open questions, one row per participant, split by whether the points were raised positively or as friction. Entries paraphrase what participants wrote. A request for more of new features counts as friction.

Capture was valued by P3 for giving words somewhere to go and by P4 for letting them be sorted, and the same two named its limits: P3 that capture is too slow to use during class, which is the one time-critical moment of the loop, and P4 that classification does not record part of speech. Resolution was raised once in each direction, P2 valuing the pronunciation audio and P5 reporting translations that were not correct, which touches the depth of functionality deliberately held low (Figure 6.6). Rehearsal was valued by P7, for getting used to newly encountered words, and the friction at this stage concerns coverage rather than the exercises themselves: P2 wanted more exercise types, and P7 found near-synonyms hard to disambiguate.

The autonomy loop drew a different kind of comment. Three of the five reported that it had changed something, and in each case what changed was a matter of noticing rather than of doing. P4's learning became more structured and P2 came to see the language as deserving more time, while P7 located the change in what the application had put in front of them:

> *"did make me consider that I needed to focus on more complicated comprehension and sources to expand my vocabulary"*

The friction came down to a single complaint, made independently by two participants. P2 reported simply forgetting about the application, and P7 gave the fullest account of it:

> *"The app doesn't have a visual that is captivating enough to be noticeable*

above the other apps … there was nothing to remind me about the app after"

This is the sharpest confirmation of a limit accepted before the build. The application remains browser-bound, without push notifications or any reminder beyond what the interface shows when opened (Subsection 6.7.3), so every prompt it offers reaches only a learner who has already opened it, and nothing in the system acts on a learner who has not.

6.8.3 Summary of the final evaluation

Five learners used the application unsupervised for seven days in their own everyday context. Self-regulation was measured before and after on the same eleven-item instrument, usability once at the end, and everything participants did in between was recorded by the application itself, so that what they reported could be read against what they actually did.

6.9 Solution Space Conclusion

The solution space was given one question to answer: how can a system be designed to support learner autonomy in retaining self-encountered vocabulary? What it produced is a built and evaluated answer, in two parts, a structure and the evidence of what learners did with it.

The structure is the dual loop. The constraints derived from the problem space settled the vocabulary learning loop of capture, resolution, and rehearsal, and left open whether that loop runs at all (Section 6.2). The open half was designed as an autonomy loop, drawn from Zimmerman's phases of self-regulated learning and composed into three candidate concepts, of which the goal-regulation assistant was selected (Subsection 6.4.5). The build cycles then revised it: a goal attached to a single collection was set once and forgotten, so the loop was rebuilt as a recurring weekly cycle that resurfaces commitments and holds the next week's targets behind a review of the last (Section 6.6). The system carried into evaluation is that revision rather than the concept as selected, and under Research through Design the revision is itself a finding.

The evidence is what five learners did with that system over seven days of unsupervised use in their own everyday context (Section 6.8). All five entered the weekly cycle and stayed in it, completing 75 % of the practice sessions they opened, on an interface four of the five rated above the published usability norm. Four reported higher self-regulation afterwards and none reported lower, and the largest item-level movement sits on setting goals, planning, and evaluating whether goals were met, which is where the autonomy loop intervenes. Learners also took the route back from rehearsal into capture that the dual loop predicts, saving words met inside an exercise rather than only words brought from outside.

With that, both diamonds are complete, and what remains is to examine what they rest on. The project made commitments at every stage that its own evidence cannot settle: reading the problem as one of autonomy rather than of fragmented tools, dividing the system into two loops and holding one fixed while the other was designed, letting feasibility bound what was built, and drawing language content from external services at a cost paid in accuracy and control. Chapter 7 takes up those commitments together with the method that produced them, and sets out where the work should go next.

7 Discussion

This chapter reads the thesis as a whole. The work traversed the full process once, end to end: four interviews narrowed an open problem into a precise problem statement, the design space it opened was inventoried and composed into three concepts, one concept was carried through three build cycles into a deployed application, and five learners used that application for a week on their own. Section 7.1 follows that sequence as a chain of decisions, separating those the evidence forced from those taken under constraint. Section 7.2 places the resulting system among existing tools, Section 7.3 discusses what learners did with it over the week, and Section 7.4 puts the final problem statement to that evidence and answers it. Section 7.5 sets out which learners the system is for, and the last two sections state what the thesis does not establish and what should follow from it.

7.1 From an open problem to one built system

The thesis opened on a problem with almost no edges. Section 1.1 asked how adult learners manage vocabulary across their everyday contexts and where that process breaks down, and how a system could reduce the friction of vocabulary acquisition and support long-term retention. Between them those questions span every setting in which a learner meets a word, every tool they might reach for, and every stage between meeting a word and knowing it. The thesis closed on one system, built around one behaviour, tested for a week by five learners on their own phones.

This section sets out the chain that connects the two. Three things were fixed before any of it began, and none was concluded from evidence. The subject was vocabulary, on the ground that vocabulary is the most immediate constraint on a learner [6]. The outcome was to be a working application learners could run themselves. And each diamond was to be traversed once and in sequence, because a full traversal produces an artifact and a field evaluation while two partial traversals produce neither, and the second research question can only be answered by a working system under real conditions (Section 3.1).

The third decision is why the chain is linear, and it divides in two. In the **problem space** each step follows from what the last one turned up. In the **solution space** each step follows from what was available.

7.1.1 The problem space

The domain was wide enough that what to look for in it only became clear through talking to learners. The work began with four interviews, structured on Gu's tetrahedral model [18], with Distributed Cognition as a second lens asking what carried the cognitive work at each stage besides the learner [9]. The model made the four accounts comparable. It also decided what was asked, and so what could be found: it organises person, task, context and strategy, and is silent on the metacognitive conditions surrounding a task.

That cost surfaced during analysis. Framework analysis codes against a structure fixed in advance [48], which is what makes accounts comparable and also what limits them: a breakdown that does not belong to a single block is recorded separately under each of them, and its shape is not visible from inside the frame. One breakdown did exactly that, appearing under person as a personal failing, under context as circumstance, and under tasks and strategies as a missing tool. The propositions were therefore detached from the blocks that produced them and regrouped by conceptual similarity [49], giving five clusters (Figure 4.2).

Three of the five clusters were set aside and two were carried forward as design targets: the absence of a system connecting capture to rehearsal, and external regulation as the primary study behaviour. Taken together they moved the work across the division between the strategies by which learners discover, record and consolidate words and the metacognitive strategies by which they manage that work [17, 18]. The first of the two sits on the cognitive side, and is what the initial problem statement had anticipated. The second sits on the metacognitive side, and is the finding the rest of the thesis is built on: what participants described as their own laziness or lack of motivation is the absence of a structure that would prompt and sustain study. Carrying both is what widened the work from the handling of words to the regulation of studying them.

The focused literature review named that behaviour (Section 4.4), and two narrowings then closed the problem space (Chapter 5). Autonomy was operationalised as self-regulated learning, because a capacity cannot be designed for or measured directly while SRL names observable activities (Subsection 5.1.2), and vocabulary was narrowed to self-encountered vocabulary (Subsection 5.1.1). Together they set the technical problem the rest of the work turns on: a system accepting an arbitrary word met yesterday cannot hold its content in advance.

That problem is also where the capability the thesis opened on ends up. The introduction presented large language models as what brings a tool organised around the words a learner actually meets within reach, since content for a word the system has never seen can be produced the moment it is entered (Section 2.3). What the interviews returned as the breakdown was not the supply of that content but the management of study around it, so the language model was moved out of the central investigation. It remains what makes the narrowing to self-encountered vocabulary technically possible, and it is therefore what the system is built on rather than what the thesis puts to the test.

7.1.2 The solution space

The solution space opens where the design space was divided. By this stage it was settled that the system would carry a vocabulary learning loop; what remained to explore was how that system helps a learner regulate their study. The dual loop made the division explicit, fixing the vocabulary learning loop as a baseline system and leaving the autonomy loop open (Section 6.2, Section 6.3). Within a single traversal there was time to explore one half and not both. That division is the largest single dependency in the thesis: what was held constant came out of four interviews, and the space explored afterwards was settled before any concept existed.

Sixteen mechanisms were then inventoried across the three phases of the autonomy loop (Table 6.7) and composed into whole concepts, because the inventory is interdependent and selecting from it piece by piece would have produced support in which each part lacked what the others were to supply. Three concepts were generated, one led by forethought, one by performance and one by self-reflection (Subsection 6.4.4), so the open half was covered by phase and not by preference. Only one could be built. Concept A was selected on weighted objectives, scoring 4.20 against 4.00 for both alternatives (Subsection 6.4.5). That is a narrow margin on which to settle what gets built, but with only one concept to carry forward the selection had to be made on it; Section 7.6 returns to what it rests on.

Concept A was then built as a progressive web application over Flask and PostgreSQL on a managed host [129, 130, 131], across three build cycles. Round 1 implemented the per-collection goal; participants set it without difficulty and then saw little reason to maintain it, judging their progress by their vocabulary learning as a whole, not one collection at a

time. Round 2 rebuilt around a recurring weekly cycle over that whole activity, one that resurfaces goals, adjusts targets and ends in a review that gates the next week, and the breakdown did not recur (Table 6.9). This is the only point in the process where something built came back and changed a decision already made, which is the form of knowledge Research through Design licenses (Section 3.1). The traversal then closed, as Deliver requires, in a single-group pretest–posttest over seven days.

The chain is linear because the Double Diamond was traversed once and not iterated (Section 3.2). Nothing in the model requires that. Its four phases can be run again on what the previous traversal produced, and each narrowing described above is a point at which a second pass would have reopened a decision this one had to fix: the interview structure that shaped what could be found, the division that held the vocabulary learning loop constant, the objectives on which one concept of three was selected. Running the process once was a decision about what a single thesis can produce, and the dependencies traced through this section are the consequences of it.

7.2 Where the system sits

What came out of that process is an application a learner installs on their phone and fills with their own words (Section 6.7). A word met anywhere, in class, in conversation, or in something being read, is saved into it; the system offers a translation, a part of speech, an example sentence and a spoken form, which the learner approves or edits before anything is kept; and the words a learner has collected come back as short practice sessions written for those words. Around that runs a weekly cycle. The learner commits to a target in each of four categories, one for adding words and three for practising them, progress accrues from what they actually do, and the week ends in a review they have to complete. Nothing in it decides which words a learner studies, and nothing studies on their behalf.

It is a support tool, and that is the first thing that places it. Duolingo and Babbel supply a course, so a learner can start from nothing and let the platform decide what comes next. This system has nothing to offer until a word has been met somewhere else, so it works only for a learner who is already active outside it, which is the learner Section 2.4 describes: enrolled, meeting Danish in class and around it, and left to consolidate most of it alone. What it adds to that arrangement is practice detached from any schedule, in sessions short enough to fit the time a learner actually has.

Against the tools that also hold learner-chosen words, two things place it. Quizlet and Anki are built for study in any subject, so the learner authors every card; this system is specific to vocabulary, which is what lets it produce the translation, the example sentence and the exercises for a word on its own. Closest to it are Vocabuo and SnapWords, built for self-encountered vocabulary and capturing a word where it is met (Table 2.1), and against that small group the difference is the second loop. Those tools hold the words and leave the returning to the learner. This one scaffolds the returning, through a weekly cycle of committed targets and a review that closes each week before the next opens.

The position this leaves is a narrow one: a tool that holds only the words a learner supplies and scaffolds the returning to them, for a learner already meeting the language elsewhere. Placing it there is an argument from what was built and from what the tools around it do, and it settles nothing about whether the difference matters once a learner has the system on their phone. The week with five learners is the first evidence on that.

7.3 What learners did with it

This section discusses the findings of the final evaluation.

How much the five used it varied widely. At one end a learner opened the application on all seven days for close to two hours and captured 64 words; at the other, one used it on three days for fourteen minutes and captured 22. Across the group it was opened 41 times for 289 minutes and 179 words were captured (Table 6.11). Moreover, usability returned a mean of 80.5 and a median of 85 against a published norm near 68, with four of the five above it [140]. Everything below rests on these records.

The application also logged what learners did with it. All five committed a full set of four weekly goals and three adjusted the targets proposed for the following week; 27 of the 36 practice sessions opened were finished; and 31 of the 179 words were captured from inside an exercise rather than from outside the system, by three of the five.

The statements participants wrote about the handling of words fall either side of a clean line, and four of the five wrote on both sides of it (Table 6.13). What they valued was that the system gave them somewhere to keep and sort words, a spoken form to hear, and practice that made new words familiar. What they faulted was the execution of those same things: capture too slow to use during class, a classification that does not record part of speech, translations that were wrong, too few exercise types, and distractors close enough in meaning to the target to be hard to separate. Each complaint names a fault in the build rather than a doubt about the concept, and no participant wrote that keeping their own words was not worth the effort.

The statements about regulating study are distributed differently. Three participants reported that the application changed how they thought about their learning, one toward more complex material, one toward a structure they had not had before, and one toward giving the work more time. Two reported the opposite condition, that nothing brought them back to the application once they had closed it, which is the failure a browser-bound prototype predicts (Subsection 6.7.3). One found the Achievements page unclear in purpose.

What none of the statements describes is the weekly cycle the design is built around. The three positive remarks concern noticing a need rather than setting a goal, and no participant mentioned committing to a weekly target, adjusting one, or the end-of-week review, though all five committed a full set and three adjusted. Goal setting and goal evaluation are the two elements participants did not describe and the two on which the questionnaire moved most. Either they worked without being available to report, or the movement reflects something other than what participants experienced, and a week with five learners cannot separate the two.

One further absence is worth recording. No participant questioned the exercises being produced by a language model or asked where the material came from, across 648 generation requests made unsupervised on their own phones (Section 6.8); the one complaint touching generated content concerns the quality of distractors and not their provenance.

What this record answers is taken up next.

7.4 Answer to the final problem statement

The final problem statement asks how a system can be designed to support learner autonomy in retaining self-encountered vocabulary (Chapter 5). One way to design it is two coupled loops. The vocabulary loop carries a word met anywhere from capture, through resolving its meaning, to retrieval practice [78]. The autonomy loop around it, a weekly cycle of goal setting and self-reflection, is what keeps that inner loop going: retaining a self-encountered word takes returning to it, and returning to it is what the weekly cycle asks for. Regulation is therefore a loop of its own, and the learner drives it.

Three lines of evidence say the system did that work for the users.

1. **What participants did.** They committed to a set of weekly targets and three out of five adjusted the next week's at the review. They returned on three to seven days unprompted and finished three quarters of the sessions they opened, and three of them saved words met inside an exercise.
2. **What participants rated.** Four reported higher self-regulation after the week and none lower, and the item-level movement lands where the outer loop acts: setting one's own learning goals ($+1.2$), developing daily or weekly learning plans ($+1.0$), and evaluating whether learning goals have been achieved ($+0.8$).
3. **What participants described.** Three of the five reported a change in how they went about learning Danish, toward more structure, more complex sources, or more time given to the language (Section 7.3). Each concerns a decision the learner made about their own study rather than the practising itself, and that the two loops drew different kinds of comment is the case for keeping them apart (Table 6.13).

What the autonomy loop keeps running is retrieval practice on the learner's own words, which is where retention is claimed rather than measured. Retrieval practice improves retention over restudy and is a mechanism of learning rather than only a measure of it [16, 71], and the effect strengthens when retrieval is repeated over time [72].

The week leaves two things open. No participant wrote about the weekly cycle itself, so the questionnaire movement on goal setting and review stands without a statement beside it, and how much Danish the design retains would take a delayed post-test (Section 7.7). Both concern how far the answer reaches rather than whether the design ran: for a week, on their own phones and unprompted, the regulative loop kept five learners returning to words they had chosen themselves.

7.5 Who this system is for

The problem-space study, the build-cycle evaluations and the final evaluation each drew on different people, since Danskuddannelse 3 turns over its cohorts continually. What holds them together is that all of them were enrolled in that program, so the population sampled stayed the same throughout even as the individuals did not.

That population is not uniform in how it learns, and four people interviewed in the problem space were enough to show it. Their preferences and expectations from learning process tracked their prior educational experience more closely than their proficiency, and they differed in whether they leaned external or internal regulation. The final evaluation sample spread the same way on a different axis: pretest self-regulation composites ran from 2.27 to 4.09, which suggest significantly different levels of self-regulation learners have.

This system commits to one strategy, deliberate retrieval practice over words the learner collected, which Section 4.1 adopts as the most direct route from intentional study to retention while stating that it is not the only one that works.

It is built for learners who are already meeting the language in classes or in material they seek out themselves, who find retrieval practice effective, and whose difficulty is staying consistent with it. It is not for learners who acquire vocabulary through immersion alone, or who want a syllabus and the system to take them through it.

7.6 Limitations

Three limits bound what has been claimed.

Autonomy shows in behaviour over time, and the study had one week. Self-regulation was measured as a change in self-report across seven days, a short window for a construct defined by what a learner does when nobody is asking, and the cycle was observed once rather than as the recurring obligation it was designed to be. Two free-text boxes are also a thin instrument, and interviews would have returned more, particularly on the two mechanisms the item movement points at and nobody mentioned (Section 7.4).

The outcome measure is self-report, from people who knew what it was for. Everyone who took part had answered a call describing the application (Section 7.5), and the same participants rated their own self-regulation before and after. Four composites rose and none fell, which is what a design that works produces and also what participants who can tell what a study hopes for produce; with five respondents and no comparison group the questionnaire cannot settle between those readings. What it can carry is the direction of change read against the usage record, which is why the two are never reported apart.

Each diamond was traversed once. This limit addresses overall methodology of the thesis. Iterative double diamond would have meant returning from the solution space to the problem space, putting what the week with five learners returned back to the problem statement and to the constraints it produced (Table 6.1) before a second pass through concepts. Traversing once instead bought an artifact and a field evaluation, and it left the four interviews setting the trajectory of everything after: their propositions became the design constraints, which became the dual-loop split, which settled which half would be explored before any concept existed.

7.7 Future work

The order below is a dependency order: the artifact should be finished before it is put in front of learners for longer.

1. **Add prompting to the goal cycle.** Notifications tied to a learner's own plan sit in the mechanism inventory (Table 6.7) and give the weekly goals something to act through between sessions, which is the absence two participants named directly. For example, Push API delivers them to an installed progressive web application, so it does not require leaving the platform [141].
2. **Make meaning resolution contextual.** Making the language model primary instead of a fallback, so a word is resolved together with the sentence it appeared in and more than one sense is offered where the word carries more than one, delivers what 12 asks for within a call the system already makes.
3. **Raise depth of functionality.** Depth was held at three of five deliberately (Subsection 6.7.3). Carrying the exercise progression through to production answers the request for more exercise types, and surfacing the part of speech already stored at Resolution costs almost nothing.
4. **Test usefulness of separate features.** For example, ask users directly how useful they found weekly cycle of goals,
5. **Rebuild natively if it goes further.** A native application would remove the remaining platform-determined frictions together: capture from a share target or a lock-screen entry at the moment of encounter, and prompting that does not depend on the learner having already opened the application.
6. **Measure retention against a control group.** What the design claims about retention rests on the mechanism it applies, not on anything measured here (Section 7.4). Closing that gap needs a delayed post-test of vocabulary gain, set far enough after practice to test retention rather than recall of a recent session, against a control group drawn from the same Danskuddannelse 3 classes.

This closes the discussion, which has traced the chain from the open problem to the built system, placed that system among existing tools, read what learners did with it over the week, answered the final problem statement, and set out what bounds that answer and what should follow from it.

8 Conclusion

This thesis asked how a system can be designed to support learner autonomy in retaining self-encountered vocabulary. The question came from opening a broad concern with adult vocabulary learning, reviewing what research establishes about how words are learned, and interviewing four adults enrolled in Danskuddannelse 3 about how they handle the words they meet. Those learners understood the words they met. What they lacked was anything that carried a word past that moment, or that sustained returning to it once the structure of a class was removed.

The answer is a design built on two loops. At its core is a vocabulary learning loop, carrying a word from Capture, through Resolution, to Rehearsal as retrieval practice, with the language content generated on demand so that a word the system has never seen can still be studied. What keeps that loop turning is an autonomy loop around it, a weekly cycle through the three phases of self-regulated learning, forethought, performance, and self-reflection, opened by targets the learner sets and closed by a review they complete before the next week begins. One of three candidate concepts was selected against weighted objectives, realised over three build cycles, and deployed as an installable web application.

Five learners used it for a week on their own phones, with no supervision once the study began. Every one of them committed a full set of weekly goals, 179 words were captured between them, and 27 of the 36 practice sessions opened were finished. Four reported higher self-regulation after the week and none reported lower, and the largest item movements fall on setting one's own goals, making weekly plans, and evaluating whether goals were achieved, which is where the autonomy loop acts. Three described a change in how they approached studying Danish.

A tool organised around the words a learner actually meets has only recently come within reach, since the material it practises cannot be written before the learner chooses the word. Every exercise the five answered was generated for them by a large language model, and none of them raised it, so content produced on demand proved a workable substrate for practice.

What the thesis delivers is a problem statement grounded in how adult learners handle the words they meet, a design that answers it in two coupled loops, and a working application realising that design and evaluated with learners in their own everyday context. Across that week the five kept the cycle turning without being asked, which is what the design was built to secure. Supporting learner autonomy here means the learner remains the source of direction, while the system carries the organisational work that keeps self-encountered words moving from capture through to rehearsal.

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A Interview Guide

Introduction

Speech to participant before interview begins

This interview is conducted as part of a master's thesis at the Technical University of Denmark (DTU), investigating how people learn vocabulary in a second language, specifically Danish. The interview is not a test, there are no right or wrong answers. The focus is on vocabulary acquisition, as vocabulary is crucial for comprehending listening and reading material. Other aspects of language learning, such as grammar and pronunciation, are outside the scope of this study.

The interview is semi-structured and will be conducted as a conversation, guided toward relevant topics. Audio will be recorded for transcription purposes only. All data will be anonymised; participant names will not appear anywhere in the thesis.

Questions

This is a semi-structured interview, therefore questions may vary

Block 1: Person

Goal: understand who this learner is and what drives them.

1. How long have you been learning Danish, and why did you decide to learn it?
2. How would you describe yourself as a language learner... do you feel like you are good at picking up new languages or it is more challenging to you than to others?
3. How much time in a typical week do you actively spend on Danish outside of class?

Block 2: Context

Goal: understand where and when learning happens

1. Where do you most often encounter Danish: in class, at work, on your phone, watching something?
2. How big does the load of unknown words in Danish class feel – not enough to learn from, or so many it feels overwhelming? (*Offer to rate from 1 to 5, where 1 id not enough and 5 is too much and overwhelming*
3. How would you describe environment that is offered you you to practise Danish?

Block 3: Tasks and Strategies

Goal: map what happens when person meets new words.

Deciding to engage

1. Do you try to engage with all unknown words, or do you select which ones to focus on? If you select, how do you decide?

Meaning resolution

1. What do you usually do when you encounter an unknown word – ask a teacher, use a digital tool, or guess from context?

Recording

1. How do you record new words anywhere (notes app, a notebook, something else)?

Rehearsal and repetition strategies

1. What do you do with a word after you have recorded it: do you go back to it, and if so how?
2. Do you use any specific strategies to remember words, for example flashcards or spaced repetition?

Block 4: Strategies and pain points

Goal: identify what breaks down and what participants wish existed.

1. What part of learning new vocabulary feels most effortful or frustrating to you?
2. What specific strategies work and don't work for you?

Conclusion

Thank the participant for their time and invite them to reach out if they recall anything worth adding after the interview.

B Consent form

Informed consent form (EN)

Experiment: Semi-structured interview about vocabulary learning workflow of people who attend formal Danish language education

Investigator: Alona Konstantinova

I confirm that:

- I was satisfactorily informed about the study concerned both verbally and in writing by means of the subject information letter.
- I have had the opportunity to put forward questions regarding the study and that these questions have been answered satisfactorily
- I have carefully considered my participation in the experiment
- I participate in my own free will

I agree that:

- My data will be acquired and stored for scientific purposes as mentioned in the subject information letter in anonymized form (detached from any personal identification)
- Will not be shared with anyone other than the investigator

I understand that:

- My participation is voluntary, and I have the right to withdraw from the experiment at any time without having to give a reason
- Should I withdraw from the experiment, my research data collected before my withdrawal may still be used for research purposes
- My privacy is protected according to Danish law and European guidelines (GDPR; EU 2016/679)
- My consent will be sought every time I participate in a new experiment

I give my consent to take part in this experiment:

_____ _____

Full name Date of birth

_____ _____

Date, Place Signature

C Framework matrix

	Block 1: Person		Block 2: Context		
Participant	**Danish Learning Background**	**Learning prefferences**	**Time Investment Outside Class**	**Danish class experience**	**Danish Encounter Environments**
P1	Attended PD3 classes 2021-2023, completed 4 of 5 modules. Took a two year break and returned to workplace classes, placed back at Module 2 due to attrition. Prior positive experience with structured school English using a flashcard app.	Prefers structured, test-driven learning with external accountability. Learns best through active production, typing words in both directions rather than passive recognition. Positive prior experience with bidirectional flashcard apps. Frustrated by gamification and ads that interrupt the learning flow.	No designated study time. Reads Danish articles opportunistically. Estimates some hours per week but cannot quantify.	Curernt classes is conversation based. Notes, that doesn't learn a lot because they don't learn new content. No homework assigned by current teacher. Previous class experience was opposite: lot of content but no practise, which led to lexical attrition in future.	Workplace Danish classes and self-initiated reading of Danish news and online content. Almost no ambient spoken Danish outside class due to entirely non-Danish-speaking social circle. Frustration of people switching to English
P2	Began Module 1 in February 2025, progressed to Module 3. Almost continuous attendance for roughly one year. Knows multiple languages, enjoys language learning and word origins. Motivation is both instrumental (job market) and integrative (cultural belonging).	Prefers exposure-driven learning through real-world interaction over systematic study. Learns through interest and etymological curiosity rather than deliberate rehearsal. Values speaking and conversation over written exercises. Treats classroom as complementary to everyday exposure rather than the primary learning environment.	Low to class related activities. Opportunistic exposure through rommmate, shopping, Danish-speaking classmates, and passive media. Frames even small efforts as valuable.	No particular comment, treats class as a just complementary tool, learns the most in real life	Classroom texts, conversations with Danish-native landlady, grocery store clerks, coworkers, and self-selected reading. Unusually high ambient exposure due to living with a Danish native speaker.
P3	Began Danish two years ago in a university course in Germany, then continued in Denmark through Modules 3, 4, and 5 of the free language programme. Currently on Module 5. Motivation is integrative: wants to potentially stay in Denmark and speak the language. Also pragmatic: classes are free so low cost to continue.	Responds well to exam-driven deadlines as external motivation.	Low. Attends class once a week and does not actively study outside class beyond assignments.	Experience varried from Module to Module and widely depends on a teacher. No particular comment	Formal classroom texts and assigned reading. Leisure reading in Danish via Kindle which reduces lookup friction. Signs and ambient everyday text occasionally encountered but words found there are almost never recorded.
P4	Started Danish in February or March 2025, studying for just over a year. Only one teacher throughout. Self-described as naturally good with languages. Motivation is primarily social and integrative: wants to speak Danish in everyday situations, not for professional reasons as workplace operates in English. Positive prior experience with Quizlet in high school for English vocabulary	Prefers structured external curricula with progressive tracking and measurable completion.	Approximately one hour per week, almost entirely through Duolingo. Self-aware this is not enough and frames it as a priority issue. Occasionally uses AI by screenshotting content for homework support.	Notes lack of imposed structure. Wishes, for external lrarning planning, for eaxample, wish for teacher giving more tests	Classroom texts and teacher speech. Some ambient Danish in everyday life. Little structured outside-class reading or listening. Classroom videos of fluent Danish speakers are very difficult, estimates understanding around 20%. Tried apps such as Duolingo, but doesn't like slow speed of progress

	Block 3: Tasks & Strategies						Block 4: Reflection	
Participant	**Resource seeking (outside class)**	**Selective Engagement with unknown words**	**Meaning Resolution**	**Recording**	**Retention**	**Learning scheduling**	**Pain Points**	**Wishes & Unmet Needs**
P1	Seeks to read news and personal media to learn and practise Danish. Also tried content based apps such as Duolingo.	Frequency-driven: pays attention to a word when it appears multiple times across a text. Single-occurrence words that can be inferred from context are ignored. Deliberate triage to avoid overload.	Asks the teacher in class. Uses Google Translate outside class. Aware it gives direct translation without usage context, which is insufficient for productive use.	Writes words and translations in a notebook during class. Does not record anything outside class, relies on trying to remember. Wishes for automated capture directly into a flashcard app, citing manual entry as too effortful.	No active retention strategy currently. Most positively recalled experience: a flashcard app requiring typed input in both directions used in middle and high school. Tried Duolingo on returning to Danish but abandoned due to ads and UX friction (gamification). Also admits but learns best by practise	Wants a structured review system that requires minimal manual effort.	Attrition after a two-year break: lost most vocabulary. Within-session forgetting of previously known words in both directions. Lack of spoken practice outside class is the biggest self-identified barrier. Friction of manual entry into any tool.	Automated capture of words from reading or class directly into a flashcard app. Contextual practice (gap-fill, sentence production) rather than isolated word memorisation. A Duolingo-style tool without ads and disruptive gamification.
P2	Choose to speak in Danish with people around them, and retrieves new words through conversations. Consumes media in Danish.	Reads quickly and does not stop for every unknown word. Selectively attends to nouns and adverbs using grammatical pattern recognition as a filter. Words that are phonetically or semantically interesting get extra attention.	Uses context inference roughly half of the time before deciding whether to look something up, mainly because checking every word takes too much time. Google Translate for direct translation, Den Danske Ordbog (ordbog.dk) for deeper information including inflections and tenses. Asks teacher or Danish speaker for contextually specific usage questions.	Maintained a notebook organised by grammatical category, grouping words such as adverbs by type (time, space, manner), with the explicit intention of reusing them later. Not consistent.	No explicit or structured rehearsal method. Relies on interest-driven retention: words that are phonetically or etymologically interesting are naturally practised more. No flashcard use or spaced repetition mentioned.	Aware that approach lacks structure but has not found a system worth maintaining. Values exposure and use over systematic study. Does not set explicit learning goals or schedules outside class.	Main friction is recording: understands its value but lacks a system that makes it easy or worthwhile. Comprehension of dense unknown texts without structural scaffolding. Lazy to look up each individual word translation.	Visual labelling of words in texts by grammatical type (colour-coded or tagged) to provide structural scaffold for reading. An interactive synonym suggestion feature similar to Word's synonym tool to explore meaning through known related words.
P3	Actively seeks as much danish content as possisble. At Module 5 level most unknown items are isolated words rather than full comprehension failures. At earlier levels had higher density and found it very disruptive, especially with printed texts	Distinguishes between words worth engaging with and words that are too context-specific to bother.	DeepL preferred over Google Translate for multi-word expressions and idiomatic phrases because it handles context better. In class writes words into a Word document. For leisure reading uses Kindle built-in dictionary, which avoids the phone distraction loop they previously experienced. Previously tried entering whole sentences when single-word lookup felt insufficient.	Single Word document used as a reusable class template: deletes previous session content and records 5-6 new words per session with translation and sometimes a usage note. Recent habit started this module. Does not record anything outside class. Aware they look up the same words multiple times (especially connective words) because they do not retain them.	Flashcard-based review using GoodNotes on tablet, but only before exams. Reports flashcards work well when actually used. Words that are frequent and useful in their daily context transfer to productive use; class-specific or low-frequency words are forgotten quickly regardless.	Frames this as a priority problem. Expects exam pressure to temporarily raise engagement, acknowledging this is reactive rather than proactive.	Repetition loop: repeatedly looks up the same words especially connective and functional words without retaining them. Gap between recording and review: writes words down but never returns to them so recording has limited downstream value. Previous friction of phone-based lookup during reading, partially resolved by switching to Kindle.	Lower-friction bridge between recording and review that does not require active scheduling. A system that flags recurring words they have already encountered and looked up before.
P4	Largely opportunistic, no deliberate strategy	No deliberate strategy by word type or frequency. Just tries to understand meaning overall without active triage of which words to engage with.	Glosbe preferred strongly over Google Translate because it provides usage examples and contextual sentences. Previously used Google Translate but found it unreliable for sentences. Also uses AI by screenshotting content for complex homework.	No recording of any kind for Danish. No vocabulary notebook, word list, or personal dictionary. Aware this is a gap. Explicitly says that externally imposed structured vocabulary lists with tests would make them use flashcards and significantly improve learning.	No current active retention strategy for Danish. Most positively recalled retention experience was Quizlet in high school: progressive study sessions tracking percentage of words known with sense of completion over time. Stopped using it when it became monetised. Duolingo is current substitute but rated as low value except for the hardest exercise type (writing a full sentence from	Wants the teacher to assign specific weekly word lists with tests. Believes external accountability would significantly improve learning.	Absence of an external vocabulary curriculum with accountability: wants the teacher to assign specific weekly word lists with tests. Mixed-level class where some classmates are near-native and teacher pitches content to the higher end. Quizlet paywall and Duolingo being too gamified.	Structured external vocabulary curriculum with accountability. A free Quizlet-style flashcard tool with progressive session tracking. Sentence-level flashcards rather than only word-to-word mapping. Has heard about AI-assisted personalised language learning tools and is interested.

D The Weighed Objective Method

The concept evaluation in this thesis uses the weighed objective method described by Cross [50]. The method supports design selection by making objectives, relative priorities, and alternative scores explicit. It does not replace design judgement, but it gives a structured basis for comparing alternatives.

Defining objectives.

The first step is to define the objectives used to evaluate the design alternatives. Each objective should be clearly worded and assessable, so that alternatives can later be scored against it. Objectives should overlap as little as possible, since overlap can lead to the same concern being counted twice.

In the solution chapter, the objectives are labelled with letters, such as A, B, C, and so on. These labels are then reused in the pairwise comparison table, the relative weighting table, and the final concept evaluation.

Ranking objectives through pairwise comparison.

The second step is to compare objectives two at a time. For each pair, the question is whether the row objective is more important than the column objective.

- Enter 1 if the row objective is more important than the column objective.
- Enter 0 if the row objective is less important than the column objective.
- Enter 0.5 in both symmetric cells if the two objectives are equally important.

The row total is calculated by summing the values in each row. The objective with the highest row total is the highest priority objective. The objective with the lowest row total is the lowest priority objective. This produces a rank order of objectives. The completed pairwise comparison used in this thesis is shown in **??**.

Assigning relative weights.

The pairwise comparison produces a rank order, but not yet a numerical weight. The third step is therefore to assign relative importance values. In this thesis, 100 points are distributed across the objectives. Giving more points to one objective means giving fewer points to another, which makes the trade-offs explicit.

The normalised weight of each objective is calculated as:

$$w_i = \frac{\text{points}_i}{100}$$

where w_i is the weight of objective i. The weights should sum to 1.

Scoring alternatives.

After weights are assigned, each design alternative is scored against each objective using the five-point scale in Table D.1. The scale runs from 1 to 5, where 1 means the alternative satisfies the objective inadequately and 5 means it satisfies the objective excellently.

Score	Meaning
1	Inadequate
2	Weak
3	Satisfactory
4	Good
5	Excellent

Table D.1: Five-point scoring scale for evaluating alternatives against each objective.

The weighted score for an alternative is calculated as:

$$\text{Weighted score} = \sum (s_i \times w_i)$$

where s_i is the score from 1 to 5 for objective i, and w_i is the normalised weight of objective i. Since the weights sum to 1, the final weighted score also falls on the 1 to 5 scale. The score should be interpreted alongside the qualitative trade-offs identified during the evaluation.

E Objective Weighting Calculation

This appendix documents the calculation used to assign relative weights to the five objectives used for concept evaluation, following the weighted objective method described in Appendix D. Each objective is compared against every other in turn, and the resulting row totals rank the objectives (Table E.1a). One hundred points are then distributed across that ranking and normalised to a weight, $w_i = \text{points}_i/100$ (Table E.1b). A hundred point total is used in place of a ten point total because it allows finer distinctions while producing identical normalised weights, so it is a reporting format rather than a different calculation.

(a) Pairwise comparison

Obj.	A	B	C	D	E	Total
A	–	1	1	1	1	4
B	0	–	0	1	1	2
C	0	1	–	1	1	3
D	0	0	0	–	1	1
E	0	0	0	0	–	0

(b) Ranking and weights

Obj.	Total	Rank	Points	Weight
A	4	1	30	0.30
C	3	2	25	0.25
B	2	3	20	0.20
D	1	4	15	0.15
E	0	5	10	0.10
		Total	**100**	**1.00**

Table E.1: Objective weighting. (a) Each objective is compared against every other, where a 1 records that the row objective was judged more important than the column objective. (b) Row totals give the ranking, across which 100 points are distributed and normalised to a weight.

The weights preserve the priority order from the pairwise comparison while avoiding a zero weight for Objective E, since prototype feasibility remains a practical constraint on the system even though it ranked last. In the concept evaluation these weights multiply concept scores on the 1–5 scale defined in Table D.1.

F Detailed Concept Evaluation

This appendix gives the score calculation behind concept A, B and C. Objectives carry the weights established in: A 0.30, B 0.20, C 0.25, D 0.15, E 0.10. The weighted score is the objective score multiplied by that weight.

Obj.	Score	Reasoning	Weighted
Concept A: Goal-regulation assistant			
A	4	Supports primarily Rehearsal.	1.20
B	4	Supports transfer from Danish class and everyday encounters into practice when goals attach to personally relevant Collections, such as class vocabulary or words captured from daily situations. Scores no higher because the Goal attaches to an existing Collection rather than to capture itself, leaving the building of a Collection uncovered.	0.80
C	5	Strongly develops autonomy as defined in Subsection 5.1.2: the learner practises taking control of vocabulary learning by setting goals, deciding what progress means, and revising direction without depending on external structure.	1.25
D	3	Moderate interaction burden, since the learner must define and maintain a Goal before it can guide practice. This supports regulation at the cost of more setup.	0.45
E	5	Highly feasible as a web-app prototype: Goal objects, progress counts, and goal-linked practice entry need no mobile-native functionality or complex AI.	0.50
		Total	**4.20**
Concept B: Consistency coach			
A	5	Supports the SRL cycle across the whole vocabulary loop, from prompting the learner to capture new words to prompting return to words already captured.	1.50
B	4	Supports transfer from Danish class and everyday encounters by prompting return to captured Word items and Collections. This depends on notifications and reminders, whose effectiveness rests on timing and relevance, and which can become disruptive if poorly fitted to the learner's context [28].	0.80
C	3	Partly develops autonomy as defined in Subsection 5.1.2, helping the learner sustain activity across time while creating an extra dependency on system prompts, so independent regulation is weaker than in a concept where goals or reflection are more learner-led.	0.75
D	5	Low interaction burden, since most support comes from passive activity recording and lightweight prompts. The learner need not define goals or complete reflection steps before practising.	0.75
E	2	Weakly feasible as a web-app prototype, because the value depends on well-timed re-engagement outside the app. Activity logs and in-app prompts are straightforward; delivering the reminders requires push subscription handling and a server-side scheduler, and on iOS requires every participant to install the web app and grant notification permission before use begins.	0.20
		Total	**4.00**
Concept C: Reflective focus coach			
A	4	Supports the complete SRL cycle: rehearsal outcomes feed into reflection, reflection creates a next focus, and that focus leads back into practice. The cycle is centred on the practising part of the system rather than spanning the whole vocabulary loop from encounter and capture to organisation and later practice.	1.20
B	4	Supports transfer from Danish class and everyday encounters by helping learners return to words captured and practised earlier. The concept starts from existing Word items and weak-word evidence, so it does not directly support transfer from new encounters into added vocabulary.	0.80
C	3	Partly develops autonomy as defined in Subsection 5.1.2: the learner reflects on practice evidence and can choose whether weak Word items guide the next session. Support is mostly limited to adjusting practice, leaving learner control over goals, capture, and broader direction less developed.	0.75
D	5	Low interaction burden, since weak Word items are generated automatically from practice outcomes and can be used directly as a focus for the next session. The learner need not define goals or maintain a separate plan.	0.75

Obj.	Score	Reasoning	Weighted
E	5	Highly feasible as a web-app prototype: weak-word sets can be generated from incorrect answers or repeated mistakes using ordinary web-app logic, without notifications, background services, or complex AI infrastructure.	0.50
		Total	**4.00**

Table F.1: Score calculation for the three concepts. Objective weights are A 0.30, B 0.20, C 0.25, D 0.15, E 0.10.

G Design Build-Cycle Log: from Concept A to the Final System

This appendix reports the detail behind the overview in Table 6.9: per round, the design question, what was built, what system testing covered, and what validation with participants found. System testing followed the same manual, black-box procedure in every round, described in Section 6.6; what follows is what that procedure covered, not the procedure itself.

G.1 Round 1: Initial Prototype

Design question

Is a goal attached to a collection, on its own, enough to scaffold forethought and make practice accountable to a learner-set target?

What was built

Round 1 established most of the system: learner accounts, with registration, login, and a login that survives closing the app; the database and its deployment on Render; capturing and storing words; translation, with the language model standing in when translation is unavailable; generated usage examples and practice content; pronunciation audio; collections that can be created, renamed, and deleted; the figures behind the progress display; the interface itself; and the ability to install the app to the home screen and open it while offline. Concept A itself was realised as a goal set on each collection, carrying learner-defined goals (M1), mastery framing (M5), entry into practice guided by that goal, and progress feedback (M12, M15). It was laid out across three tabs (Figure G.1): **Capture** (capture, enrich, organise); **Practice** (set a goal for a collection, generate practice guided by it, update the goal, and strengthen knowledge); and **Progress** (the history of goals, and reflecting on them). A practice session was assembled by the system itself, drawing words at random from the collection the learner selected and building eighteen exercises around them, with the wording of each exercise generated by the language model.

System testing

Element	What was verified
Accounts and sessions	An account can be created, the password is stored in scrambled form rather than as readable text, login persists across page reloads, and logging out clears it.
Word capture and storage	Submitting a word creates an entry with the correct text, translation, and owner.
Translation and fallback	Translations come from the live service rather than being simulated, and when that service cannot be reached the language model returns a usable translation instead.
Generated content	Usage examples and exercises target the correct word and are grammatical, with gaps falling on the intended word.
Pronunciation audio	Returned audio is playable Danish speech, including for irregular and loanword pronunciations.

Element	What was verified
Collections	A collection can be created, renamed, and deleted, with deletion carrying through to its contents.
Per-collection goal	A goal set on a collection persists, and its displayed progress matches the activity actually recorded.
Progress figures	Returned counts match recorded activity rather than only a correct-looking screen.
Interface and offline use	Moving between tabs works without the page reloading; the app installs to the home screen and still opens with the network slowed or absent.
Deployment	The deployed system starts, connects to the live database, and offers the same flows over a secure connection as it does during development.

Table G.1: Elements covered by Round 1 system testing.

Validation: answering the design question

Three participants (Table G.2), individual think-aloud on their own phone, 30 minutes, followed by an unstructured debrief [87, 54]: capture a word and review its enrichment, create a collection and add the word to it, set a goal for the collection, complete a practice session, and check progress toward the goal.

ID	Danish level	Occupation
P1	Module 3	Student
P2	Module 2	Software engineer
P3	Not a learner	Student

Table G.2: Participants, Round 1.

Answer: no. A goal attached to a collection is not, on its own, enough to scaffold forethought or hold practice accountable to a learner-set target. Capture and organisation held up, and the goal itself was set without difficulty; what failed was its scope, since a target covering one collection did not match how participants thought about their vocabulary learning. Table G.3 sets out the findings and what each redirected.

Operation	Finding	Redirect for Round 2
Capture, Enrich, Organize	No breakdowns; the smoothest part of the test.	Retained unchanged.
Set collection goal	Set without difficulty and understood, but a target bound to one collection was not seen as worth maintaining: participants judged their progress by their vocabulary learning as a whole rather than one collection at a time.	Move the goal off the individual collection and onto the whole of the learner's vocabulary activity, as a commitment that recurs and is resurfaced in the application itself (M1, M8), with non-punitive framing of lapses (M14).

Operation	Finding	Redirect for Round 2
Generate, Strengthen knowledge	Exercise formats understood, but sessions of eighteen exercises felt too long; one request for a pronunciation option.	Shorten the session.
Update goal, Reflect on goals	Progress read as a plain count rather than an accomplishment, with only one fixed-target goal type available.	Replace with an achievements page (M16); add a consistency-oriented goal type.
Cross-cutting	Participants wanted a single consolidated view of goals, questioned why capture needed its own tab, and expected a default goal to be proposed.	Consolidate goals into one tab, fold capture into Practice, and propose a default weekly target.

Table G.3: Findings and redirect, Round 1, consolidated across P1–P3.

G.2 Round 2: Weekly Goals Redesign

Design question

Does restructuring the autonomy loop around a recurring weekly goals over the learner's whole vocabulary activity, ones that resurfaces commitments and frames outcomes as accomplishments, resolve the Round 1 breakdowns?

What was built

- **Weekly goal cycle**, moving from setting goals, through the week itself, to a review: four goal categories, add words, practice words, complete sessions, and master words, each with a recommended target, and a review the learner must complete before new goals can be set (M1, M8, M14).
- **Self-adjusting targets**: a met goal raises next week's recommended target by five; a missed goal keeps it (M15).
- **Re-earnable mastery**: a word counts as mastered only if the most recent practice session was error-free (M12, M13).
- **Session tracking and achievements page**: a words-over-time chart and a grid of met goals, replacing the plain progress count (M5, M12).
- **Shorter practice sessions**: cut from eighteen exercises to twelve. The system draws six words from the selected collection and builds two exercises for each, one multiple-choice and one gap-fill in context, with the wording of both generated on request by the language model. Candidates are drawn at random, but a word whose most recent session was answered correctly is set aside and another drawn in its place, biasing the six toward what the learner has not yet answered correctly.
- **Resumable sessions**: leaving mid-session and returning offers to resume or to start again.
- **Generated exercise content**: fresh gap-fill sentences each session, and multiple-choice alternatives matched on part of speech and length, both tappable to save as new words (M4, M6).
- **Live spelling check**: typing in the add-word dialogue raises a suggestion without a button press.
- **Adding a word**: moved out of its own tab into Practice, with the correct fields enabled according to the chosen direction, English to Danish or Danish to English.

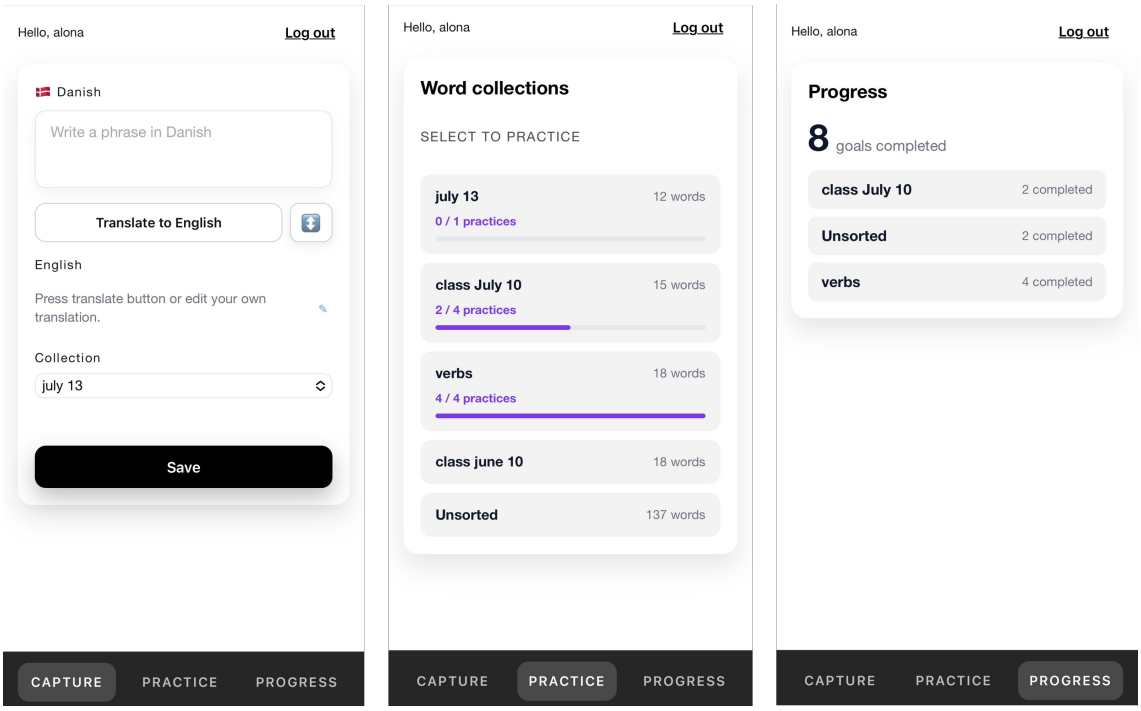


Figure G.1: Round 1 interface: Capture, Practice, and Progress tabs.

The three tabs became Current Goals, Practice, and Achievements (Figure G.2).

System testing

Element	What was verified
Weekly goal cycle	Setting goals, the week itself, and the review follow one another correctly as one week turns into the next, checked by setting the week's start and end dates by hand.
Goal progress	Live figures for each goal category match the activity recorded within that week.
Target adjustment	A met goal raises next week's recommended target by five; a missed goal leaves it unchanged.
Re-earnable mastery	A word is mastered only if its most recent session was error-free; an old mistake does not block re-mastery, and a new mistake removes it.
Session composition	A session contains twelve exercises built from six words of the selected collection, two per word, one multiple-choice and one gap-fill, with the words drawn from that collection alone.
Session tracking	Exercises answered in one sitting count as a single session, so the running total rises once per session rather than once per exercise.
Multiple-choice alternatives	Generated alternatives are plausible, matched on part of speech and length, without duplicates or giving away the answer.
Achievements chart	Chart values match the underlying cumulative counts at several points.
Resumable sessions	Closing the app part-way through a session and reopening it offers to carry on from where the learner left off.
Spelling check	Misspelled English or Danish input raises a suggestion without any further action.

Element	What was verified
Adding a word	Switching direction enables the correct field and disables the other.

Table G.4: Elements covered by Round 2 system testing.

Continued use across successive builds surfaced defects that a single straight-through walkthrough did not, since they appeared only under rapid or unusual interaction: chart axis scaling that made small values look near-empty; figures on the achievements page stuck at zero when they appeared before their data had arrived; gap-fill exercises occasionally appearing without a field to type into; tapping ahead too quickly corrupting the answer field; unfinished sessions silently discarding progress; and multiple-choice cards overflowing on long words. All six were fixed and re-checked: the chart scale was tightened to the real range of the data, the figures refresh once their data arrives, gap-fill exercises retry and fall back when generation fails, rapid tapping is now blocked until the screen has settled, sessions became resumable, and answer cards resize to fit.

Validation: answering the design question

Two participants (Table G.5), individual think-aloud, around 15 minutes, in the same format as Round 1: create a collection, add words to it, practise the collection, and reflect on progress.

ID	Danish level	Occupation
P1	Module 1	Student
P2	Module 1	Student

Table G.5: Participants, Round 2.

Answer: yes. Restructuring the autonomy loop around a recurring weekly goal did resolve the Round 1 breakdowns: a goal covering the learner's whole vocabulary activity was worth maintaining across the week, progress read as accomplishment, and reception was positive overall. Two further findings concerned finish rather than the goal mechanism. Colours across the interface did not sit well together, and a small number of interface defects surfaced during use. Both were carried into Round 3 and fixed there.

G.3 Round 3: Final System

Design question

Does the resulting system support learner autonomy in retaining self-encountered vocabulary?

What was built

The colours were revised into a consistent palette and the interface defects surfaced in Round 2 were fixed. No further design mechanisms were added. What Round 3 introduced instead was usage logging, so that real-world use could be observed. This is a backend-only layer: nothing about it is surfaced to the learner, and it exists to support the analysis rather than the learning.

- **Login events**: one entry per successful login, recording the learner and the time.
- **Visits**: app-opens rather than logins, updated whenever a logged-in learner does anything in the app. Activity more than thirty minutes after the last opens a new visit,

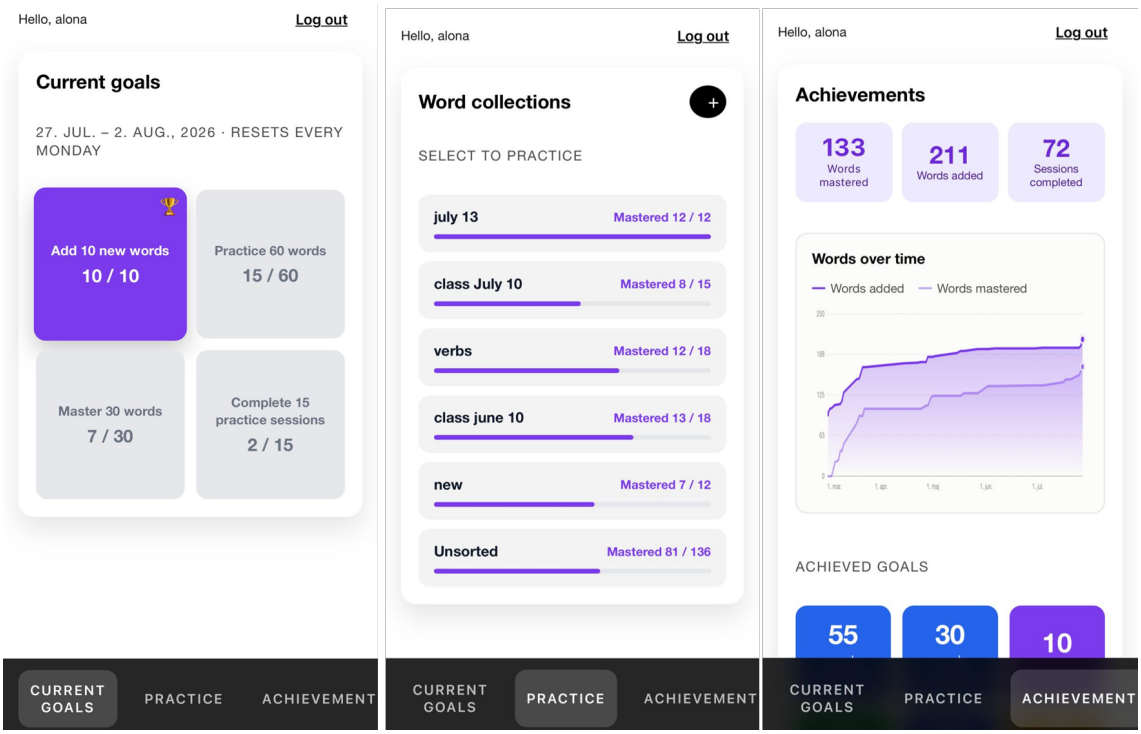


Figure G.2: Round 2 interface: Current Goals, Practice, and Achievements tabs.

which is what counts as opening the app afresh; activity within that window extends the current one, recorded at most once a minute. The distinction matters because the app is installed to the home screen, so a learner may return for days without logging in again and logins alone would understate use.

- **Report script**: run on demand outside the application, drawing the records together into per-learner lifetime figures: app opens, distinct days used, total time spent, words added and how many of them came from inside an exercise, collections created, targets adjusted, and sessions opened and completed. The per-learner figures reported in Section 6.8 are drawn from this script.

These sit on top of the learning records built in earlier rounds, which remain user-facing: one entry per answered exercise, scored by attempt quality and grouped by session, driving mastery and the weighting of word selection; one entry per practice run, driving completed-session counts; the weekly goal commitments with their commit, review, and acknowledge cycle; and the achievements timeline of cumulative words added and mastered. All of it is held in the same database, whose structure is updated automatically when the system starts. The full system is described in Section 6.7.

System testing

Element	What was verified
Login events	Repeated logins each create exactly one entry, with the correct time and learner.
Visit events	Opening the installed app on an existing session, without logging in afresh, still creates a visit entry distinct from a login; activity after more than thirty minutes opens a new visit, while activity within that window extends the current one.

Element	What was verified
Usage report	Aggregated logins, visits, and session totals match manual queries against known activity, and the app-open figure is consistent with the recorded visits.
Round 2 defects	The steps that had produced each defect were repeated after this round's interface changes, to confirm none recurred.

Table G.6: Elements covered by Round 3 system testing.

Validation: answering the design question

Whether the resulting system supports learner autonomy in retaining self-encountered vocabulary was put to the test through sustained independent use by learners in their own everyday context, with usage logs, as set out in Section 6.8. The answer is reported in Section 6.9.

H Self-Regulation Questionnaire Items

The self-regulation items used in the pre- and post-treatment questionnaires were adapted, without altering the original item wording, from the self-regulation subscale of the Language Learner Autonomy Scale (LLAS) [139]. The subscale comprises two components, *identifying needs and learning goals* and *planning and monitoring the learning process*, listed here in full as administered (Table H.1). The components are shown because they form part of the published instrument; the evaluation reported in this thesis treats the eleven items as a single scale, for the reasons given there.

Component	**Code**	**Item wording**
Identifying needs and learning goals	SRN1	I am able to set my own learning goals (in Danish vocabulary learning)
	SRN2	I am able to identify my own needs (e.g., why I want to learn Danish vocabulary)
	SRN3	I am able to identify my own learning problems and means of addressing them (in Danish vocabulary learning)
	SRN4	I am able to identify my strengths and weaknesses and structure my learning accordingly (in Danish vocabulary learning)
	SRN5	I am able to evaluate to what extent I have achieved my learning goals (in Danish vocabulary learning)
Planning and monitoring the learning process	SRP1	I am able to work with a variety of materials and resources to enhance learning (in Danish vocabulary learning)
	SRP2	I am able to find information about new topics on my own (in Danish vocabulary learning)
	SRP3	I am able to identify and develop learning strategies (e.g., learning words by association, repeating words or sentences, studying with flashcards)
	SRP5	I am able to develop the ability to study by myself (Danish vocabulary)
	SRP6	I am able to plan where I want to learn (e.g., in/outside the classroom, at home, in the library)
	SRP7	I am able to develop daily/weekly learning plans (for Danish vocabulary)

Table H.1: Self-regulation subscale items.

I Use of AI Tools

Generative AI tools were used during the preparation of this thesis. They supported brainstorming, reformulation, and software implementation.

Two categories of use are distinguished. In the written thesis and the design process, conversational assistants were used to explore ideas, question framings, locate literature for subsequent verification at the source, and improve the clarity of text drafted by the author.

In the prototype, an agentic coding assistant was used to implement the system, working from specifications written by the author.

Tool	Purpose	Example of use
Claude, ChatGPT	Brainstorming and structuring	Generating alternative framings of the problem statement and candidate directions for the design, exploring how chapters could be organised, and questioning whether a proposed framing was supported by the available evidence.
Claude, ChatGPT	Language and clarity	Reformulating passages drafted by the author for readability, shortening overlong sentences, and checking grammar and phrasing.
Claude, ChatGPT	Literature orientation	Suggesting research fields and candidate sources relevant to a topic, each of which was then located and read at the original source before citation.
Claude Code	Prototype implementation	Implementing the data model for word items, collections, and practice sessions from a specification written by the author; setting up user registration and multiple user support, Flask routes, prompts for OpenAIAPI.
Claude Code	Debugging and refactoring	Locating and correcting defects identified during the build cycles. **??**.

Table I.1: Generative AI tools used during the preparation of this thesis, the purpose each served, and an example of that use.